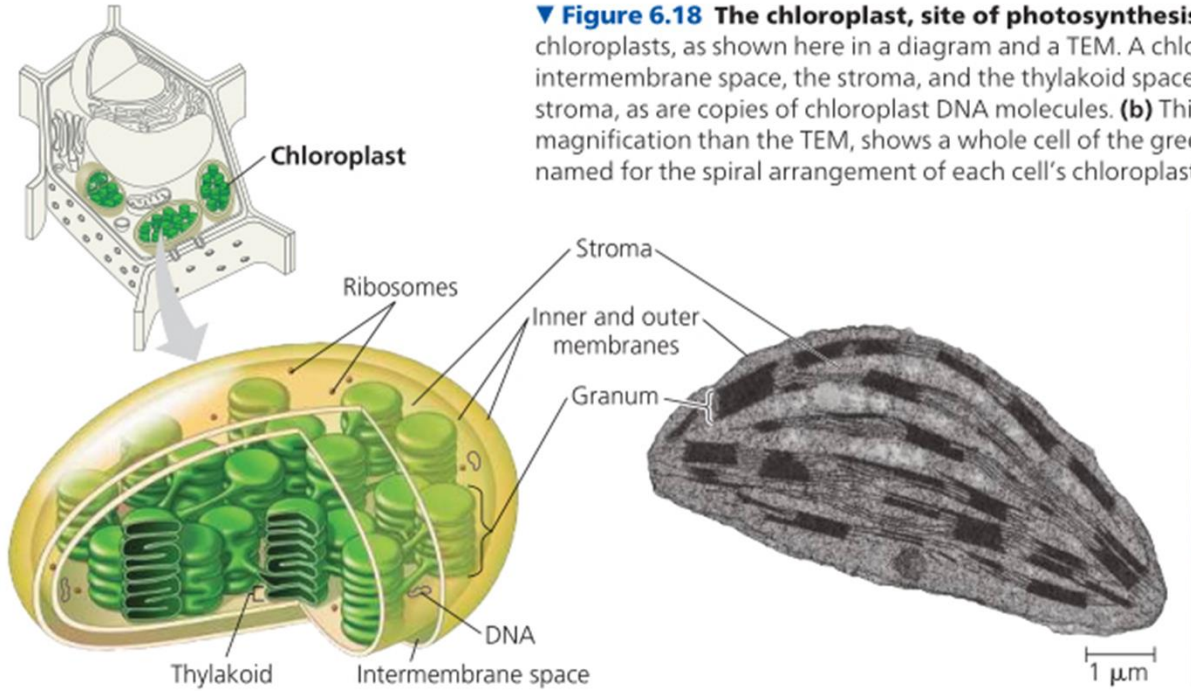
MBG 1201 Introduction to Modern Biology

Course Lecturer: Şebnem Öztürk
seb.ozturk@gmail.com
sebnemozturk@marun.edu.tr
22 October 2024

The chloroplast, site of photosynthesis

The chloroplast is a member of a family of organelles called **plastids**
Chloroplasts contain the green pigment chlorophyll, as well as enzymes and other molecules that function in the photosynthetic production of sugar.
Chloroplasts are found in leaves and other green organs of plants and in algae
Chloroplast structure includes:
Thylakoids, membranous sacs, stacked to form a **granum**

Stroma, the -fluid outside the thylakoids. Stroma contains the chloroplast DNA, ribosomes and many enzymes.
The membranes of the chloroplast divide the chloroplast space into three compartments: the intermembrane space, the stroma, and the thylakoid space. This compartmental organization enables the chloroplast to convert light energy to chemical energy during photosynthesis.



▼ **Figure 6.18 The chloroplast, site of photosynthesis. (a)** Many plants have lens-shaped chloroplasts, as shown here in a diagram and a TEM. A chloroplast has three compartments: the intermembrane space, the stroma, and the thylakoid space. Free ribosomes are present in the stroma, as are copies of chloroplast DNA molecules. **(b)** This light micrograph, at a much lower magnification than the TEM, shows a whole cell of the green alga *Spirogyra crassa*, which is named for the spiral arrangement of each cell's chloroplasts.

(a) Diagram and TEM of chloroplast

(b) Chloroplasts in an algal cell

The chloroplast, site of photosynthesis

1. Light-Dependent Reactions (Occurs in the thylakoid membranes)

These reactions capture sunlight and convert it into chemical energy in the form of ATP and NADPH.

Absorption of Light: Chlorophyl absorbs light. This energy excites electrons in the chlorophyll molecules.

Water Splitting: The energy from the light is used to split H_2O into O_2, protons (H^+), and electrons. O_2 is released as a byproduct.

Electron Transport Chain: The excited electrons travel through a series of proteins in the thylakoid membrane, called the electron transport chain. As they move, they pump protons into the thylakoid space, creating a proton gradient.

ATP Formation: The proton gradient powers an enzyme called ATP synthase, which synthesizes ATP from ADP and inorganic phosphate (Pi). This process is called photophosphorylation.

NADPH Formation: Electrons at the end of the chain combine with $NADP^+$ and H^+ to form NADPH, another energy-carrying molecule.

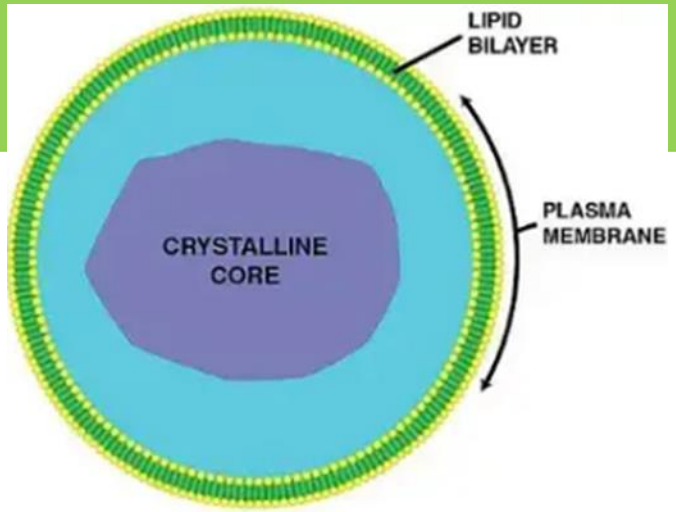
2. Calvin Cycle (Light-Independent Reactions) (Occurs in the stroma)

The ATP and NADPH produced in the light-dependent reactions are used to convert CO_2 into glucose and other sugars.

Equation of Photosynthesis:

$$6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

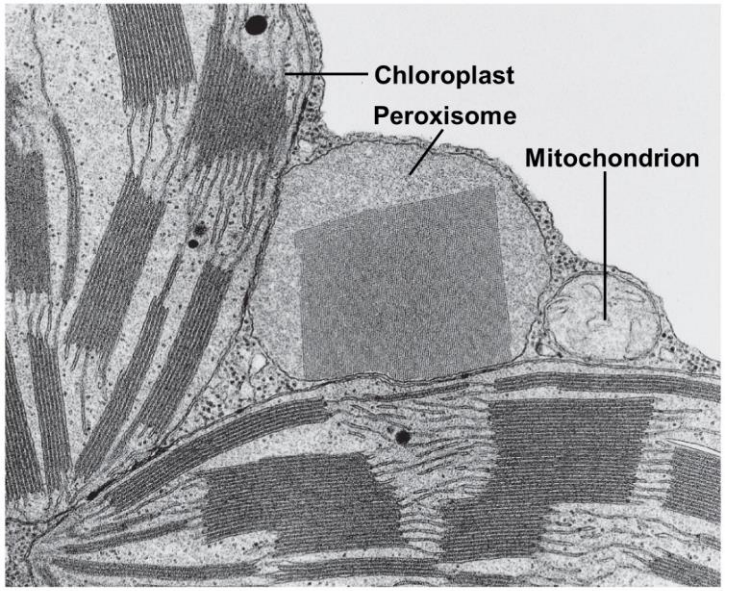
Peroxisomes are oxidative organelles



Peroxisomes are specialized metabolic compartments bounded by a single membrane

Peroxisomes perform reactions with many different functions. Oxygen is used to break down different types of molecules. (fatty acids, alcohol). They also detoxify harmful substances.

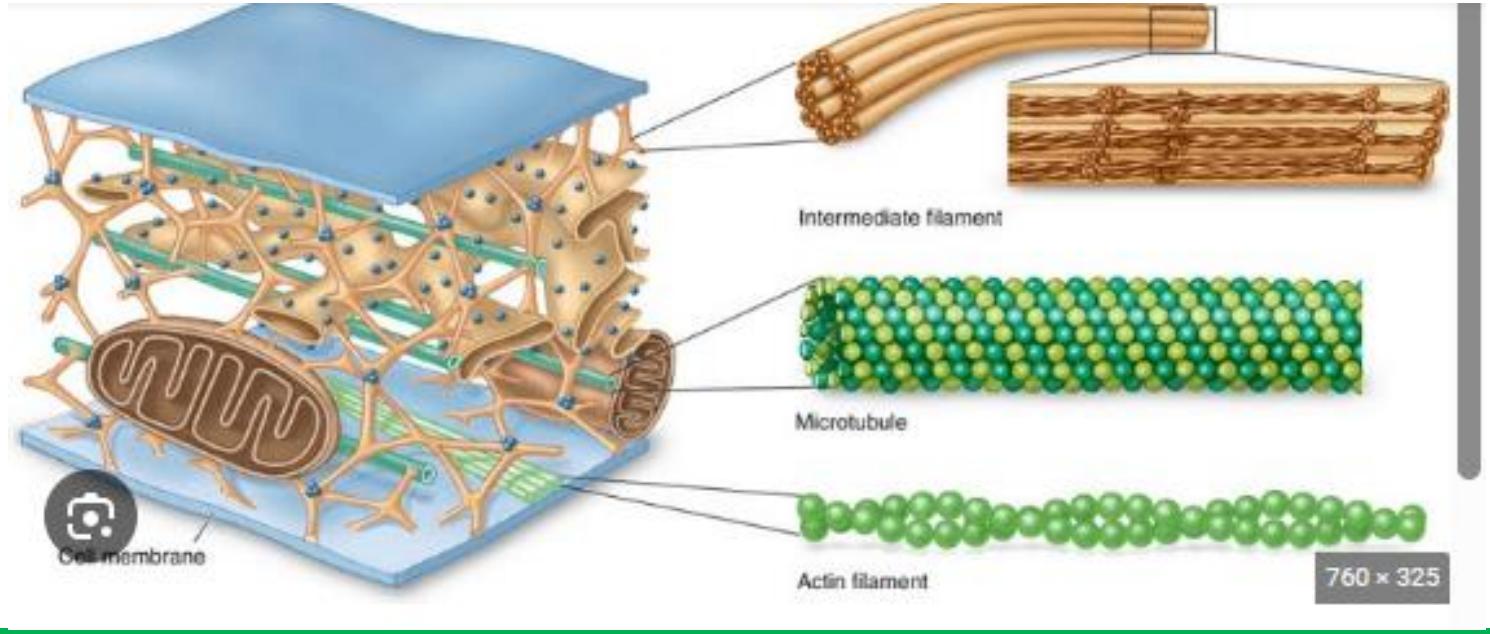
Peroxisomes produce hydrogen peroxide as a byproduct which is also toxic and convert it to water and oxygen by the enzyme catalase which is present in high concentrations.



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The cytoskeleton

- The cytoskeleton (Cell skeleton) is a network of fibers that organizes structures and activities in the cell
- The cytoskeleton extends throughout the cytoplasm
- It organizes the cell's structures and activities, anchoring many organelles



The cytoskeleton

It is composed of three types of constantly assembling and disassembling molecular structures:

Microtubules

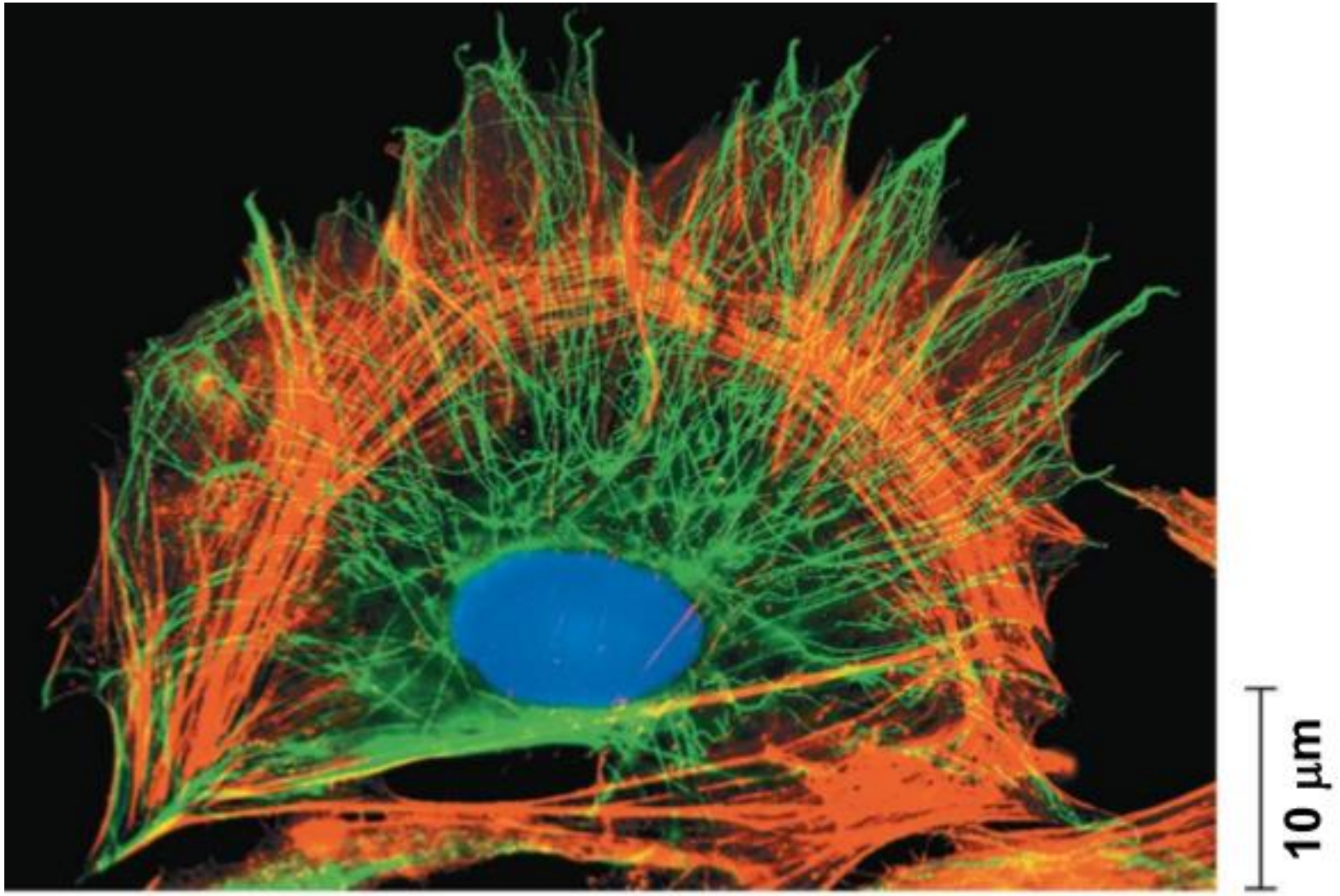
Microfilaments

Intermediate filaments

the cytoskeleton extends throughout the cell.

The cytoskeletal elements have been tagged with different fluorescent molecules: green for microtubules and orange for microfilaments.

Intermediate filaments, is not evident here. (The blue color tags the DNA in the nucleus.)



Functions of the Cytoskeleton: Support, Motility, and Regulation

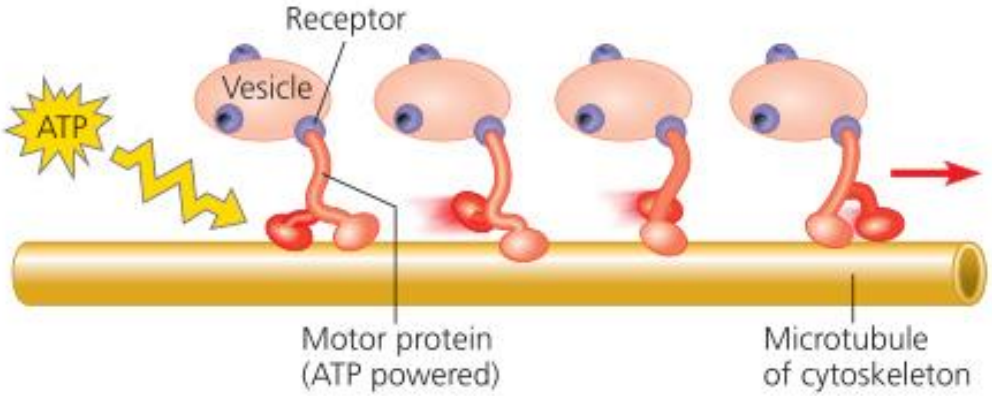
The cytoskeleton helps to support the cell and maintain its shape (Gives the cell structure)

It provides anchorage for many organelles and molecules (It holds them in place)

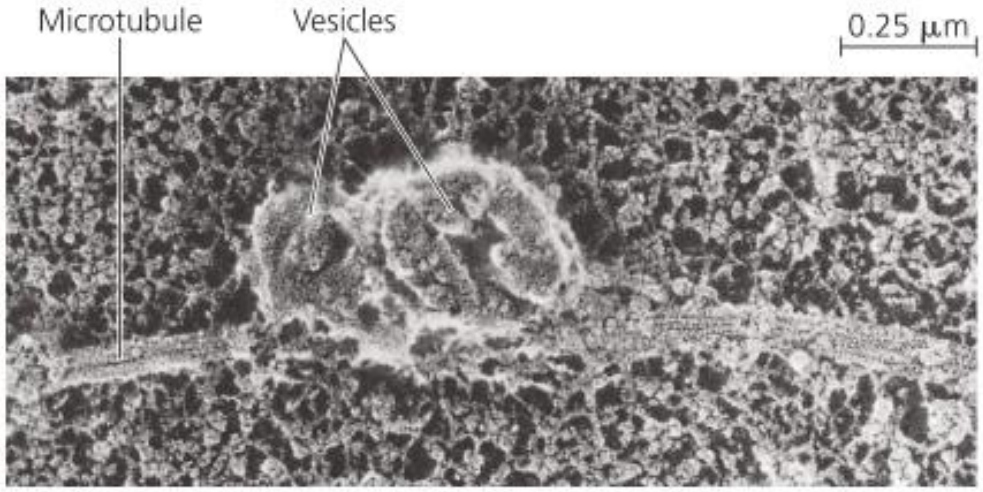
It interacts with motor proteins to produce motility. (motility includes both changes in cell location and movements of cell parts.)

Inside the cell, vesicles and other organelles can “walk” along the tracks provided by the cytoskeleton.

▼ **Figure 6.21 Motor proteins and the cytoskeleton.**

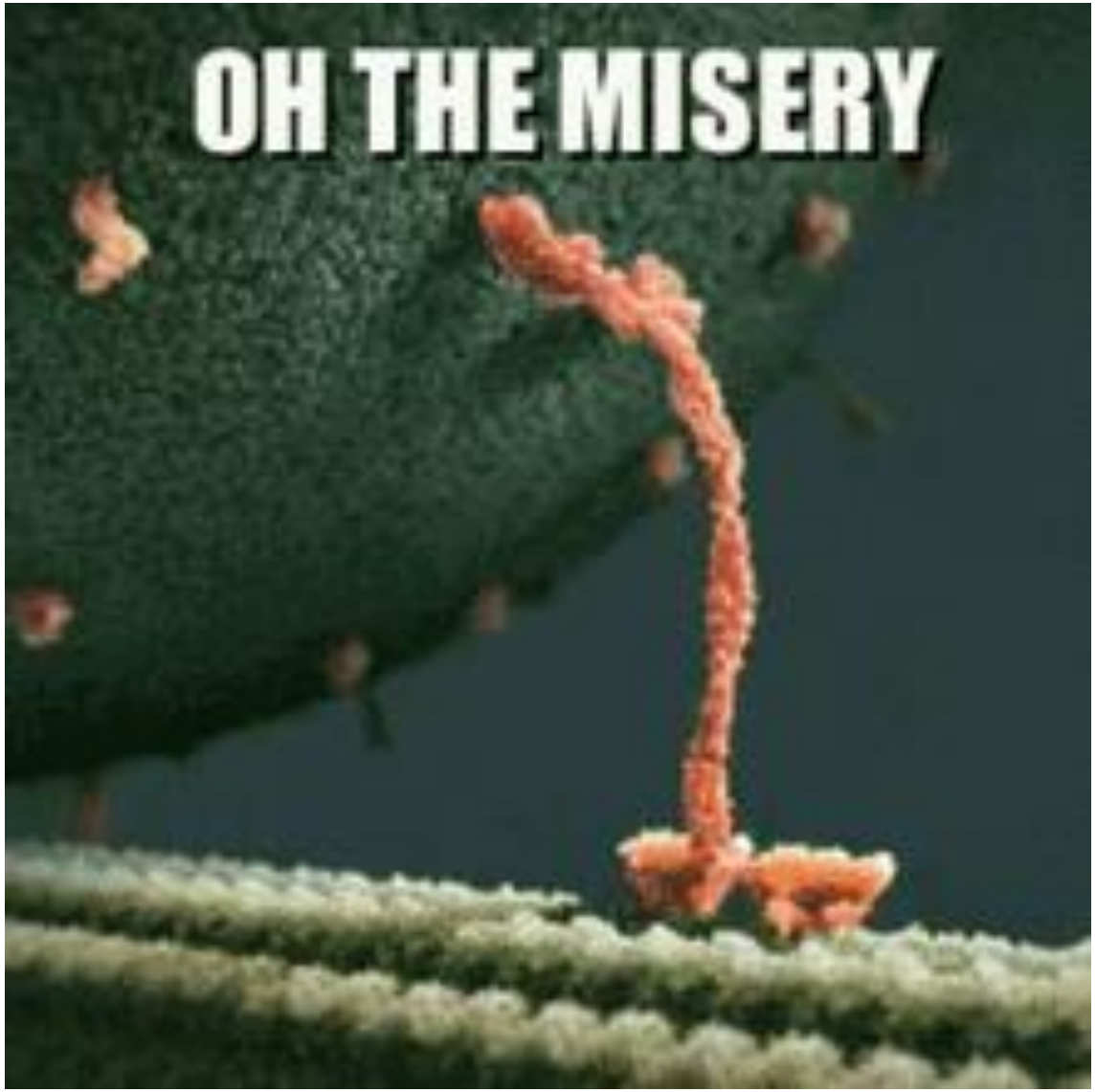


(a) A motor protein that attaches to a receptor on a vesicle can "walk" the vesicle along a microtubule or microfilament.

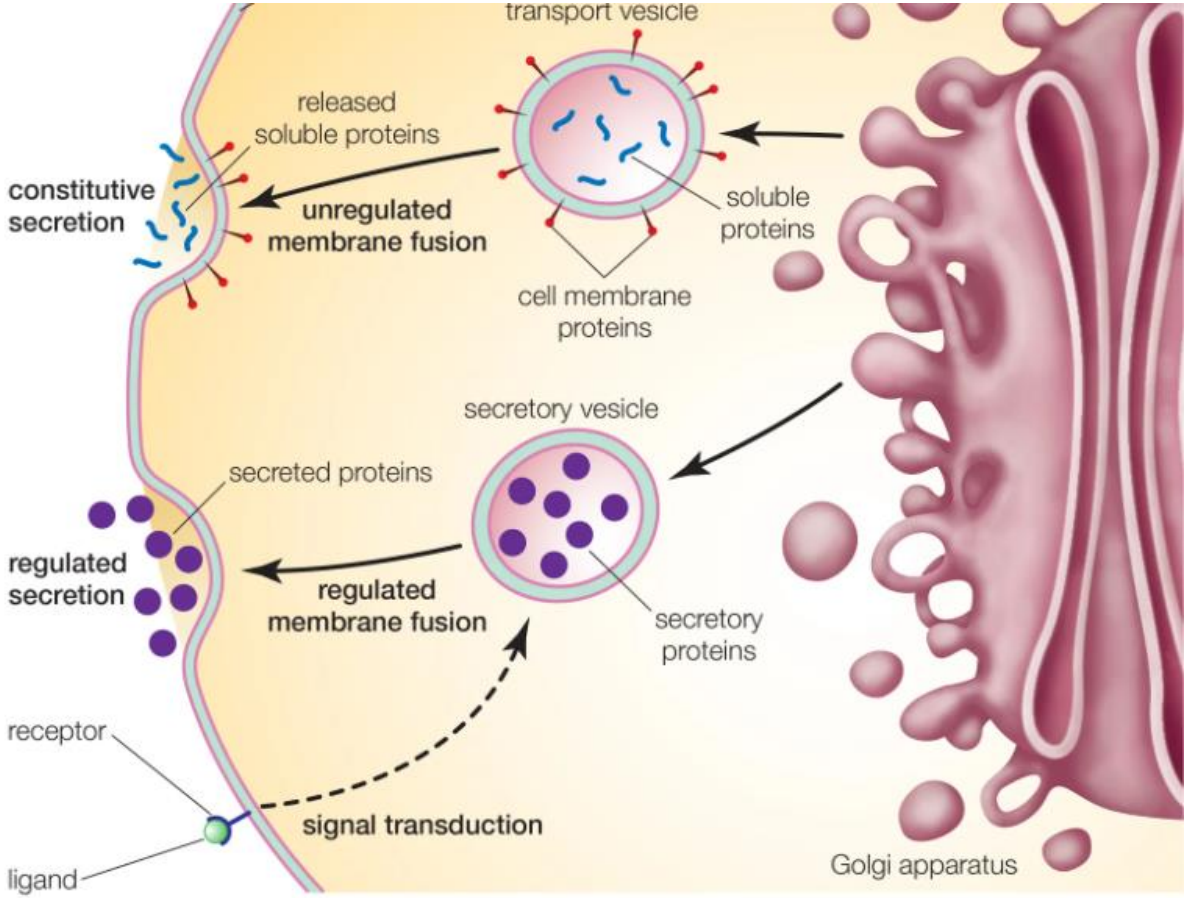


(b) Two vesicles containing neurotransmitters move along a microtubule toward the tip of a nerve cell extension called an axon (SEM).

The cytoskeleton



Kinesin (motor protein) walking on a microtubule carrying a vesicle.



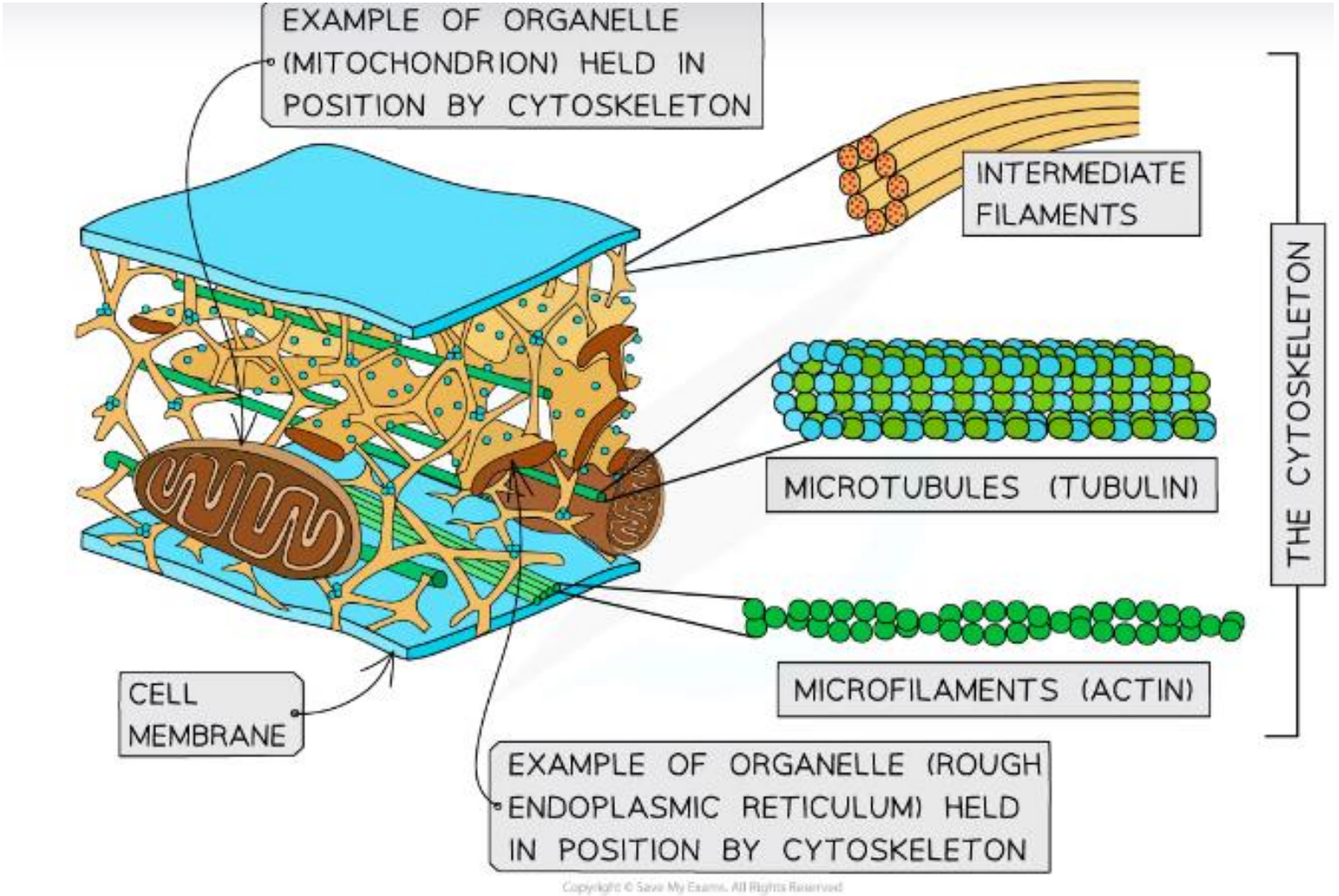
Components of the Cytoskeleton

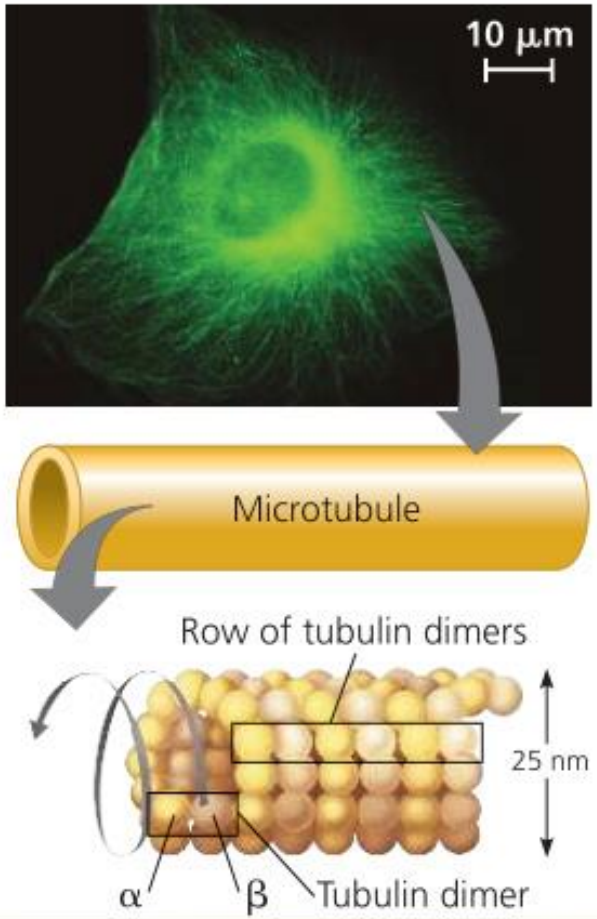
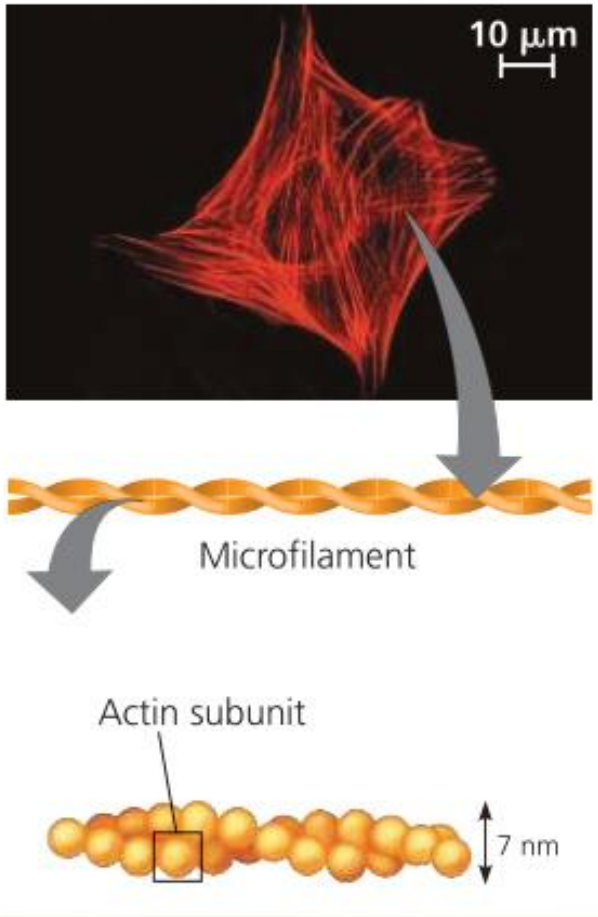
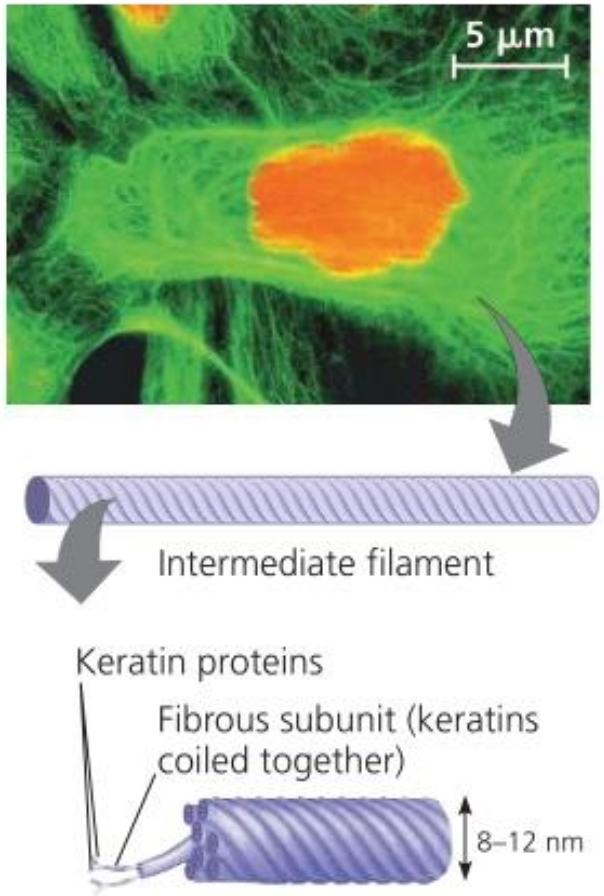
Three main types of fibers make up the cytoskeleton:

Microtubules are the thickest

Microfilaments (actin filaments) are the thinnest.

Intermediate filaments are fibers with diameters in a middle range.



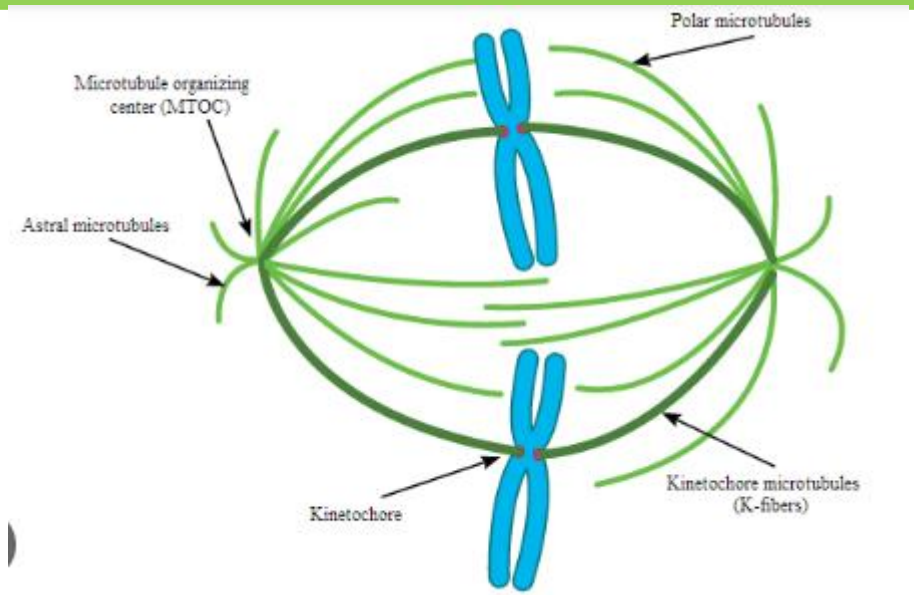
Property	Microtubules (Tubulin Polymers)	Microfilaments (Actin Filaments)	Intermediate Filaments
Structure	Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables
Diameter	25 nm with 15-nm lumen	7 nm	8–12 nm
Protein subunits	Tubulin, a dimer consisting of an α-tubulin and a β-tubulin	Actin	One of several different proteins (including keratins)
Main functions	Maintenance of cell shape; cell motility; chromosome movements in cell division; organelle movements	Maintenance of cell shape; changes in cell shape; muscle contraction; cytoplasmic streaming (plant cells); cell motility; cell division (animal cells)	Maintenance of cell shape; anchorage of nucleus and certain other organelles; formation of nuclear lamina
Fluorescence micrographs of fibroblasts. Connective tissue cells called fibroblasts are a favorite cell type for cell biology studies because they spread out flat and their internal structures are easy to see. In each fibroblast shown here, the structure of interest has been tagged with fluorescent molecules. In the third micrograph, the DNA in the nucleus has also been tagged (orange).			

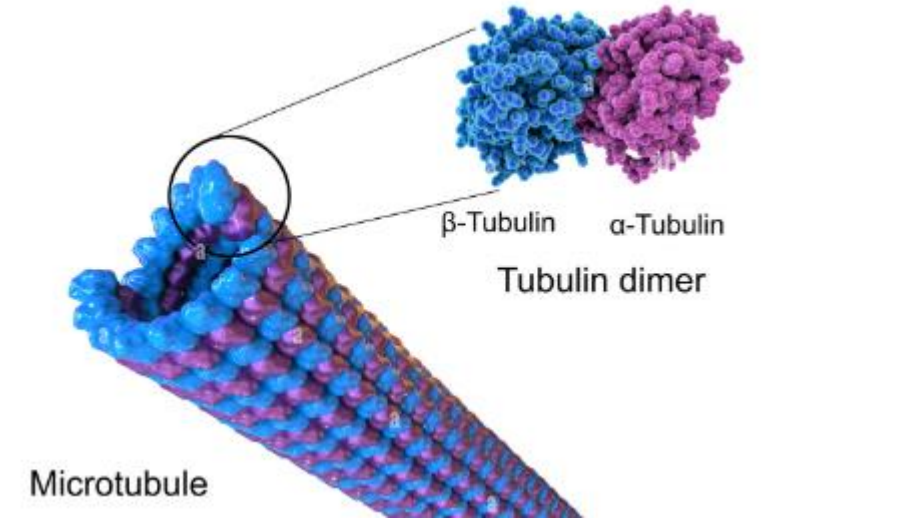
Microtubules

Microtubules are hollow rods constructed from globular protein dimers called tubulin

Functions of microtubules

- Shape and support the cell (Resist compression)
- Guide movement of organelles (microtubules guide vesicles from the ER to the Golgi apparatus and from the Golgi to the plasma membrane)
- Separate chromosomes during cell division



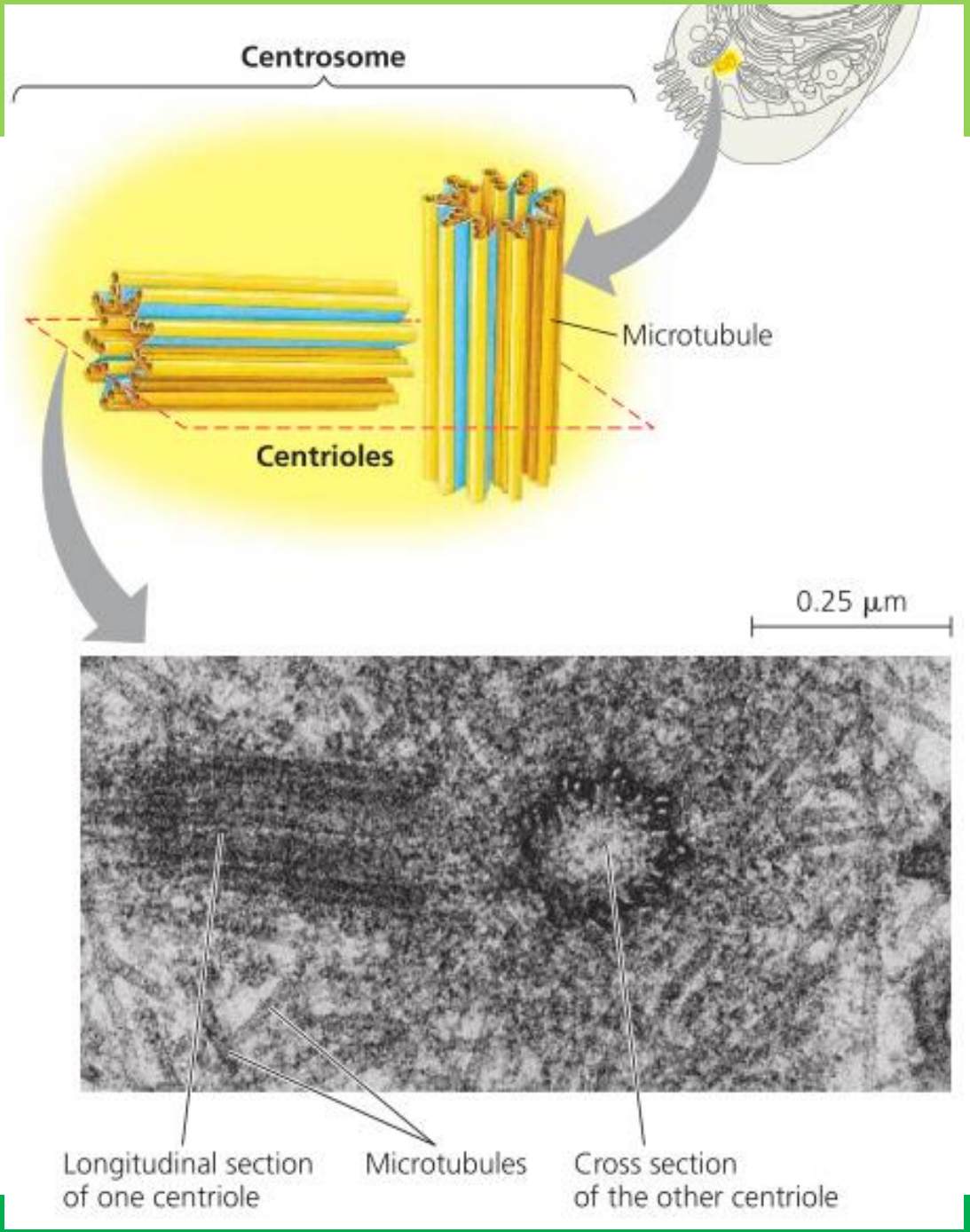


Centrosomes and Centrioles

In many cells, microtubules grow out from a centrosome a region that is often located near the nucleus.

The centrosome is a "microtubule-organizing center"

In animal cells, the centrosome has a pair of centrioles, each with nine triplets of microtubules arranged in a ring



Cilia and Flagella

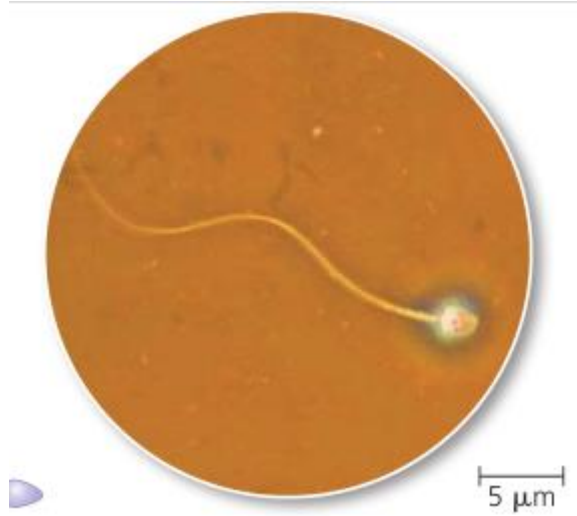
Microtubules control the beating of cilia and flagella, locomotor appendages of some cells.

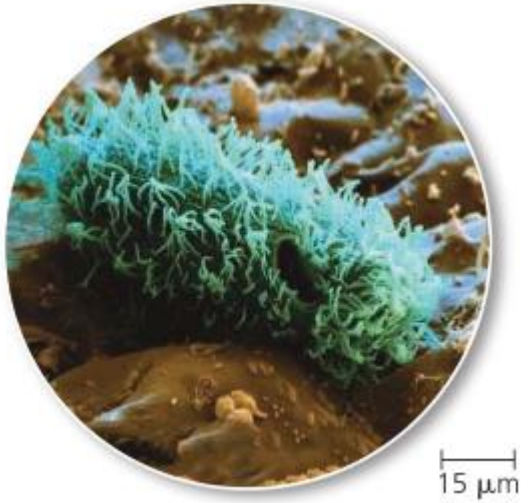
Many unicellular protists are propelled through water by cilia or flagella

the sperm of animals, algae, and some plants have flagella.

Cilia and flagella differ in their beating patterns.
A flagellum has a snakelike motion driving a cell in the same direction as the axis of the flagellum.

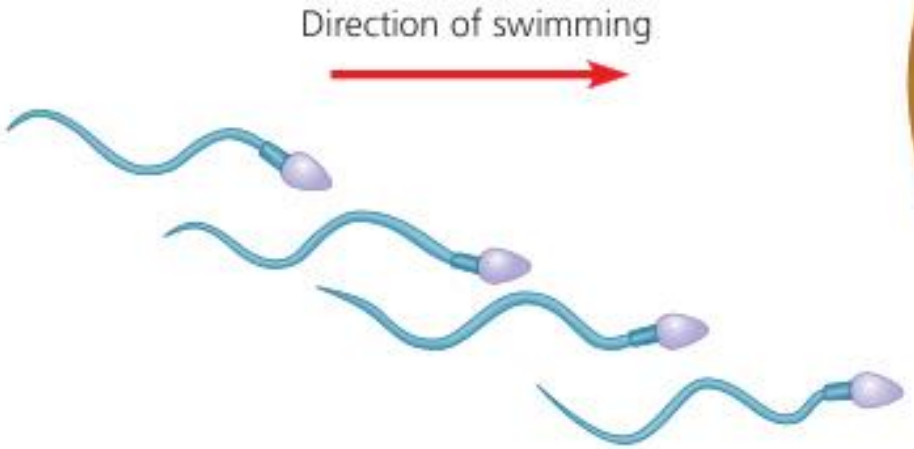
Cilia have a back-and-forth motion. The rapid power stroke moves the cell in a direction perpendicular to the axis of the cilium. Then, during the slower recovery stroke,the cilium bends and sweeps sideways, closer to the cell surface.





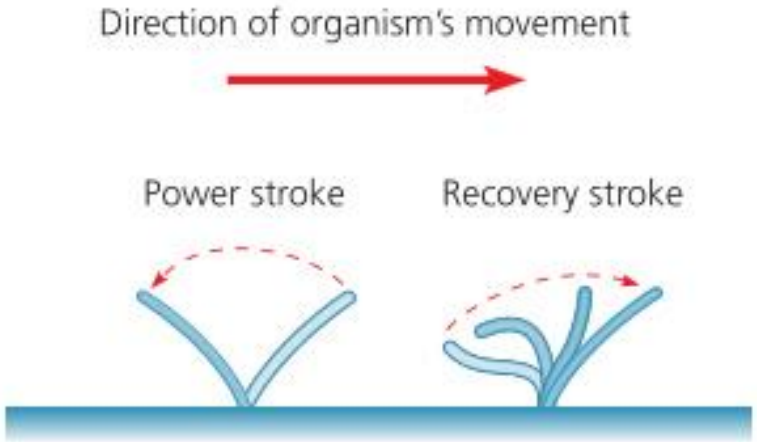
▼ Figure 6.23 A comparison of the beating of flagella and motile cilia.

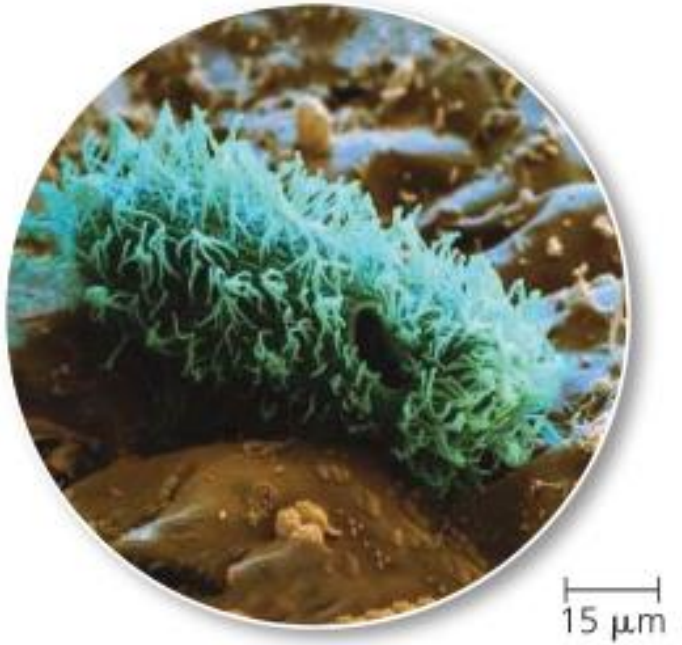
(a) Motion of flagella. A flagellum usually undulates, its snakelike motion driving a cell in the same direction as the axis of the flagellum. Propulsion of a human sperm cell is an example of flagellate locomotion (LM).





(b) Motion of cilia. Cilia have a back-and-forth motion. The rapid power stroke moves the cell in a direction perpendicular to the axis of the cilium. Then, during the slower recovery stroke, the cilium bends and sweeps sideways, closer to the cell surface. A dense nap of cilia, beating at a rate of about 40 to 60 strokes a second, covers this *Colpidium*, a freshwater protist (colorized SEM).



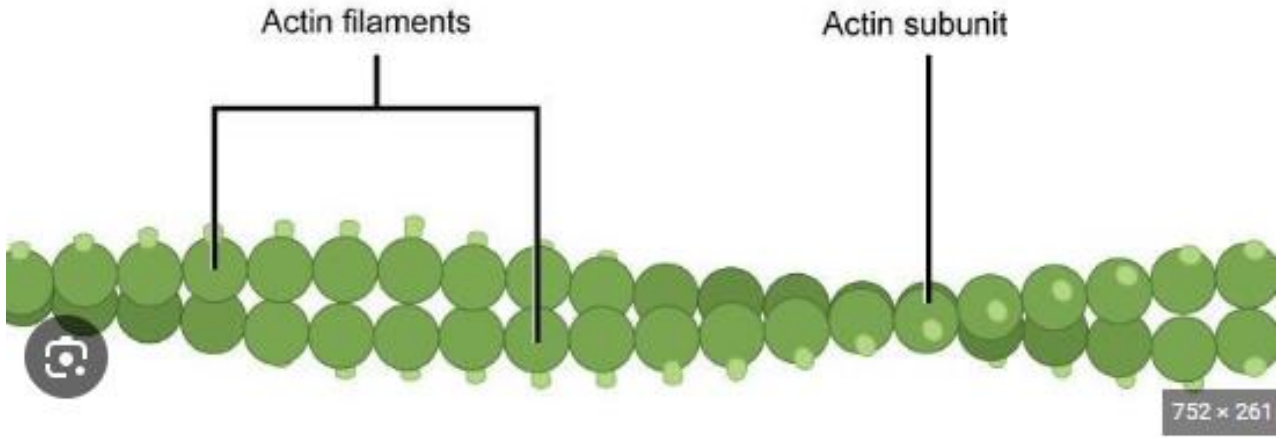


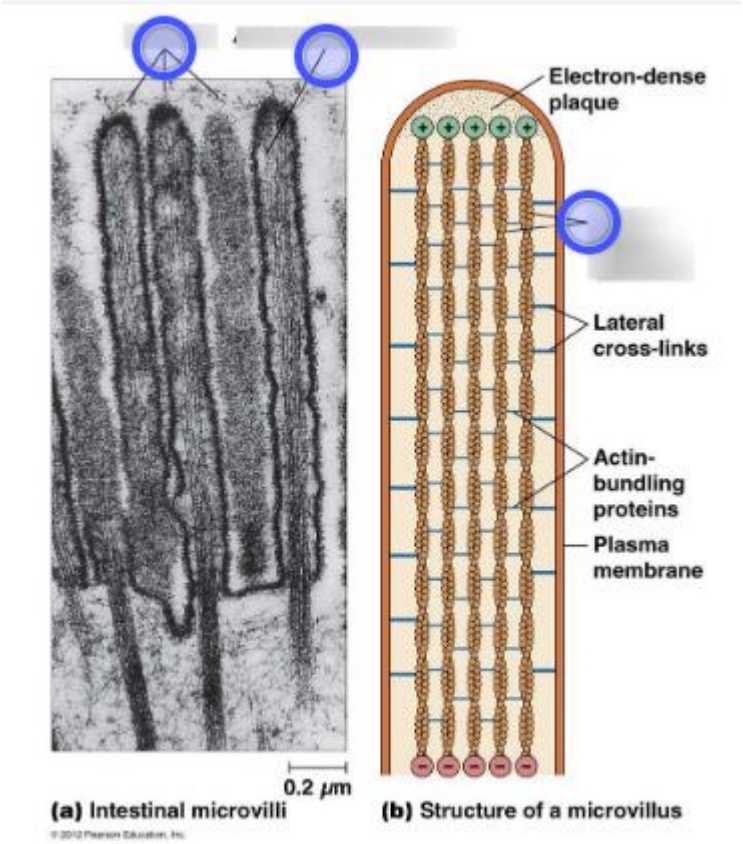
Microfilaments (Actin Filaments)

Microfilaments are thin solid rods, built from actin subunits which are globular proteins.

The structural role of microfilaments is to bear tension, resisting pulling forces within the cell.

In some kinds of animal cells, such as nutrient-absorbing intestinal cells, bundles of microfilaments make up the core of **microvilli**, delicate projections that increase the cell's surface area





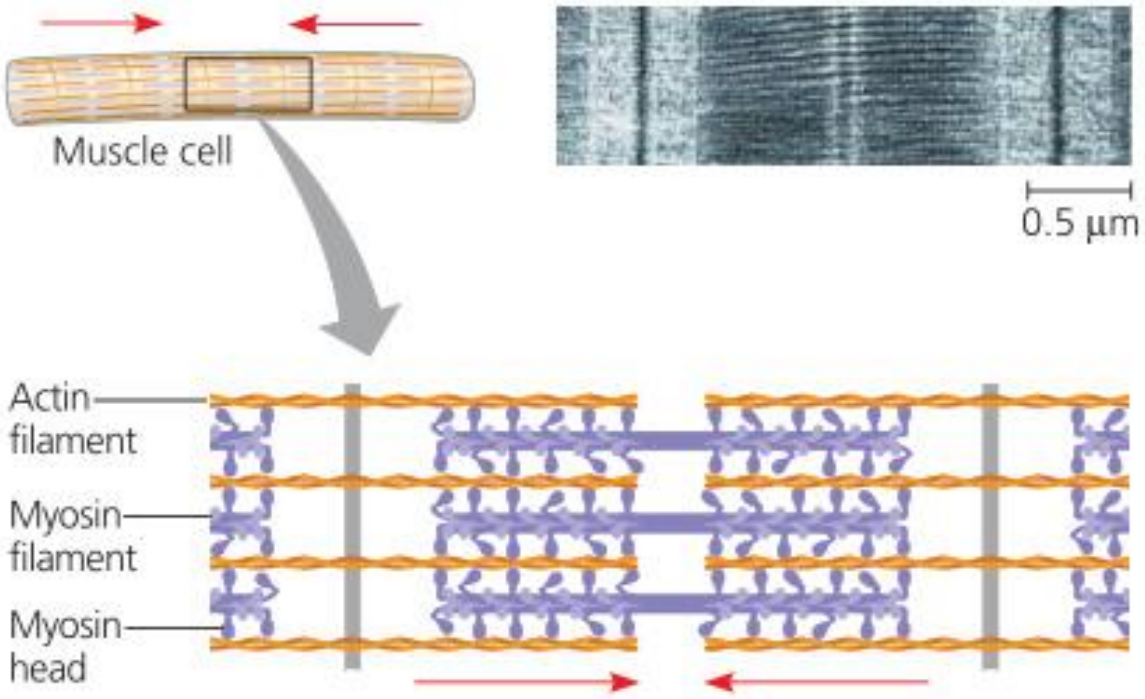
Microfilaments (Actin Filaments)

Microfilaments that function in cellular motility interact with the motor protein myosin

For example, actin and myosin interact to cause muscle contraction, amoeboid movement of white blood cells, and cytoplasmic streaming in plant cells.

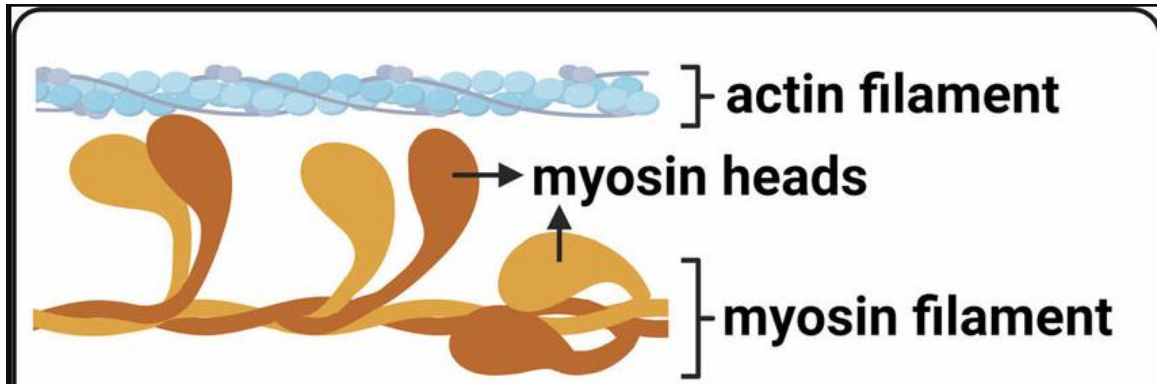
The cell crawls along a surface by extending cellular extensions called pseudopodia (from the Greek pseudes, false, and pod, foot) and moving toward them.

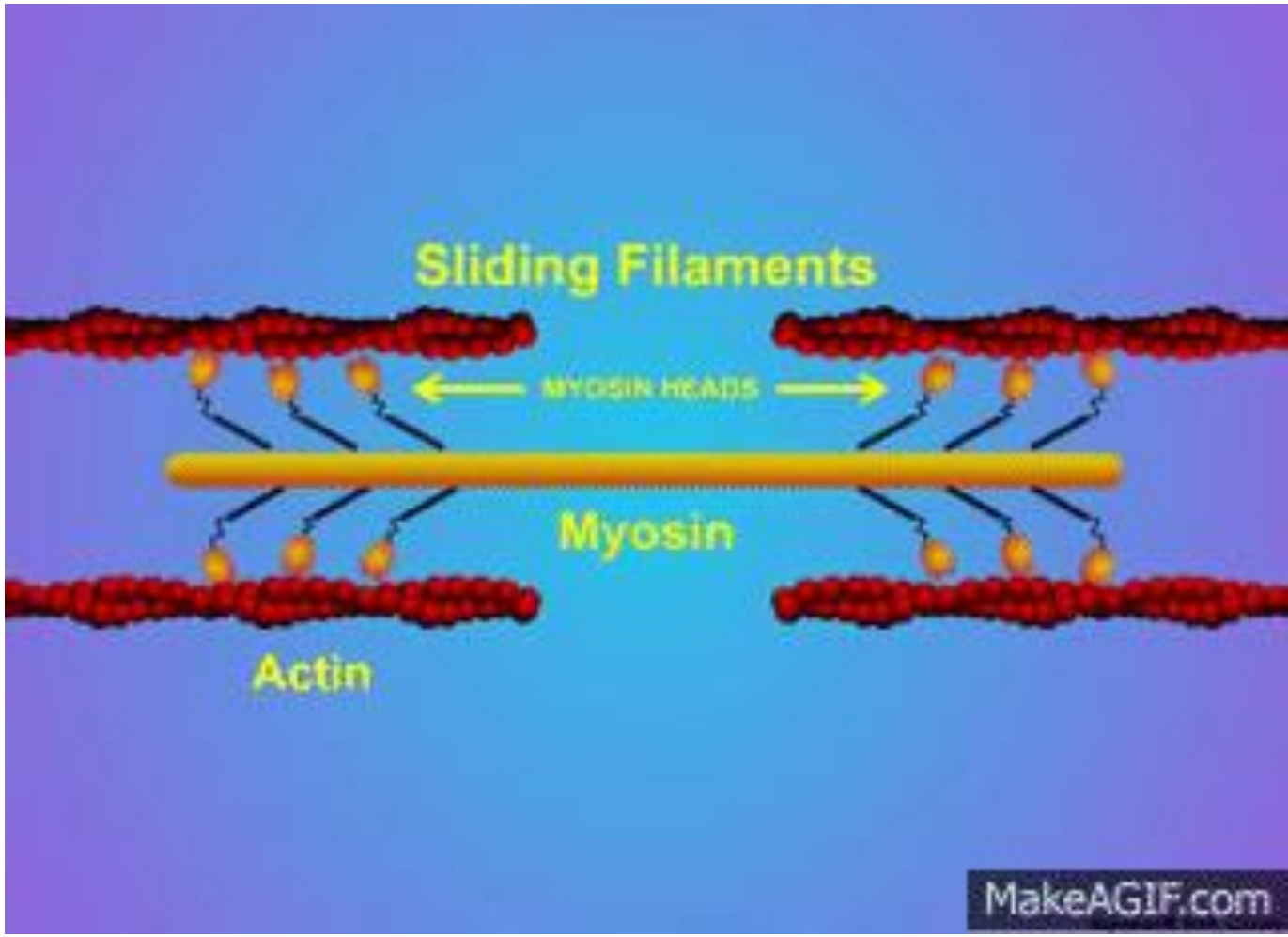
▼ Figure 6.26 Microfilaments and motility. In these three examples, interactions between actin filaments and motor proteins bring about cell movement.



(a) Myosin motors in muscle cell contraction. The "walking" of myosin projections (the so-called heads) drives the parallel myosin and actin filaments past each other so that the actin filaments approach each other in the middle (red arrows). This shortens the muscle cell. Muscle contraction involves the shortening of many muscle cells at the same time (TEM).

Microfilaments (Actin Filaments)





Intermediate Filaments

Intermediate filaments are larger than microfilaments but smaller than microtubules

Intermediate filaments are only found in the cells of some animals, including vertebrates

They support cell shape and fix organelles in place

Intermediate filaments are more permanent cytoskeleton elements than the other two classes

The nucleus typically sits within a cage made of intermediate filaments.
Other intermediate filaments make up the nuclear lamina, which lines the interior of the nuclear membrane.

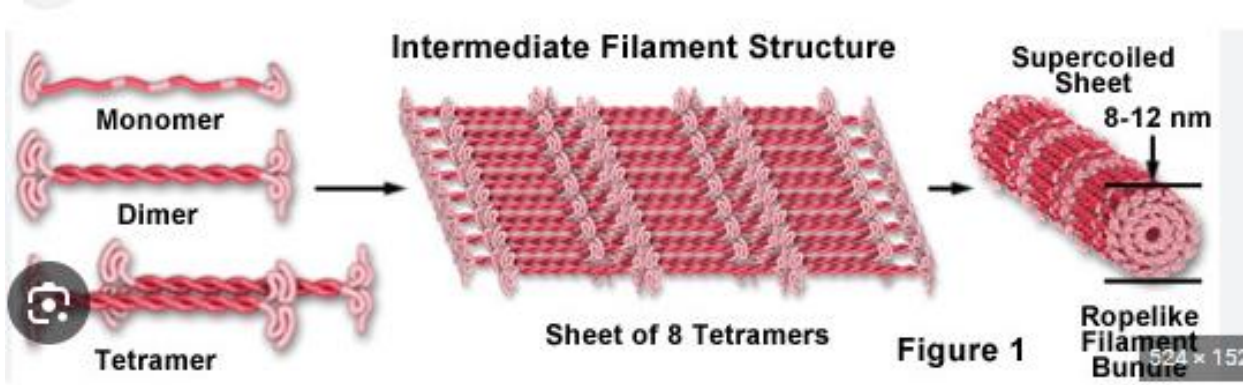


Figure 1

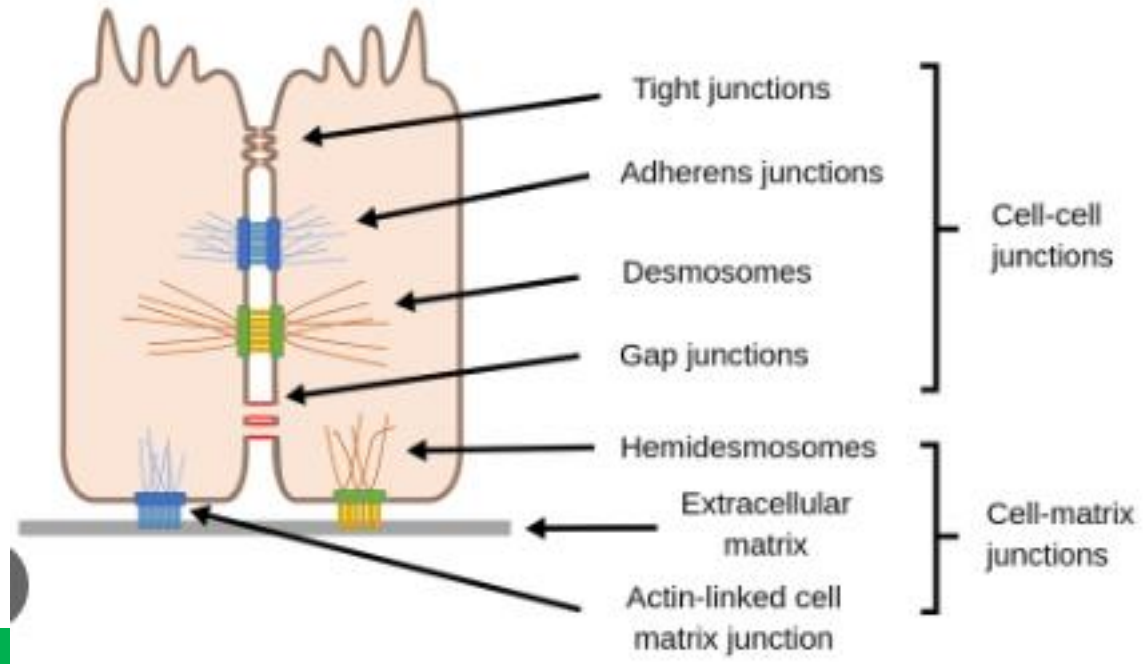
Extracellular components and connections between cells

Most cells synthesize and secrete materials that are external to the plasma membrane

These extracellular structures include:
Cell walls of plants

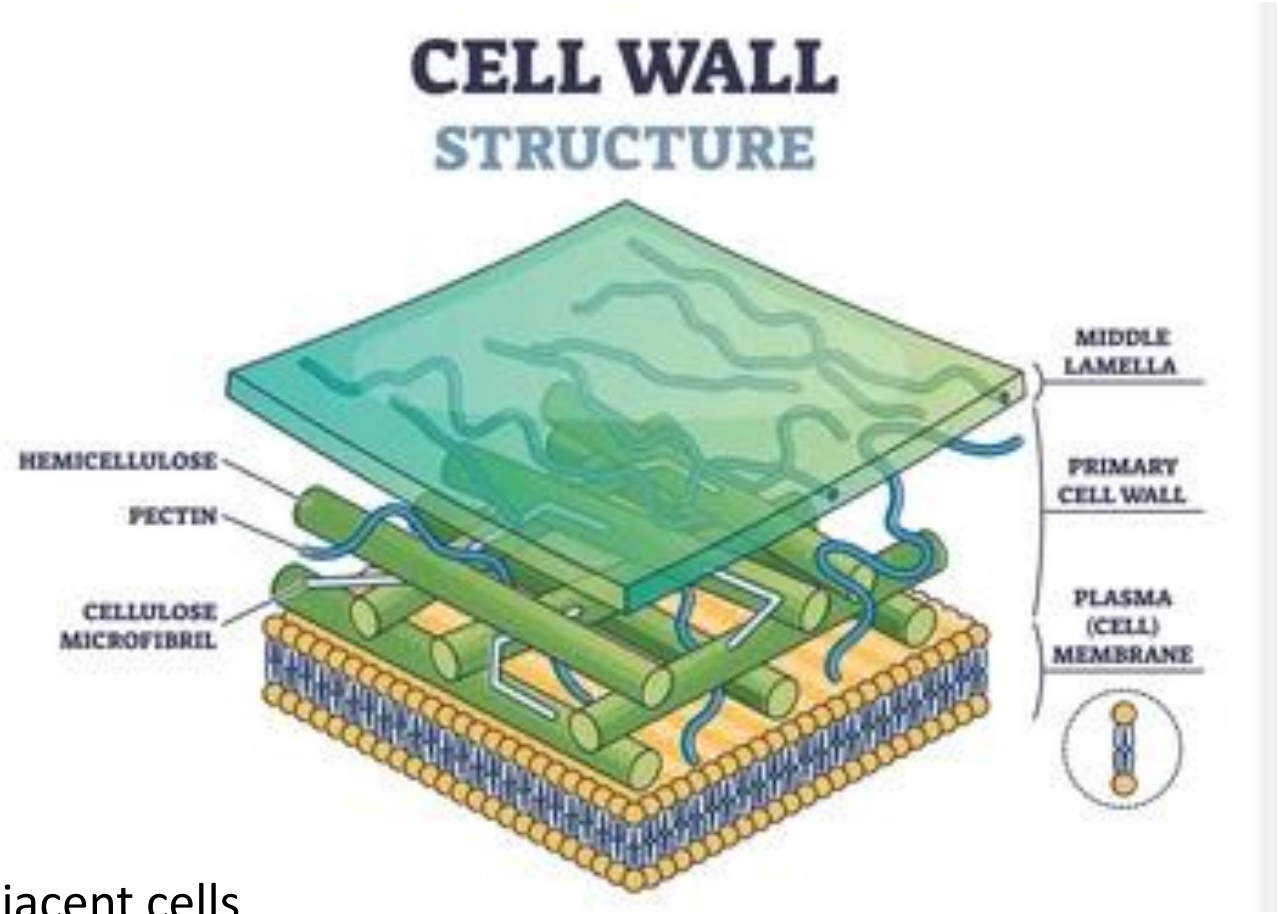
The extracellular matrix (ECM) of animal cells

Intercellular junctions



Cell Walls of Plants

The cell wall is an extracellular structure that distinguishes plant cells from animal cells
Prokaryotes, fungi, and some protists also have cell walls.
The cell wall protects the plant cell, maintains its shape, and prevents excessive uptake of water.
Plant cell walls are made of cellulose fibers embedded in other polysaccharides and protein.



Plant cell walls may have multiple layers:
Primary cell wall: relatively thin and flexible
Middle lamella: thin layer between primary walls of adjacent cells
Secondary cell wall (in some cells): between the plasma membrane and the primary cell wall. often deposited in several laminated layers, has a strong and durable matrix that provides the cell with protection and support. (Wood)

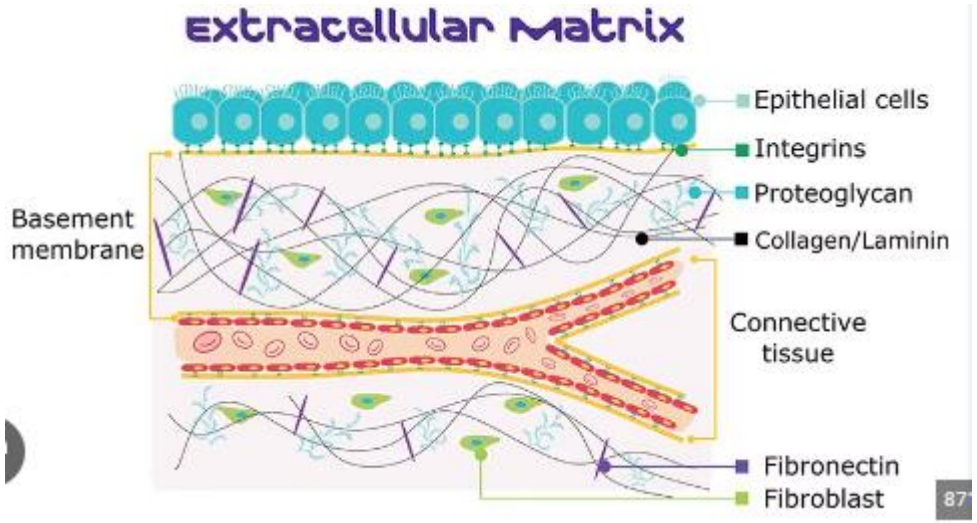
The Extracellular Matrix (ECM) of Animal Cells

Animal cells lack cell walls but are covered by an elaborate extracellular matrix (ECM).

The ECM is made up of glycoproteins (proteins with covalently bonded carbohydrates) such as collagen, proteoglycans, and fibronectin secreted by the cells.
(collagen accounts for about 40% of the total protein in the human body.)

ECM proteins bind to receptor proteins in the plasma membrane called integrins.
Integrins transmit signals between the ECM and the cytoskeleton and thus integrate changes occurring outside and inside the cell.

By communicating with a cell through integrins, the ECM can regulate a cell's behavior.



The Extracellular Matrix (ECM) of Animal Cells

Functions of the ECM:
structurally supports and organizes tissues

Support: provides mechanical support to tissues and organs. It is especially important in connective tissues, such as bone, cartilage, and tendons, where it gives strength and elasticity.

Adhesion: allows cells to adhere to the surrounding tissue, facilitating cell migration, organization, and maintenance. This is important in processes like wound healing and tissue development.

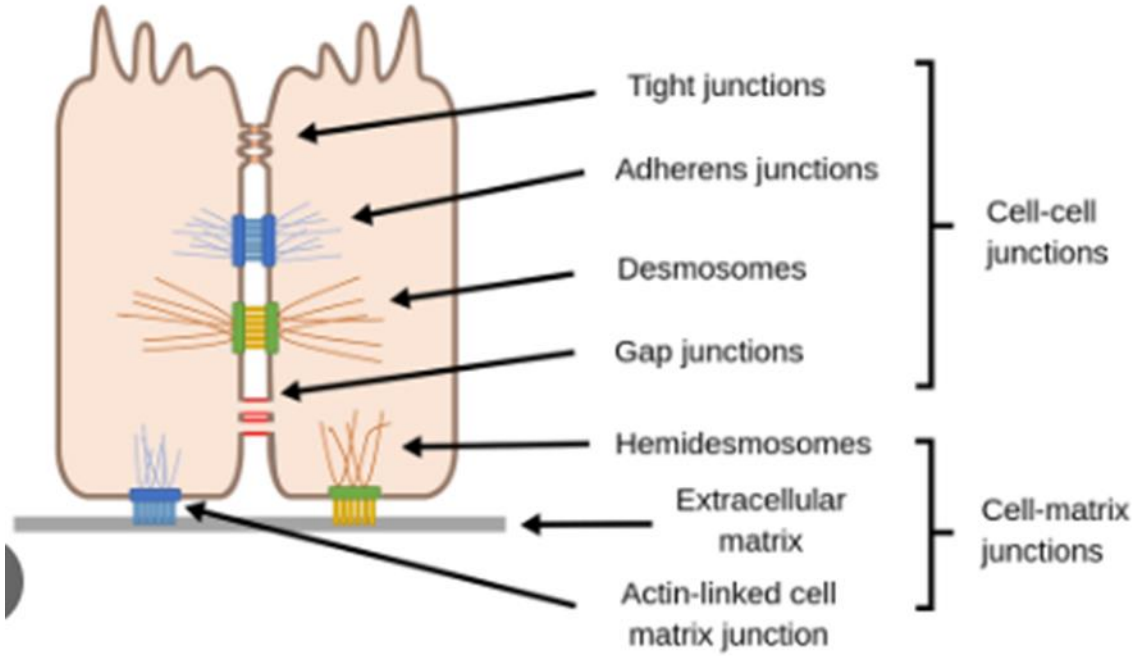
Regulation: The ECM contains signaling molecules and receptors that can influence cell behavior. These signals can regulate cell growth, differentiation, and migration.

Intercellular Junctions

Neighboring cells in tissues, organs, or organ systems often adhere, interact, and communicate through direct physical contact.

Intercellular junctions facilitate this contact.

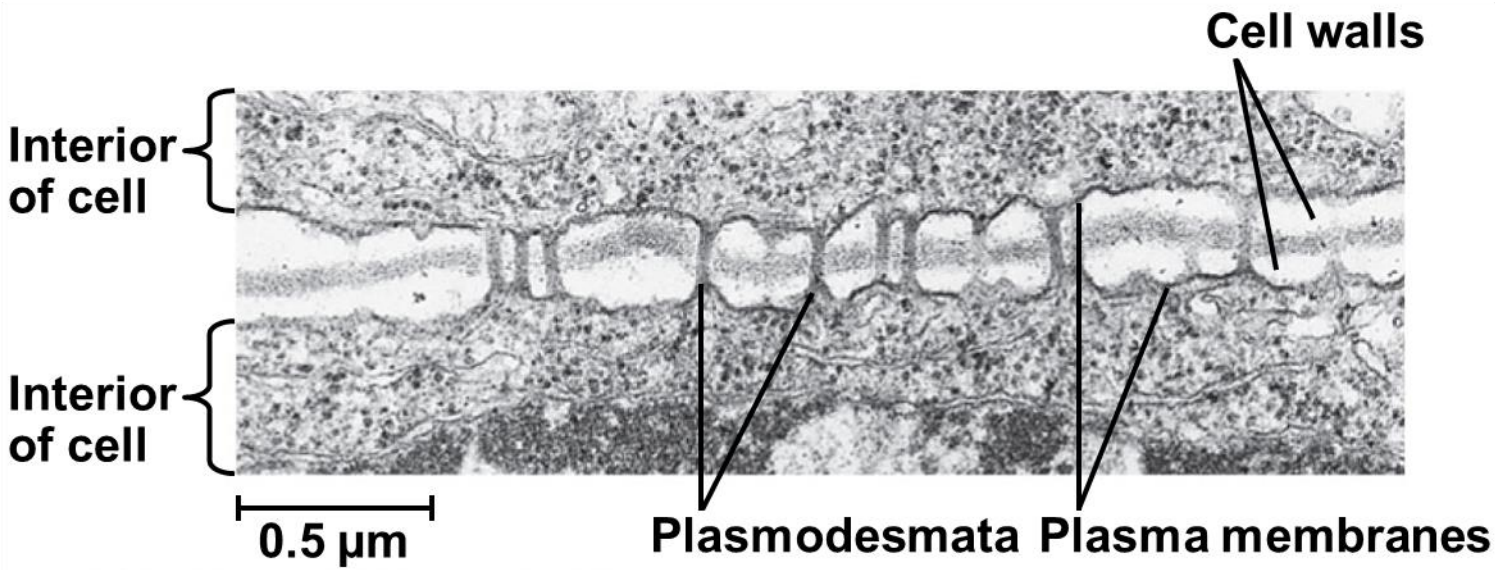
There are several types of intercellular junctions
Plasmodesmata
Tight junctions
Desmosomes
Gap junctions



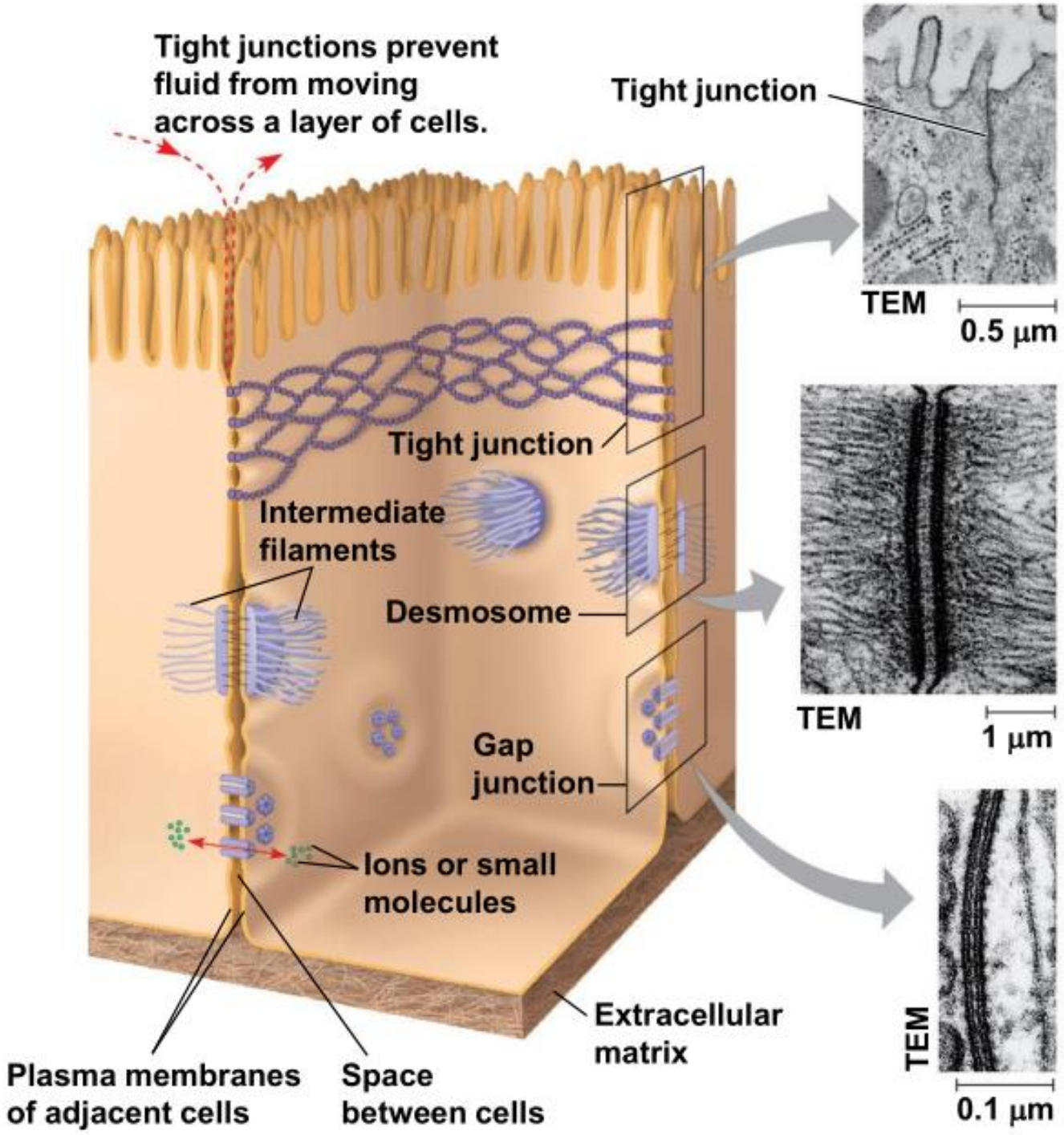
Plasmodesmata in Plant Cells

Plasmodesmata are channels that connect adjacent plant cells.
They are “tunnels” in the cell wall.

Water and small molecules can move from one cell to another via a thin strand of cytoplasm that passes through each channel.

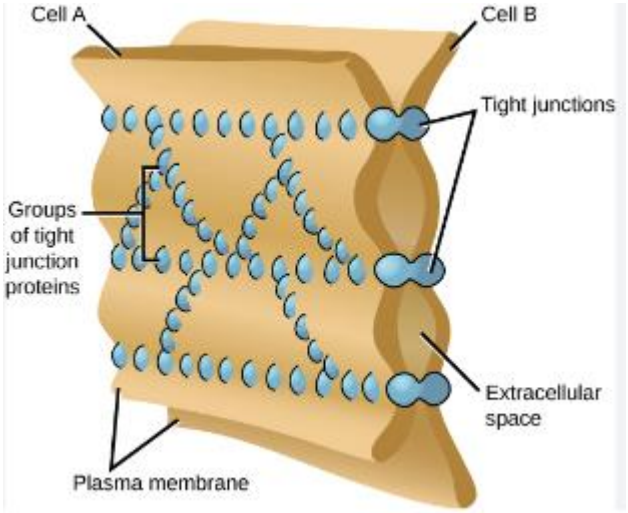


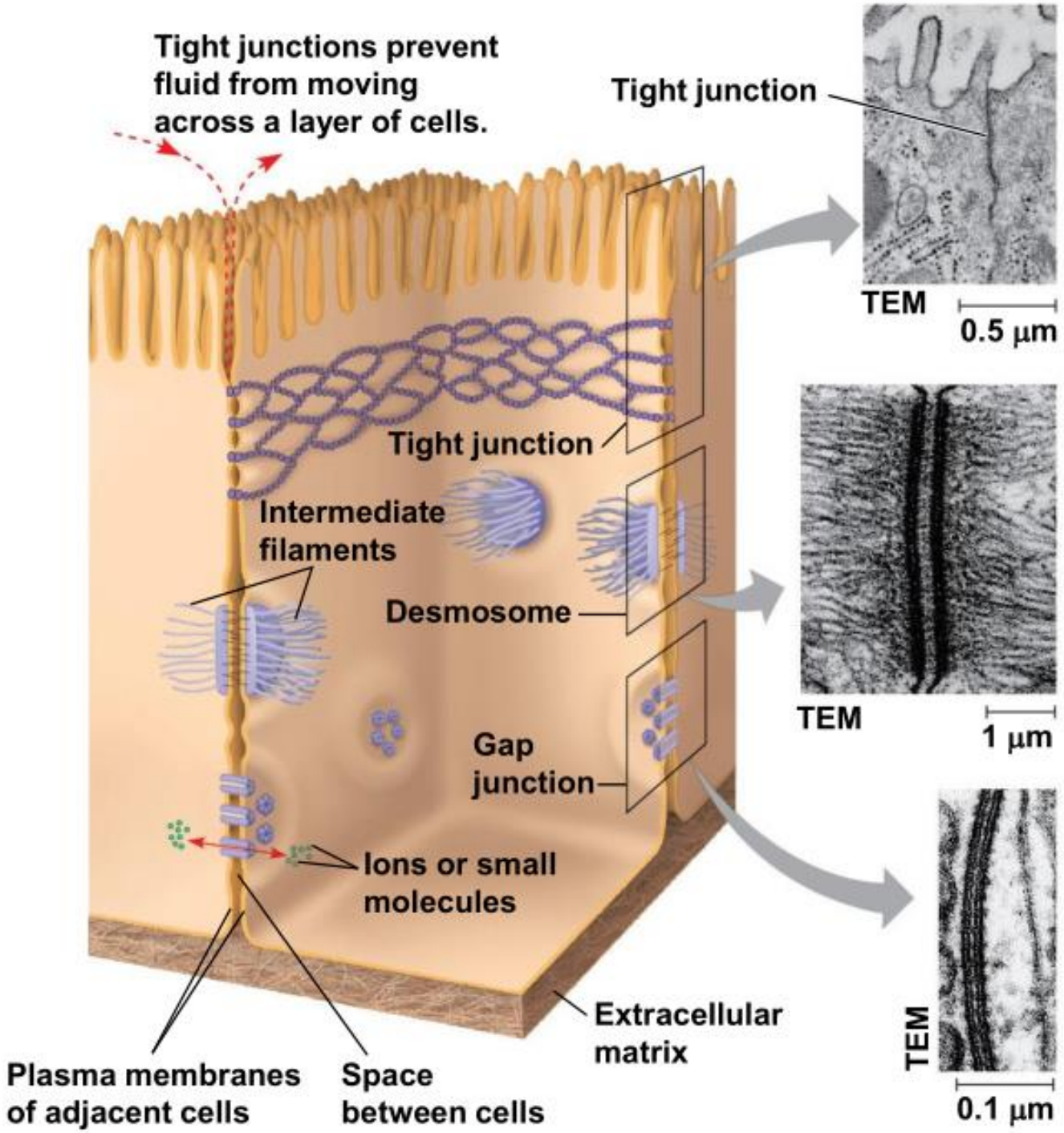
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Tight junctions, desmosomes and gap junctions are especially common in epithelial tissue which lines the external and internal surfaces of the body.

At **tight junctions**, membranes of neighboring cells are pressed together, preventing leakage of extracellular fluid. For example, tight junctions between skin cells make
us watertight.
tight junctions create a barrier that controls the movement of substances between cells

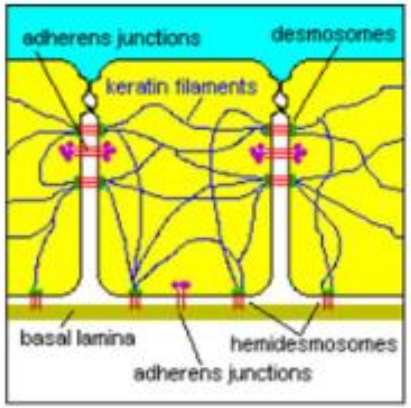


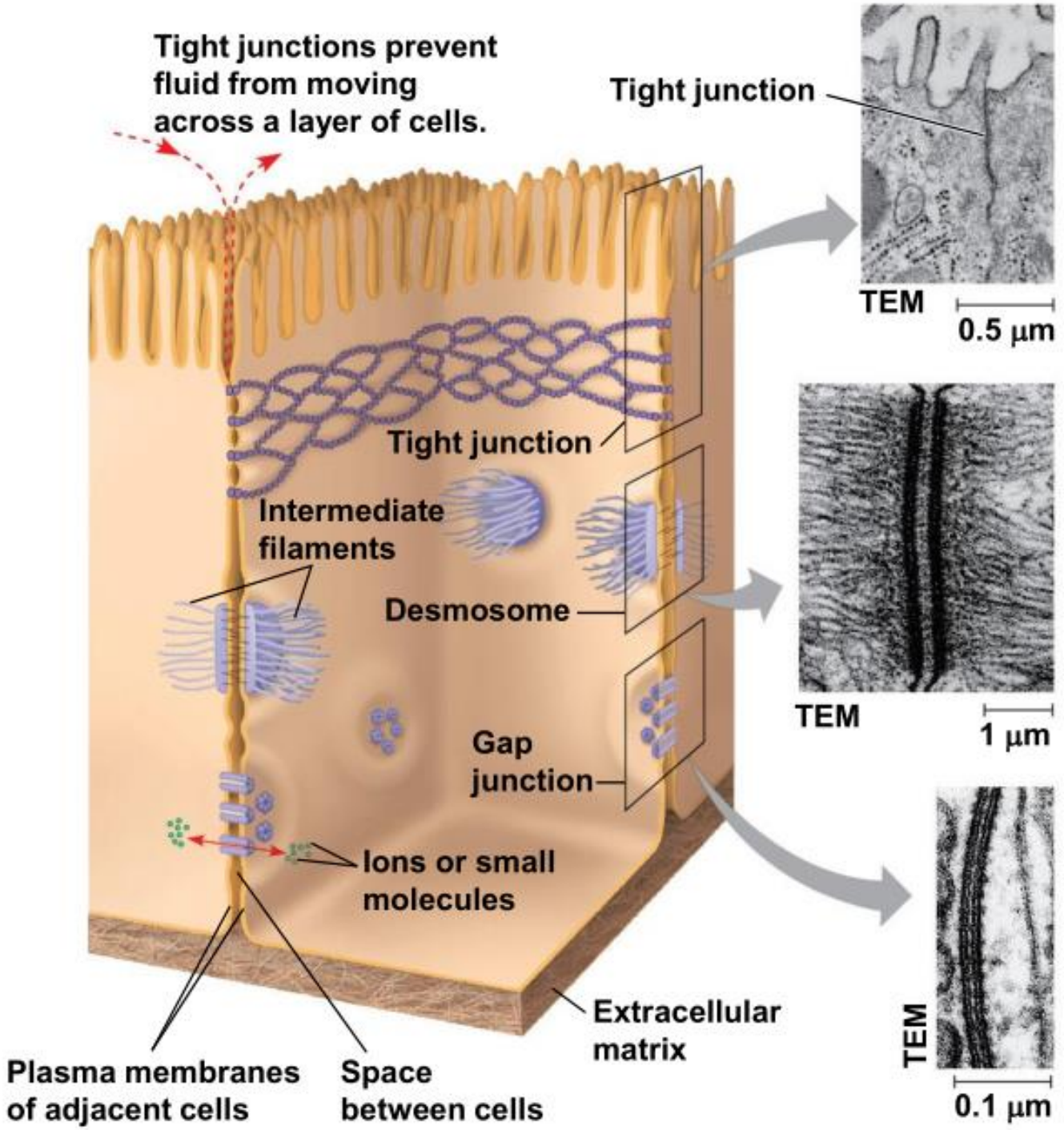


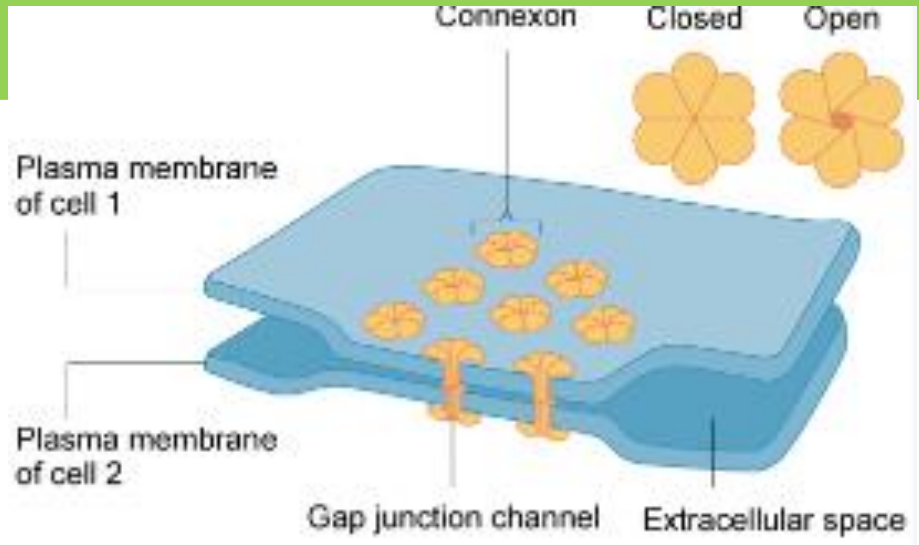
Desmosomes (anchoring junctions) fasten cells together into strong sheets. **desmosomes** provide physical strength and adhesion between cells
Intermediate filaments made of sturdy keratin proteins anchor desmosomes in the cytoplasm. Desmosomes attach muscle cells to each other in a muscle.

Anchoring Junctions

- Rivet cells together with cytoskeleton structures, but still allow materials to pass along the spaces between cells







Gap junctions (communicating junctions) provide cytoplasmic channels between adjacent cells Gap junctions consist of membrane proteins extending from the membranes of the two cells. These proteins create pores through which ions, sugars, amino acids, and other small molecules may pass.

The Cell Cycle: The Key Roles of Cell Division

The greatest difference between living things and non-living matter is that living things can produce more of their own kind.

The continuity of life depends on the reproduction of cells, or cell division.

Cell division is an integral part of the cell cycle, the life of a cell from formation to its own division.

The cell cycle is the series of events that a cell undergoes as it grows, replicates its DNA, and divides to produce two daughter cells. It is essential for growth, development, tissue repair, and reproduction in organisms. The cycle consists of several phases that ensure proper cell division and the distribution of genetic material.

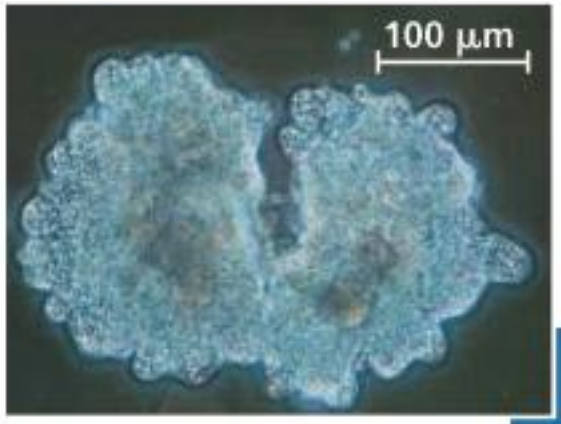
The Key Roles of Cell Division

In unicellular organisms, division of one cell reproduces the entire organism.

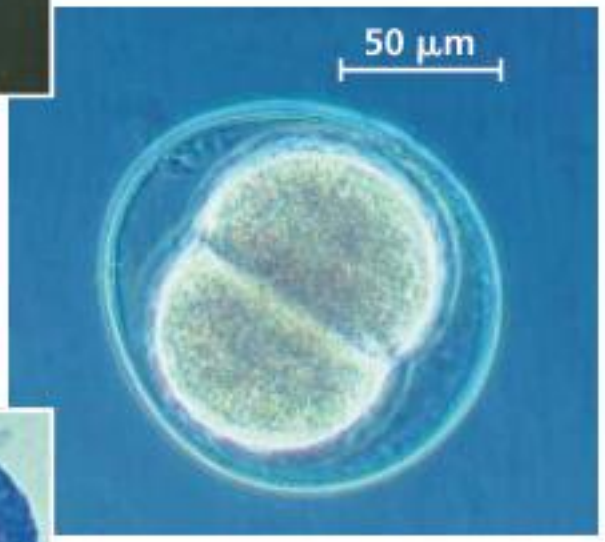
Multicellular organisms depend on cell division for:

- Development from a fertilized cell
- Growth
- Repair

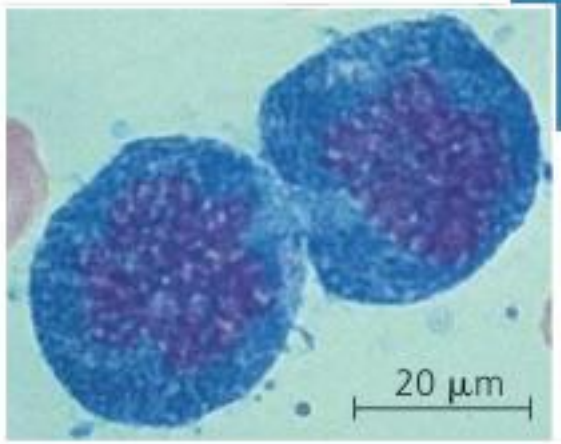
▼ **Figure 12.2 The functions of cell division.**



◀ **(a) Asexual reproduction.** An amoeba, a single-celled eukaryote, is dividing into two cells. Each new cell will be an individual organism (LM).



▶ **(b) Growth and development.** This micrograph shows a sand dollar embryo shortly after the fertilized egg divided, forming two cells (LM).

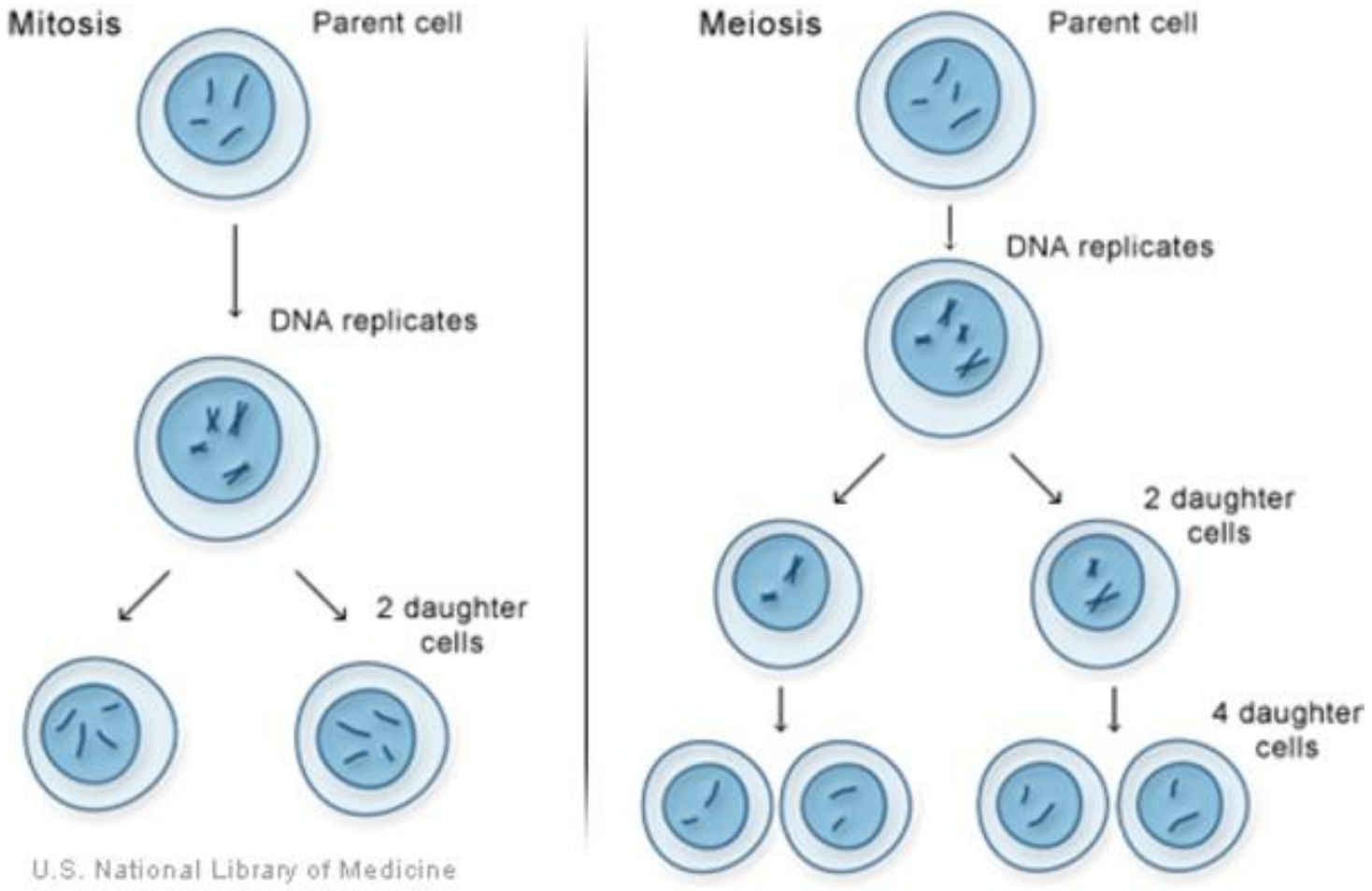


◀ **(c) Tissue renewal.** These dividing bone marrow cells will give rise to new blood cells (LM).

Most cell division results in genetically identical daughter cells

Mitosis = In somatic cells (nonreproductive cells), most cell division results in two sets of chromosomes with identical genetic information (DNA).

Meiosis = The exception is meiosis, a special type of division that can produce sperm and egg cells (nonidentical daughter cells=gametes=reproductive cells) and have half as many chromosomes as somatic cells.



Most cell division results in genetically identical daughter cells

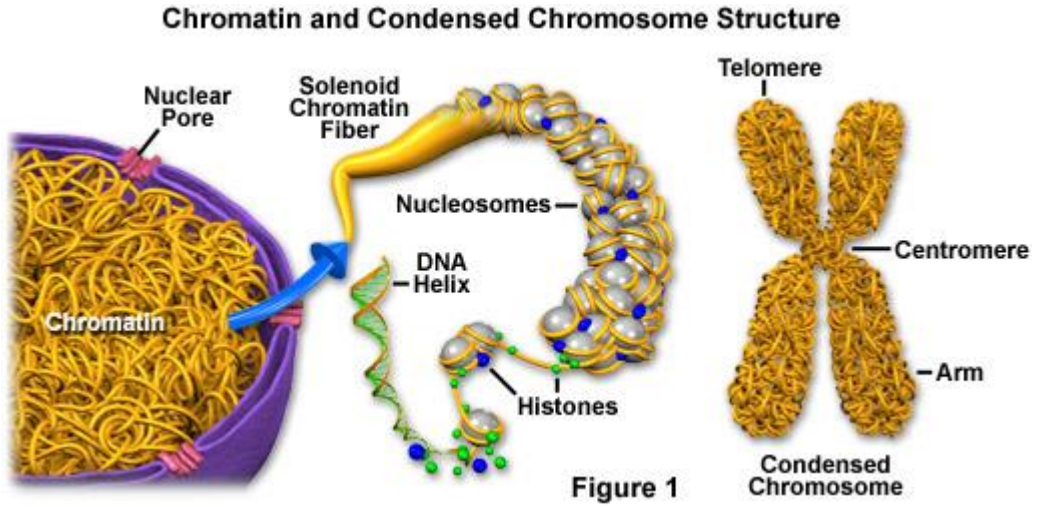
Genome - all the DNA in a cell

In prokaryotic cells the genome can consist of a single DNA molecule.

In eukaryotic cells the genome can consist of more than one DNA molecule.

DNA molecules in a cell are packaged into **chromosomes**

Eukaryotic chromosomes consist of **chromatin**, a complex of DNA and protein that condenses during cell division.



Most cell division results in genetically identical daughter cells

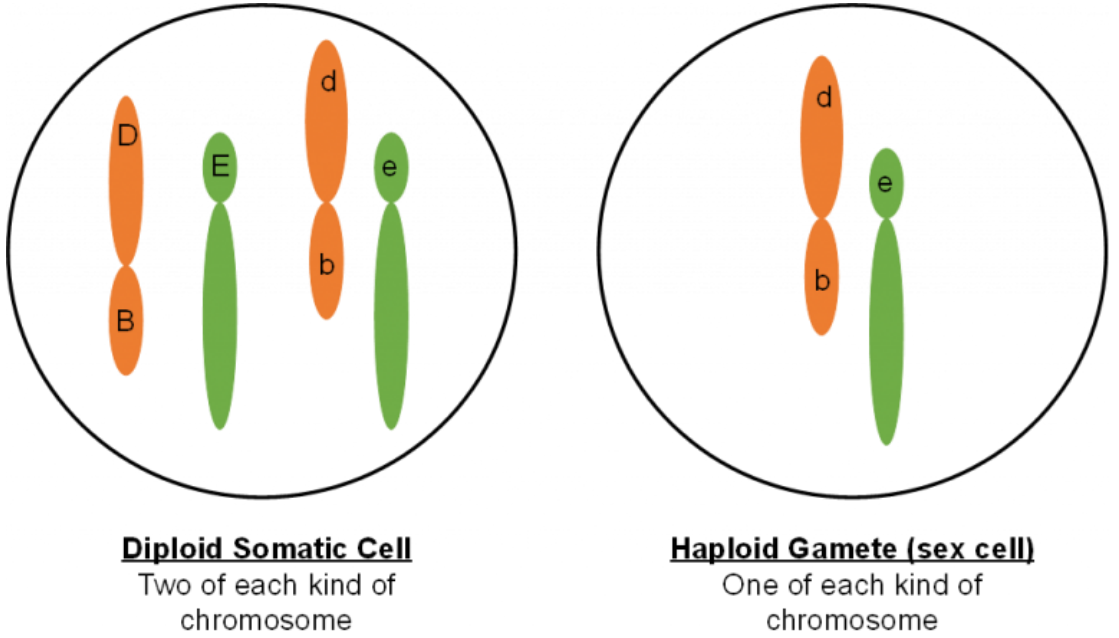
- Every eukaryotic species has a characteristic number of chromosomes in each cell nucleus

- Somatic cells (nonreproductive cells) have two sets of chromosomes

For example, the nuclei of human somatic cells contain 46 chromosomes, made up of two sets of 23, one set inherited from each parent.

- Gametes (reproductive cells: sperm and eggs) have half as many chromosomes as somatic cells

For example, human gametes have one set of 23 chromosomes.



Diploid Somatic Cell
Two of each kind of chromosome

Haploid Gamete (sex cell)
One of each kind of chromosome

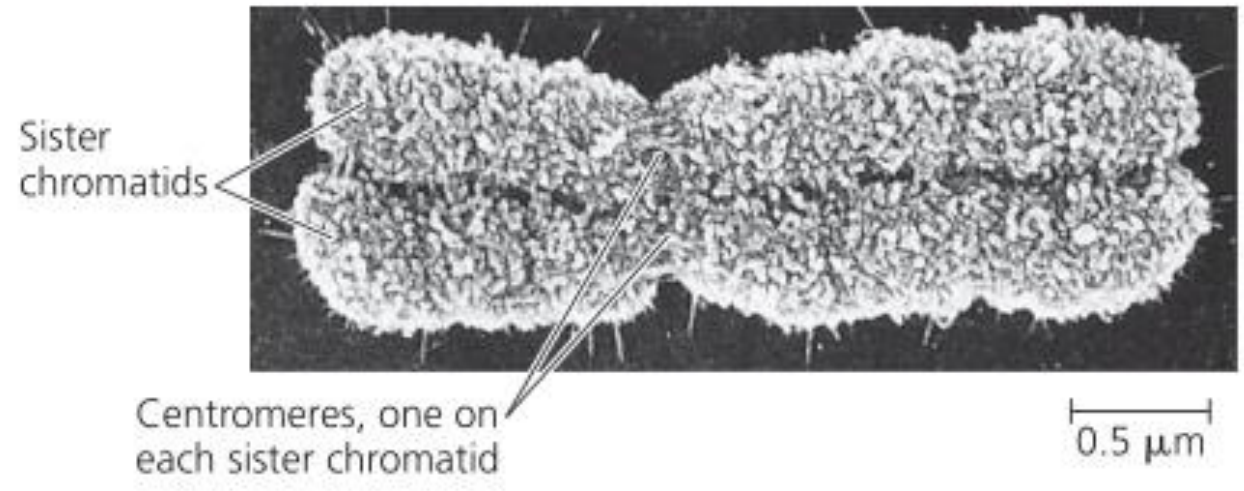
Distribution of Chromosomes During Eukaryotic Cell Division

- In preparation for cell division, DNA is replicated and the chromosomes condense.

- Each duplicated chromosome has two sister chromatids (joined copies of the original chromosome), attached along their lengths by proteins called cohesins.

- The centromere is the narrow "waist" of the duplicated chromosome, where the two chromatids are most closely attached.

▼ Figure 12.4 A highly condensed, duplicated human chromosome. (SEM)

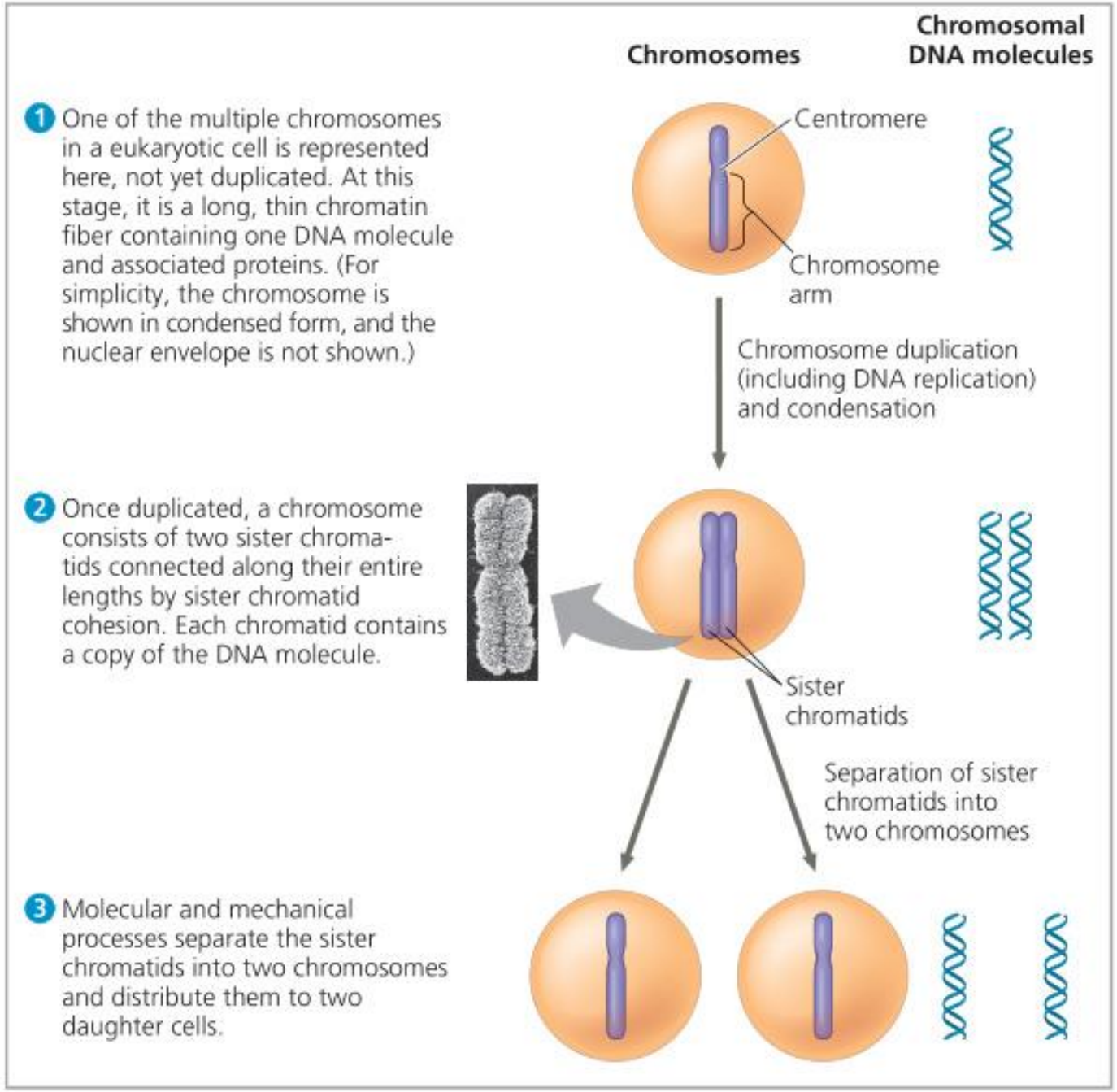


Each sister chromatid has a centromere, a region made up of repetitive sequences in the chromosomal DNA. The portion of a chromatid to either side of the centromere is referred to as an arm of the chromatid. (An unduplicated chromosome has a single centromere, distinguished by the proteins that bind there, and two arms.)

- During cell division, the two sister chromatids of each duplicated chromosome separate and move into two nuclei. Thus, each new nucleus receives a collection of chromosomes identical to that of the parent cell.

- Once separate, the chromatids are called chromosomes

▼ **Figure 12.5** **Chromosome duplication and distribution during cell division.**



Distribution of Chromosomes During Eukaryotic Cell Division

- Eukaryotic cell division consists of

    1. **mitosis**, the division of the genetic material in the nucleus
    2. **cytokinesis**, the division of the cytoplasm

- Gametes are produced by a variation of cell division called **meiosis**
- Meiosis yields nonidentical daughter cells that have half as many chromosomes as the parent cell

The mitotic phase alternates with interphase in the cell cycle

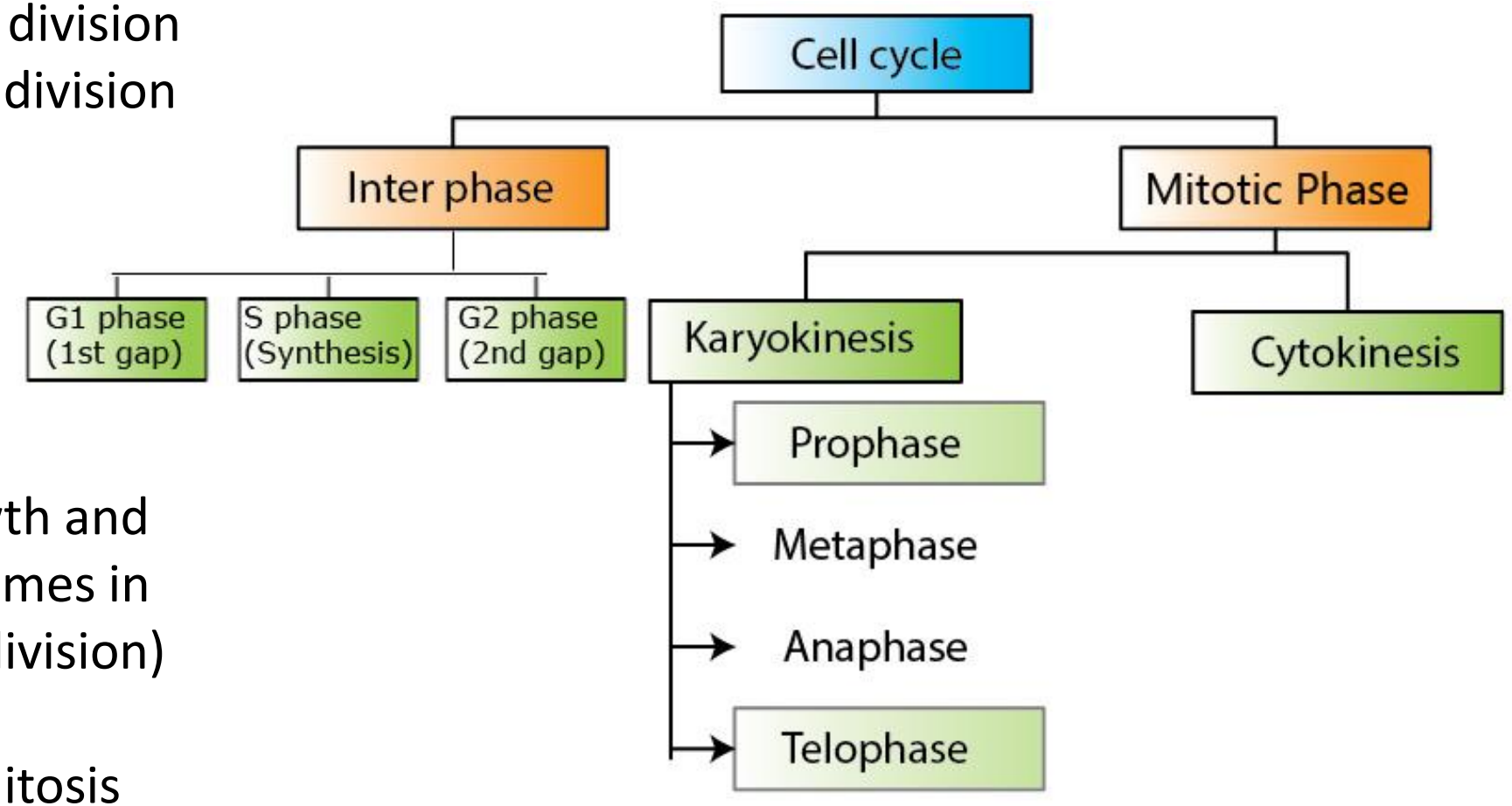
Cell cycle: the life of a cell from the time it is first formed during division of a parent cell until its own division into two daughter cells.

Phases of the Cell Cycle

- The cell cycle consists of

1. interphase (cell growth and copying of chromosomes in preparation for cell division)

2. mitotic (M) phase (mitosis and cytokinesis)



The mitotic phase alternates with interphase in the cell cycle

- Interphase (about 90% of the cell cycle) can be divided into three phases:

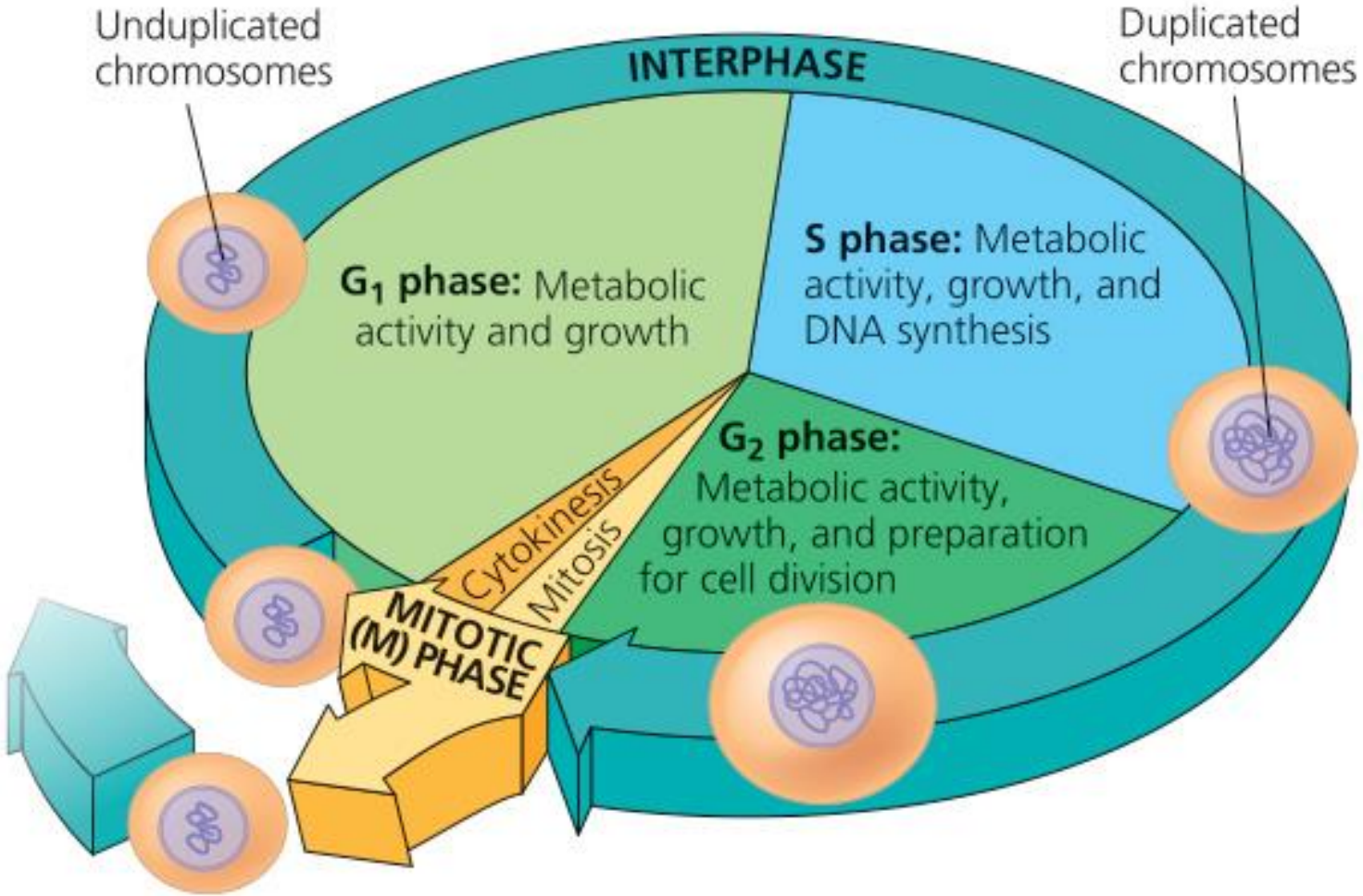
1. ▪G1 phase (“first gap”)
2. ▪S phase (“synthesis”)
3. ▪G2 phase (“second gap”)

- The cell grows during all three phases producing proteins and cytoplasmic organelles such as mitochondria and endoplasmic reticulum.
But duplication of the chromosomes, crucial for eventual division of the cell, occurs entirely during the S phase.

The cell cycle

A particular human cell might undergo one division in 24 hours. Of this time, the M phase would occupy less than 1 hour, while the S phase might occupy 10–12 hours, or about half the cycle. The rest of the time would be apportioned between the G1 and G2 phases. The G2 phase usually takes 4–6 hours; in our example, G1 would occupy about 5–6 hours. G1 is the most variable in length in different types of cells. Some cells in a multicellular organism divide very infrequently or not at all. These cells spend their time in G1 doing their job in the organism—a cell of the pancreas secretes digestive enzymes, for example.

▼ **Figure 12.6 The cell cycle.** In a dividing cell, the mitotic (M) phase alternates with interphase, a growth period.

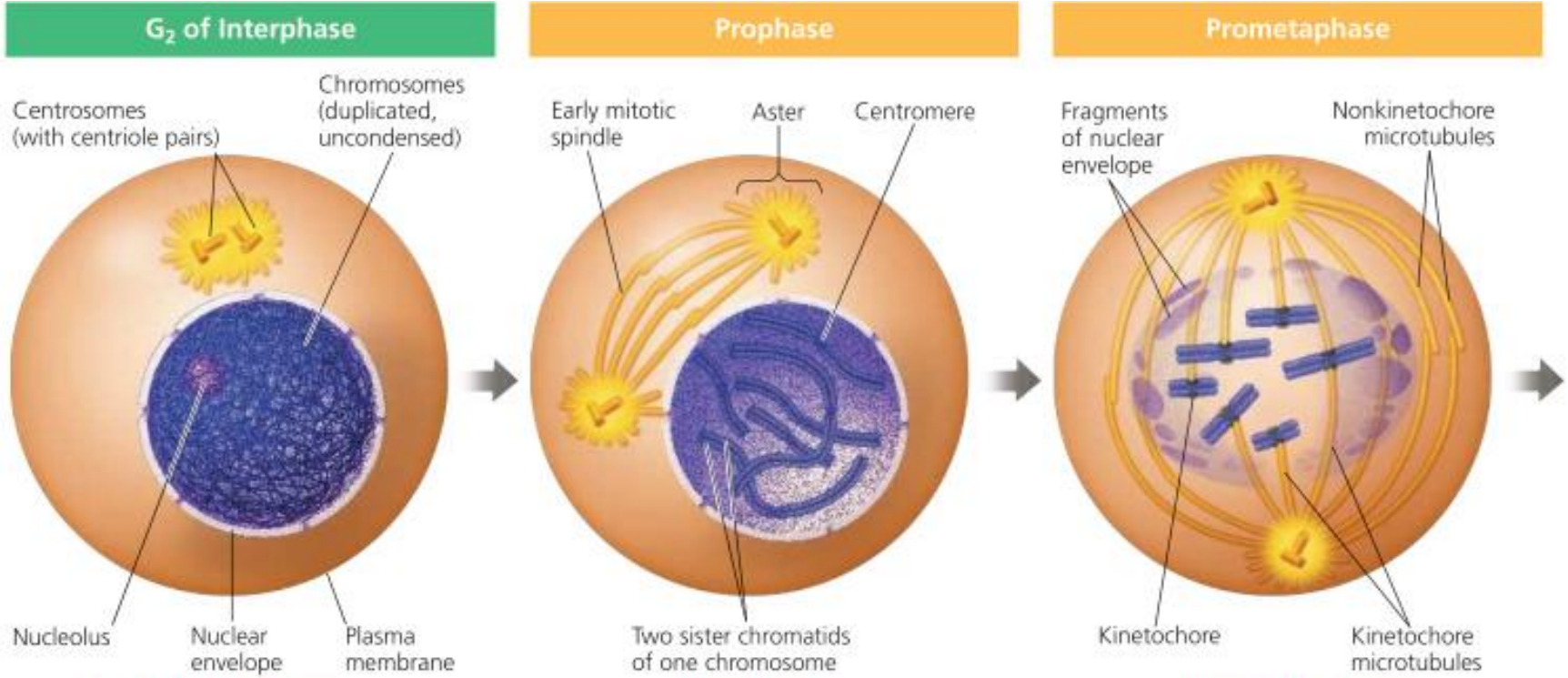


MITOTIC (M) PHASE:
Mitosis: Distribution of chromosomes into two daughter nuclei
Cytokinesis: Division of cytoplasm, producing two daughter cells. Each daughter cell can start a new cell cycle.

Mitosis: the division of the genetic material in the nucleus

- Mitosis is conventionally broken down into five stages:
    1. prophase
    2. prometaphase
    3. metaphase
    4. anaphase
    5. Telophase

- Overlapping with the later stages of mitosis, cytokinesis (the division of the cytoplasm) completes the mitotic phase.



G_2 of Interphase

- A nuclear envelope encloses the nucleus.
- The nucleus contains one or more nucleoli (singular, *nucleolus*).
- Two centrosomes have formed by duplication of a single centrosome. Centrosomes are regions in animal cells that organize the microtubules of the spindle. Each centrosome contains two centrioles.
- Chromosomes, duplicated during S phase, cannot be seen individually because they have not yet condensed.

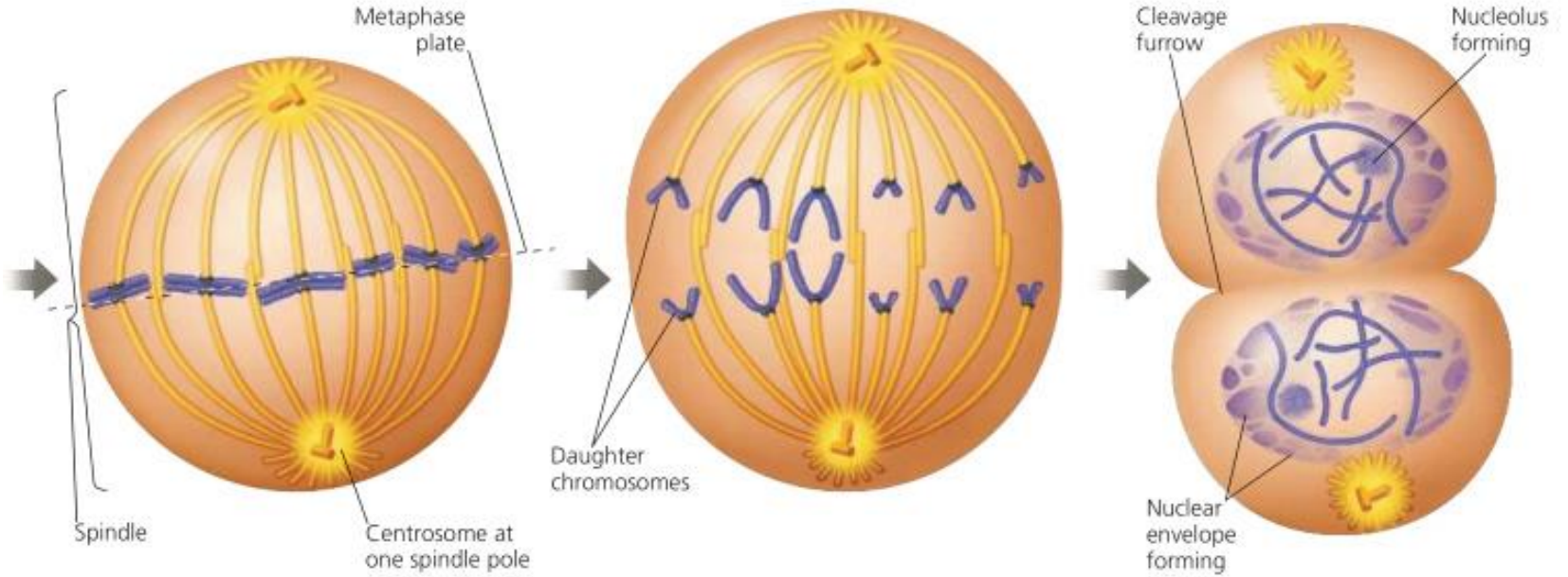
Prophase

- The chromatin fibers become more tightly coiled, condensing into discrete chromosomes observable with a light microscope.
- The nucleoli disappear.
- Each duplicated chromosome appears as two identical sister chromatids joined at their centromeres and, often, all along their arms by cohesins, resulting in sister chromatid cohesion.
- The mitotic spindle (named for its shape) begins to form. It is composed of the centrosomes and the microtubules that extend from them. The radial arrays of shorter microtubules that extend from the centrosomes are called asters ("stars").
- The centrosomes move away from each other, propelled partly by the lengthening microtubules between them.

Prometaphase

- The nuclear envelope fragments.
- The microtubules extending from each centrosome can now invade the nuclear area.
- The chromosomes have become even more condensed.
- A kinetochore, a specialized protein structure, has now formed at the centromere of each chromatid (thus, two per chromosome).
- Some of the microtubules attach to the kinetochores, becoming "kinetochore microtubules," which jerk the chromosomes back and forth.
- Nonkinetochore microtubules interact with those from the opposite pole of the spindle, lengthening the cell.

Metaphase | Anaphase | Telophase and Cytokinesis



Metaphase

- The centrosomes are now at opposite poles of the cell.
- The chromosomes have all arrived at the *metaphase plate*, a plane that is equidistant between the spindle's two poles. The chromosomes' centromeres lie at the metaphase plate.
- For each chromosome, the kinetochores of the sister chromatids are attached to kinetochore microtubules coming from opposite poles.

Anaphase

- Anaphase is the shortest stage of mitosis, often lasting only a few minutes.
- Anaphase begins when the cohesin proteins are cleaved. This allows the two sister chromatids of each pair to part suddenly. Each chromatid thus becomes an independent chromosome.
- The two new daughter chromosomes begin moving toward opposite ends of the cell as their kinetochore microtubules shorten. Because these microtubules are attached at the centromere region, the centromeres are pulled ahead of the arms, moving at a rate of about 1 μm/min.
- The cell elongates as the nonkinetochore microtubules lengthen.
- By the end of anaphase, the two ends of the cell have identical—and complete—collections of chromosomes.

Telophase

- Two daughter nuclei form in the cell. Nuclear envelopes arise from the fragments of the parent cell's nuclear envelope and other portions of the endomembrane system.
- Nucleoli reappear.
- The chromosomes become less condensed.
- Any remaining spindle microtubules are depolymerized.
- Mitosis, the division of one nucleus into two genetically identical nuclei, is now complete.

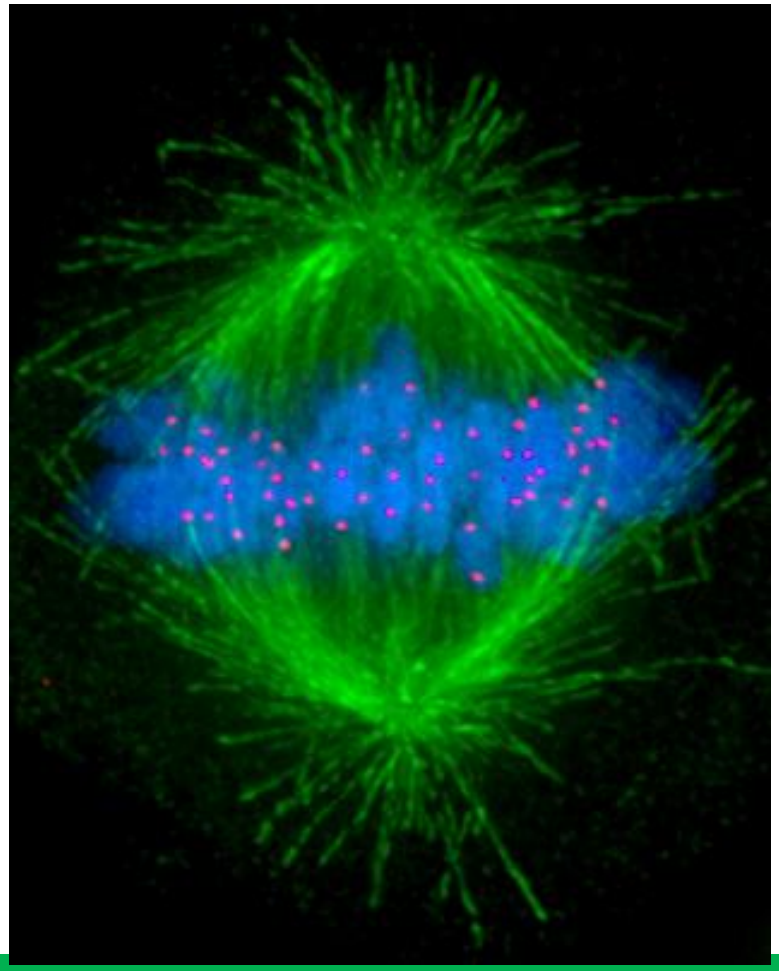
Cytokinesis

- The division of the cytoplasm is usually well under way by late telophase, so the two daughter cells appear shortly after the end of mitosis.
- In animal cells, cytokinesis involves the formation of a cleavage furrow, which pinches the cell in two.

The Mitotic Spindle: A Closer Look

While the mitotic spindle assembles, the other microtubules of the cytoskeleton partially disassemble, providing the material used to construct the spindle. The spindle microtubules elongate (polymerize) by incorporating more subunits of the protein tubulin and shorten (depolymerize) by losing subunits.





The Mitotic Spindle: A Closer Look

- **The mitotic spindle** is a structure made of microtubules that controls chromosome movement during mitosis

- In animal cells, assembly of spindle microtubules begins in the **centrosome**, the microtubule-organizing center

- The centrosome replicates during interphase, forming two centrosomes that migrate to opposite ends of the cell during prophase and prometaphase

- An aster (a radial array of short microtubules) extends from each centrosome

- The spindle includes the centrosomes, the spindle microtubules, and the asters

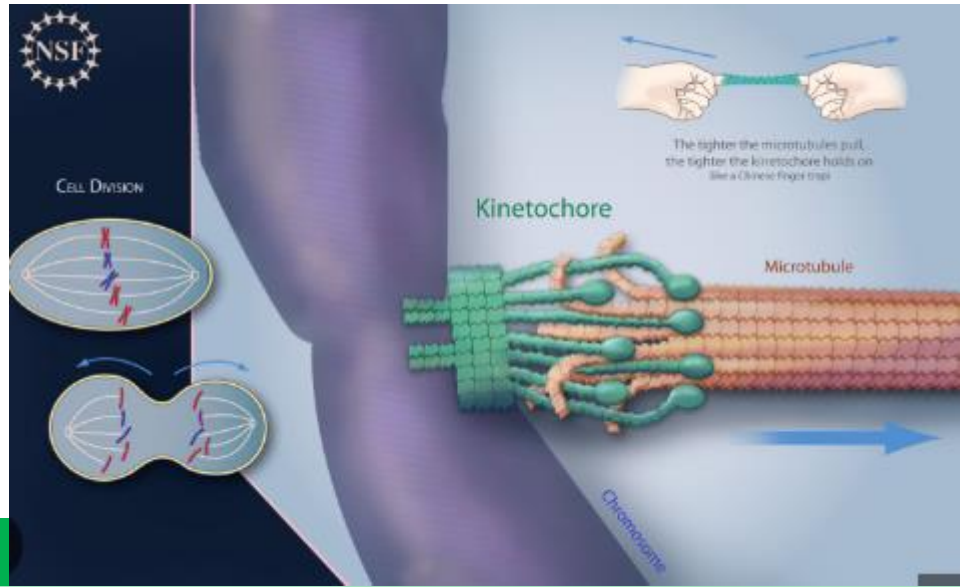
The Mitotic Spindle: A Closer Look

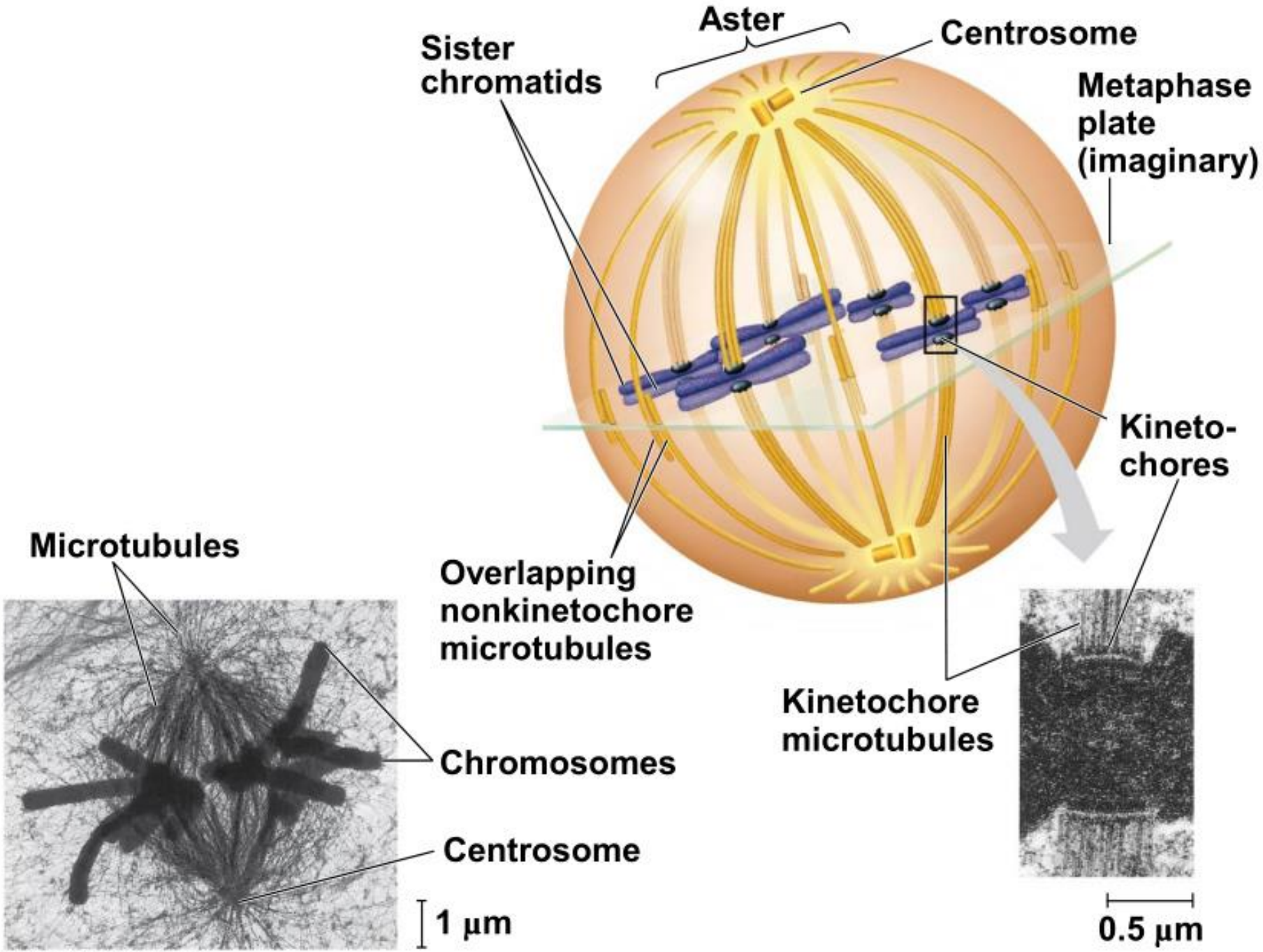
▪ Kinetochores are protein complexes associated with centromeres. They attach the chromosomes to the spindle.

▪ During prometaphase, kinetochore microtubules attach to the **kinetochores** of chromosomes and begin to move the chromosomes

- **Nonkinetochore microtubules** push against each other elongating the cell

▪ At metaphase, the chromosomes are all lined up at the **metaphase plate**, a plane midway between the spindle's two poles

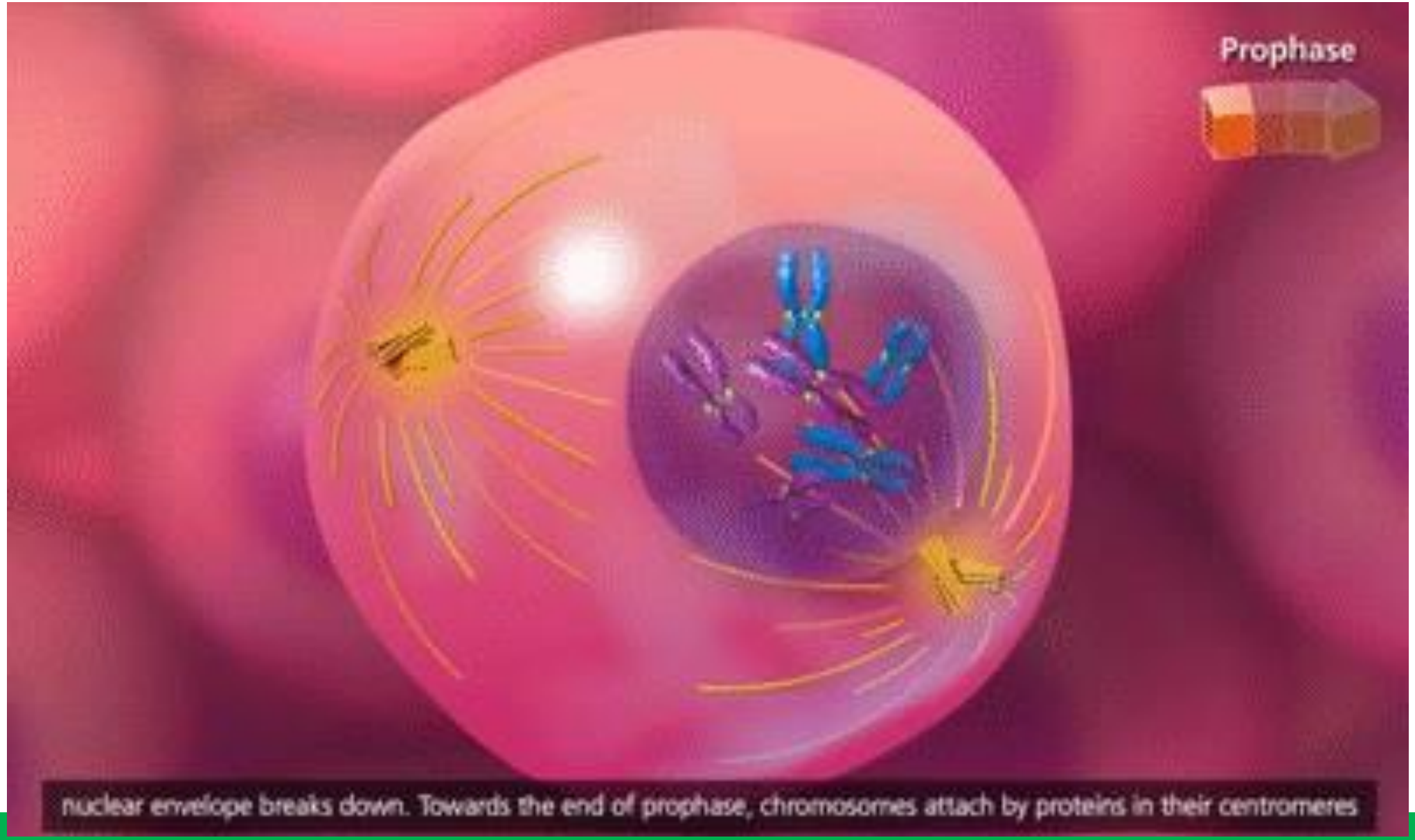




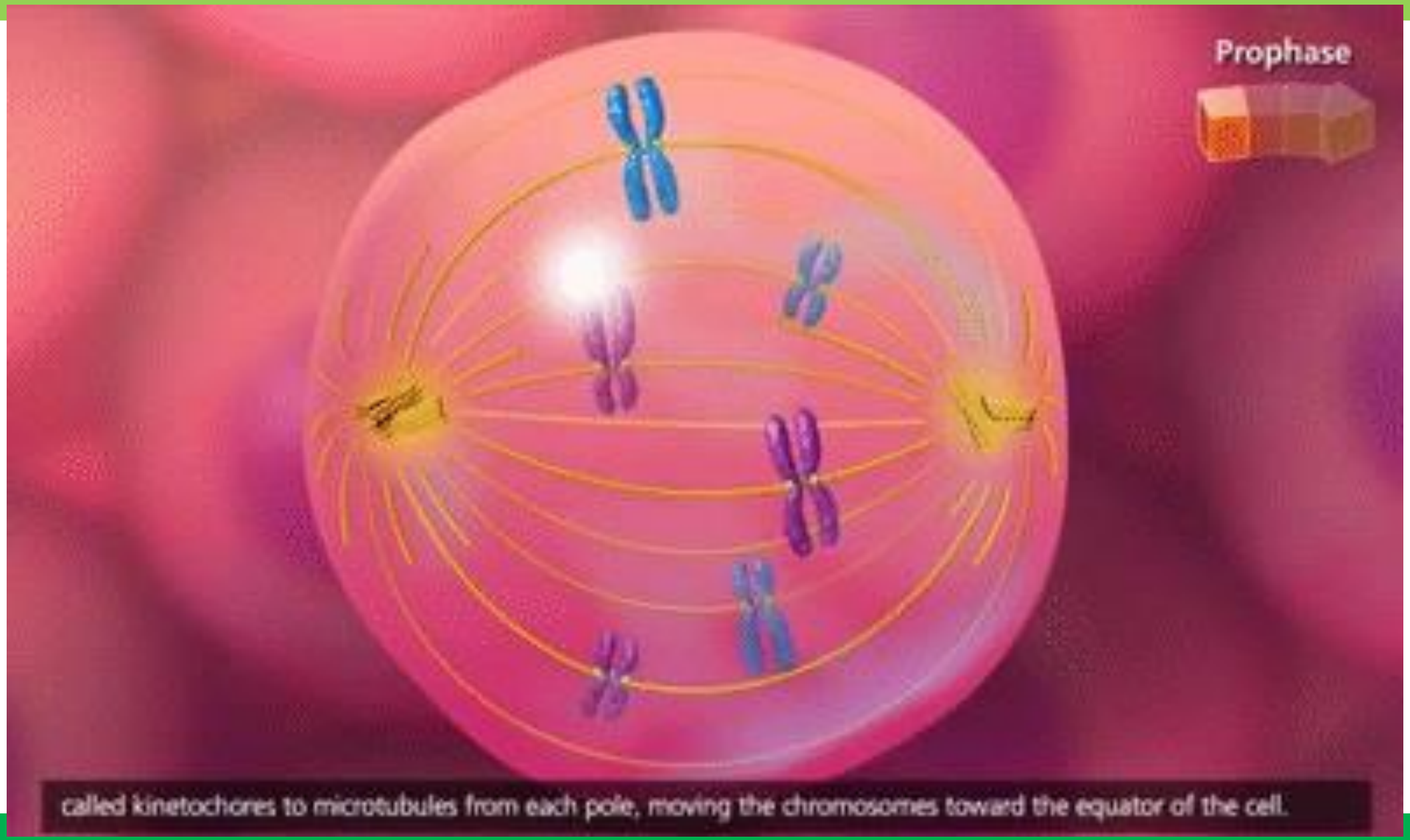
The mitotic spindle at metaphase. The kinetochores of each chromosome's two sister chromatids face in opposite directions. Here, each kinetochore is attached to a cluster of kinetochore microtubules (see TEM) extending from the nearest centrosome. Nonkinetochore microtubules overlap at the metaphase plate.

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Prophase


Prophase
nuclear envelope breaks down. Towards the end of prophase, chromosomes attach by proteins in their centromeres

Metaphase

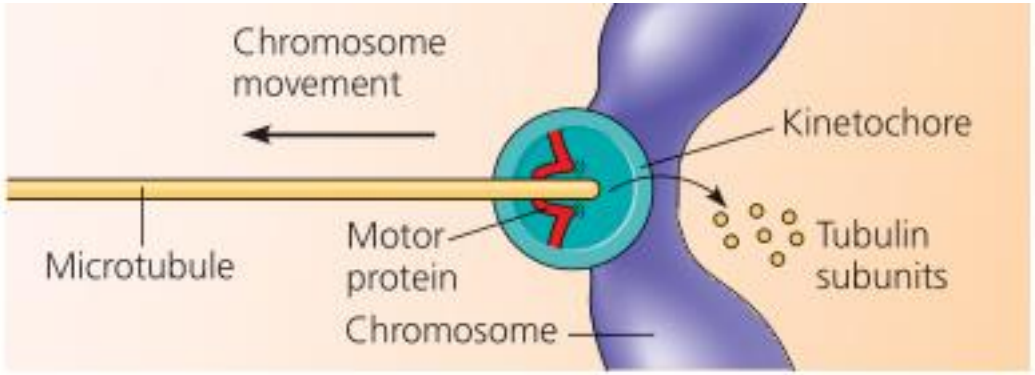

Prophase
called kinetochores to microtubules from each pole, moving the chromosomes toward the equator of the cell.

Anaphase

- In anaphase the cohesins are cleaved by an enzyme called **separase**

- Sister chromatids separate and move along the kinetochore microtubules toward opposite ends of the cell

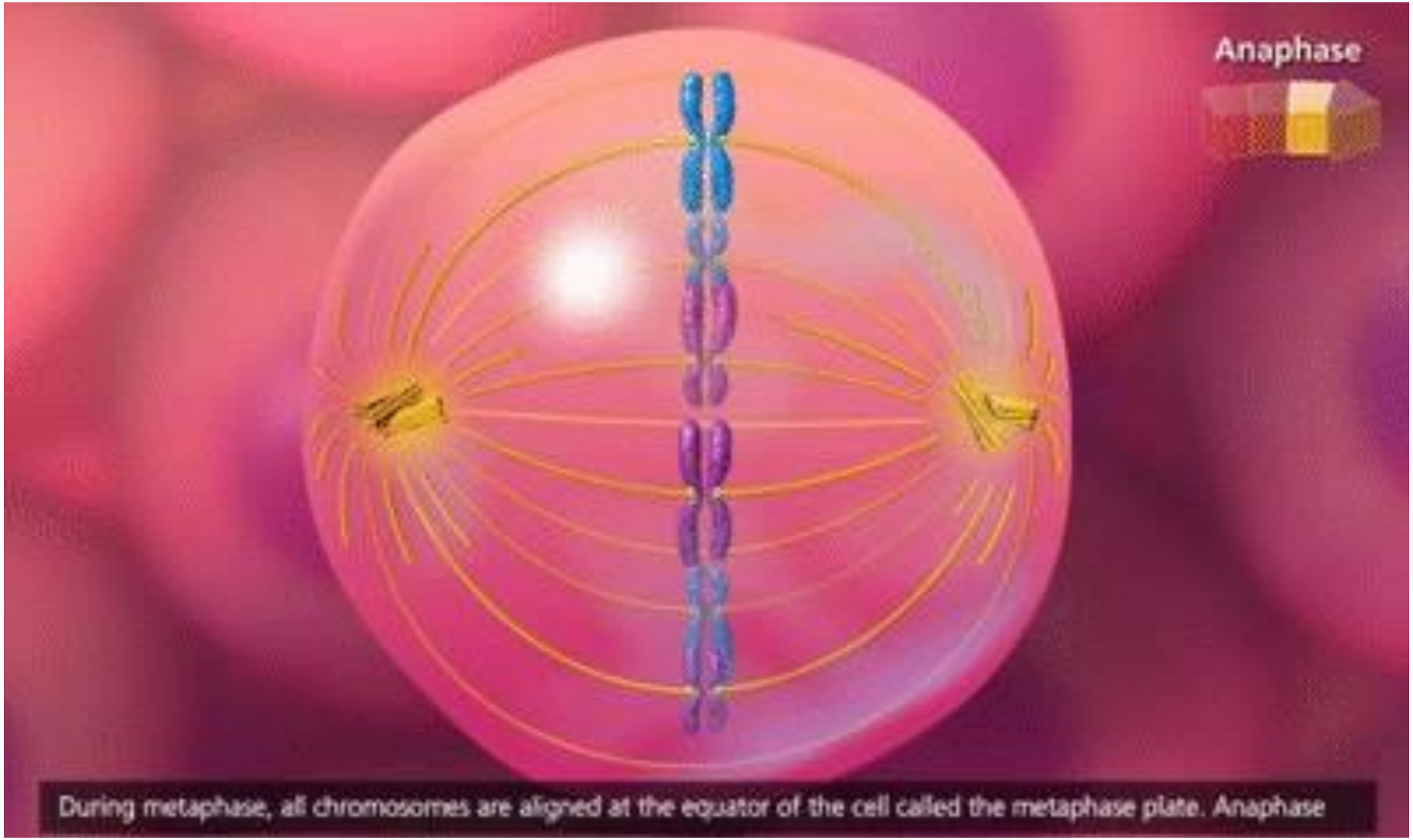
- The microtubules shorten by depolymerizing at their kinetochore ends



Data from G. J. Gorbsky, P. J. Sammak, and G. G. Borisy, Chromosomes move poleward in anaphase along stationary microtubules that coordinately disassemble from their kinetochore ends, *Journal of Cell Biology* 104:9–18 (1987).

- Results of a clever experiment suggest that motor proteins on kinetochores "walk" the chromosomes along the microtubules during anaphase
- The depolymerization of the microtubules at the kinetochore ends occurs after the motor proteins have passed
- This is called the "Pac-man" mechanism

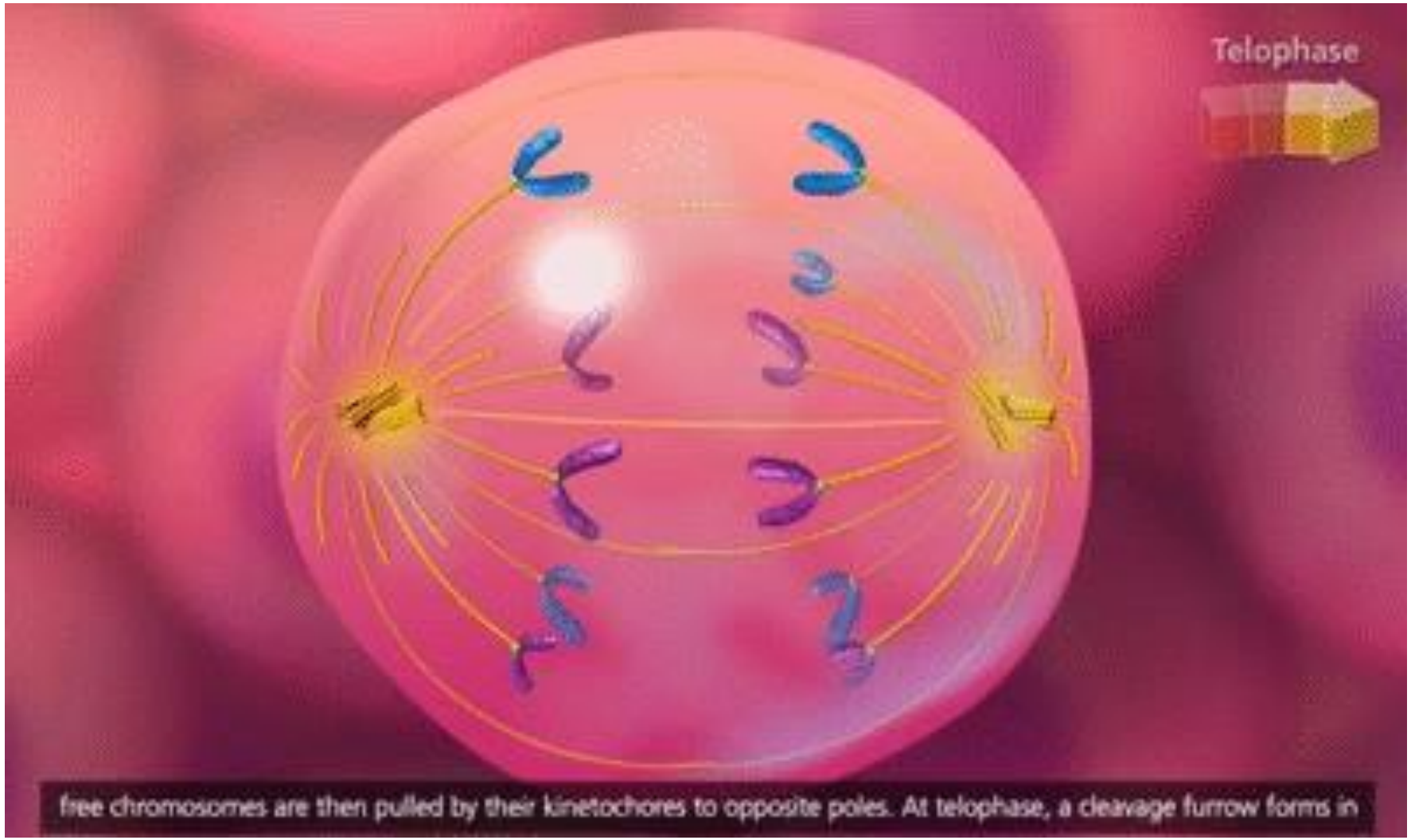
Anaphase



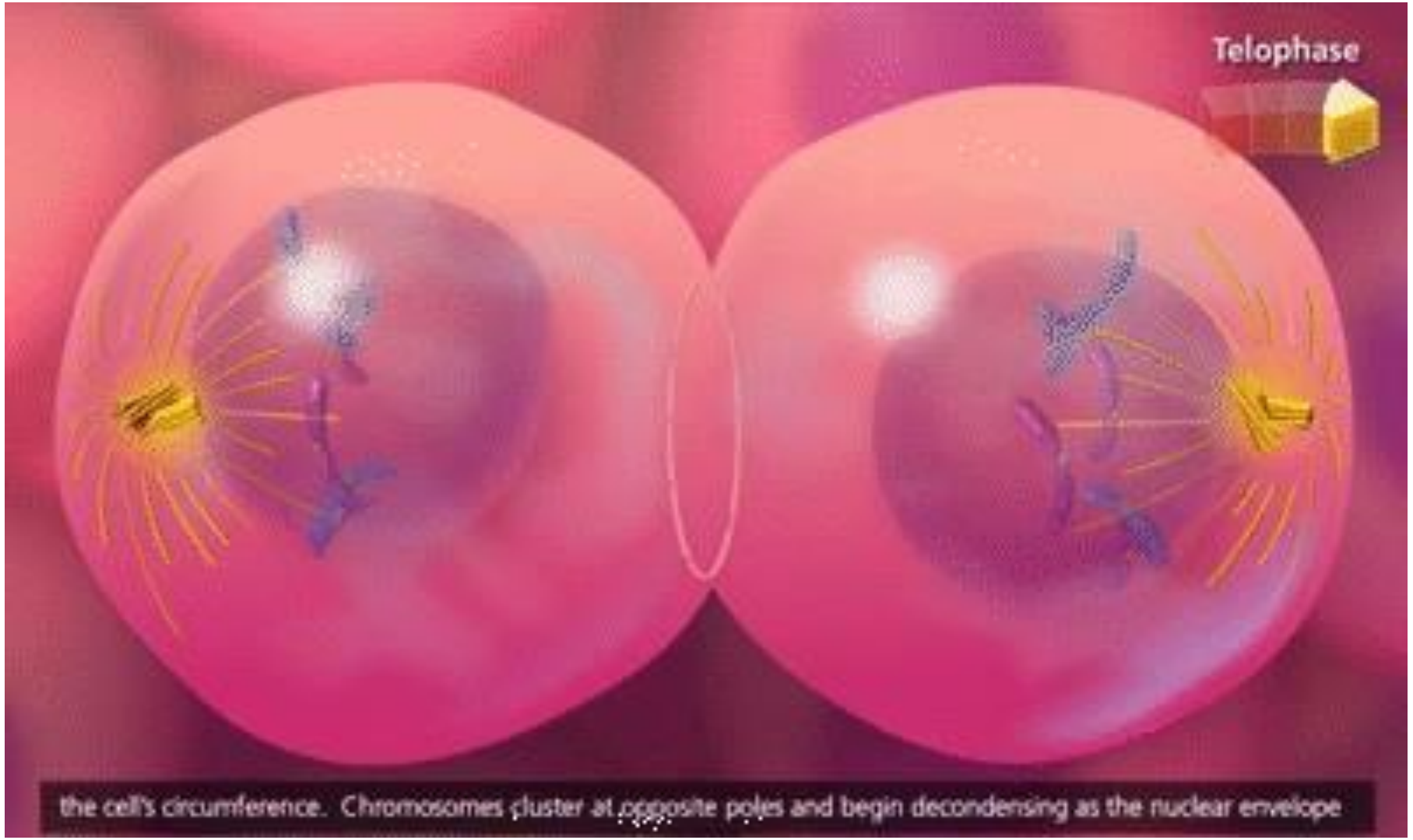
- Nonkinetochore microtubules from opposite poles overlap and push against each other, elongating the cell

- At the end of anaphase, duplicate groups of chromosomes have arrived at opposite ends of the elongated cell

Telophase


Telophase
free chromosomes are then pulled by their kinetochores to opposite poles. At telophase, a cleavage furrow forms in

Telophase & Cytokinesis



At telophase nuclear envelope and nucleolus form at each pole. Chromosomes decondense and the spindle eventually disassembles.

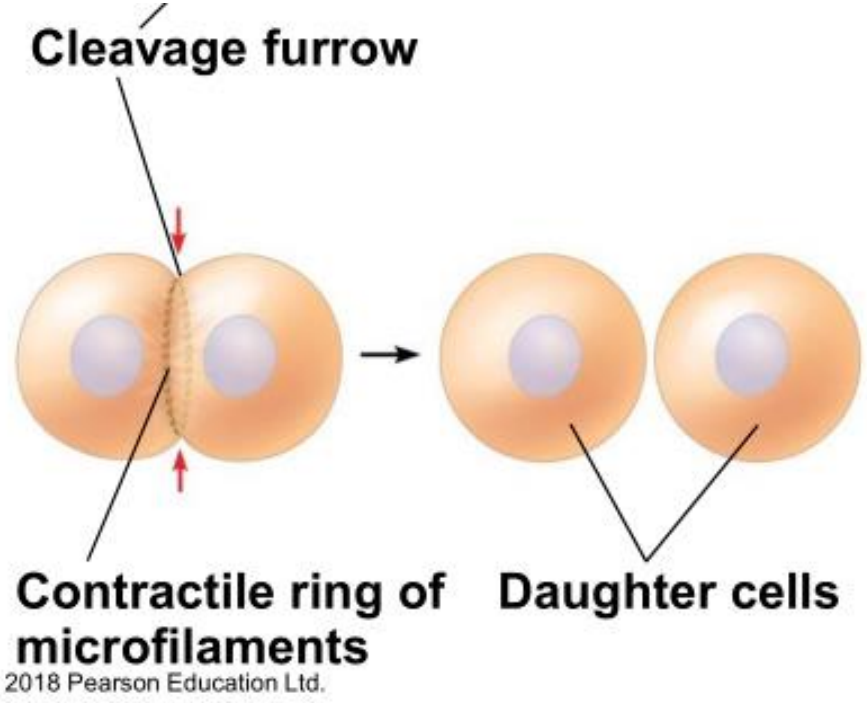
- Cytokinesis begins during anaphase or telophase.

Cytokinesis: A Closer Look

- In animal cells, cytokinesis occurs by a process known as **cleavage**, forming a **cleavage furrow** (a shallow groove in the cell surface near the old metaphase plate)

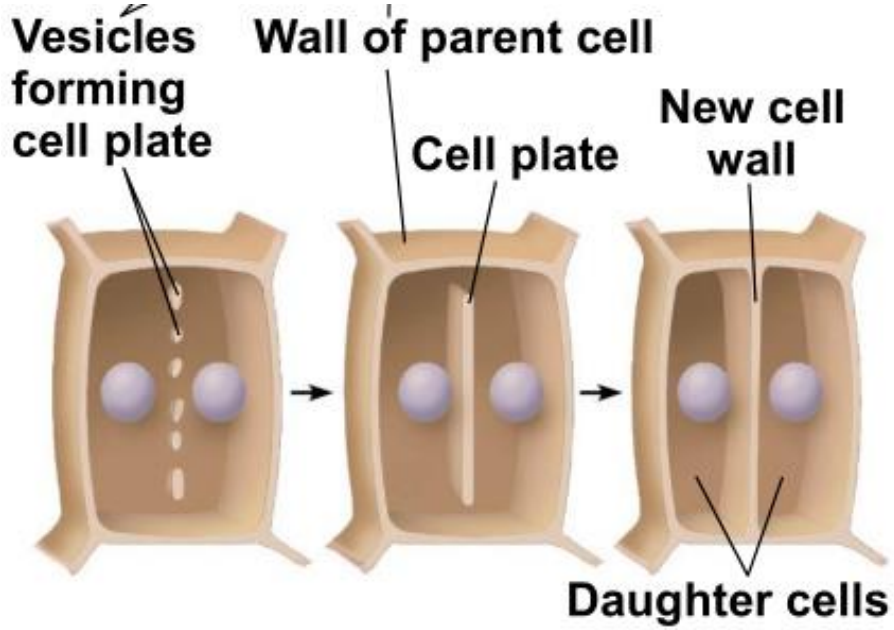
- In plant cells, a cell plate forms during cytokinesis

Cleavage of an animal cell



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Cell plate formation in a plant cell



Binary Fission (division in half)

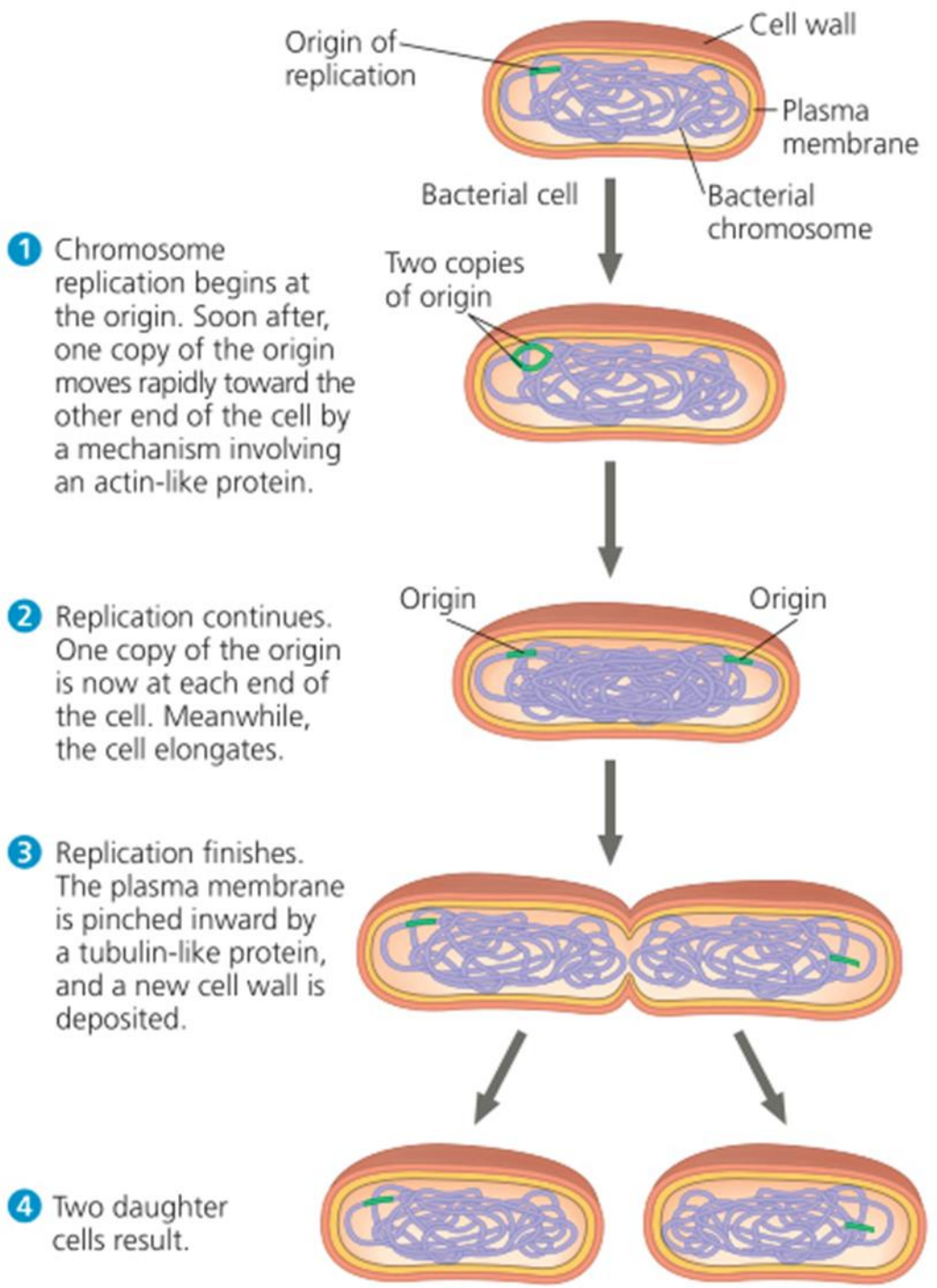
- Prokaryotes (bacteria and archaea) reproduce by a type of cell division called binary fission

- In binary fission, the chromosome replicates (beginning at the origin of replication), and the two daughter chromosomes actively move apart toward the opposite end of the cell .

While the chromosome is replicating, the cell elongates.

When replication is complete and the bacterium has reached about twice its initial size, proteins cause its plasma membrane to pinch inward, dividing the parent bacterial cell into two daughter cells.

In this way, each cell inherits a complete genome.



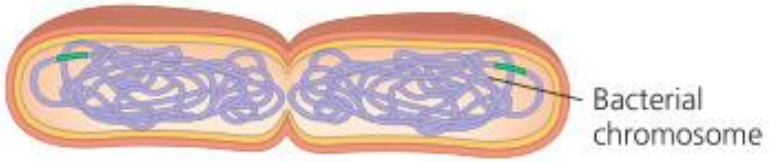
The Evolution of Mitosis

- Because prokaryotes evolved before eukaryotes, mitosis probably evolved from binary fission
- Certain protists exhibit types of cell division that seem intermediate between binary fission and mitosis

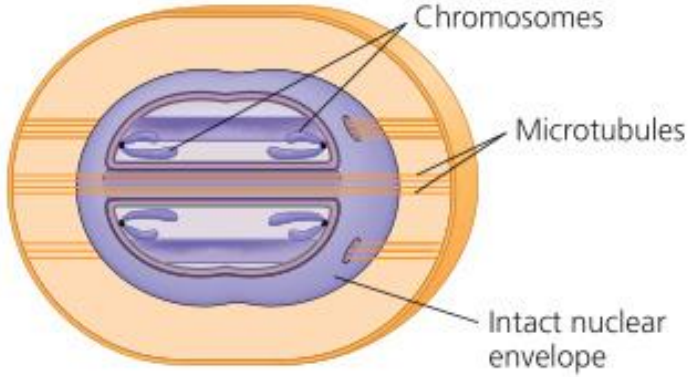
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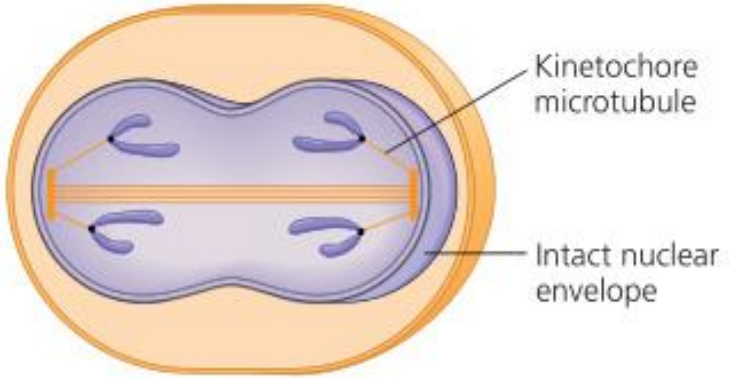
▼ Figure 12.13 Mechanisms of cell division in several groups of organisms. Some unicellular eukaryotes existing today have mechanisms of cell division that may resemble intermediate steps in the evolution of mitosis. Except for **(a)**, cell walls are not shown.



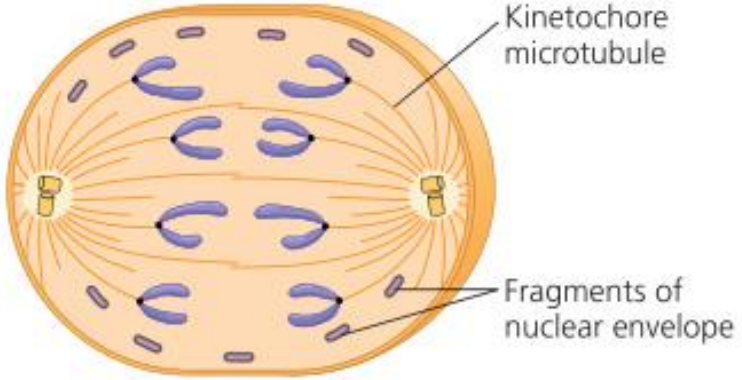
(a) Bacteria. During binary fission in bacteria, the origins of the daughter chromosomes move to opposite ends of the cell. The mechanism involves polymerization of actin-like molecules and possibly proteins that may anchor the daughter chromosomes to specific sites on the plasma membrane.



(b) Dinoflagellates. In unicellular eukaryotes called dinoflagellates, the chromosomes attach to the nuclear envelope, which remains intact during cell division. Microtubules pass through the nucleus inside cytoplasmic tunnels, reinforcing the spatial orientation of the nucleus, which then divides in a process reminiscent of bacterial binary fission.



(c) Diatoms and some yeasts. In these two other groups of unicellular eukaryotes, the nuclear envelope also remains intact during cell division. In these organisms, the microtubules form a spindle *within* the nucleus. Microtubules separate the chromosomes, and the nucleus splits into two daughter nuclei.

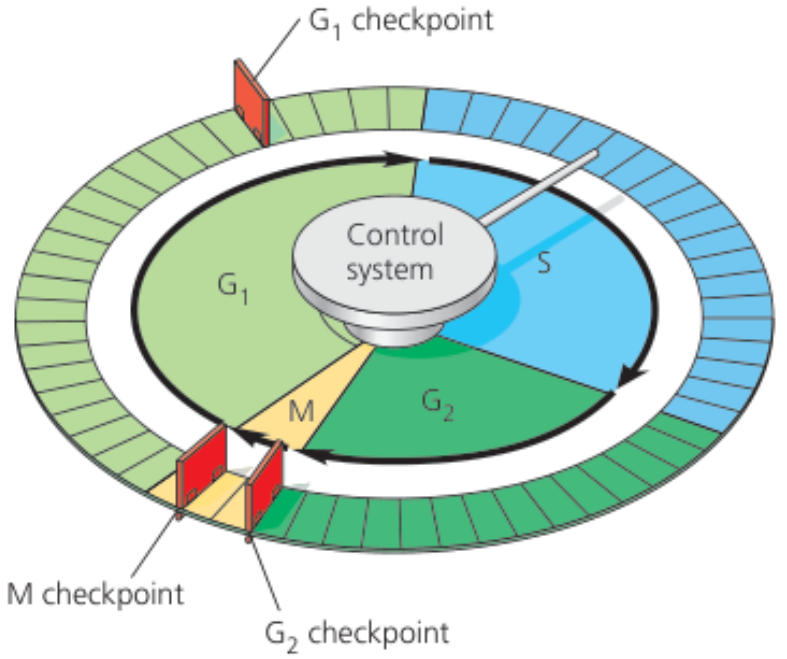


(d) Most eukaryotes. In most other eukaryotes, including plants and animals, the spindle forms outside the nucleus, and the nuclear envelope breaks down during mitosis. Microtubules separate the chromosomes, and two nuclear envelopes then form.

The Cell Cycle Control System

The eukaryotic cell cycle is regulated by a molecular control system

▪ The timing and rate of cell division in different parts of a plant or animal are crucial to normal growth, development, and maintenance. The frequency of cell division varies with the type of cell. For example, human skin cells divide frequently throughout life, whereas liver cells maintain the ability to divide but keep it in reserve until an appropriate need arises—say, to repair a wound. Some of the most specialized cells, such as fully formed nerve cells and muscle cells, do not divide at all in a mature human.



▪ These differences result from regulation at the molecular level

▪ Cancer cells manage to escape the usual controls on the cell cycle

The Cell Cycle Control System

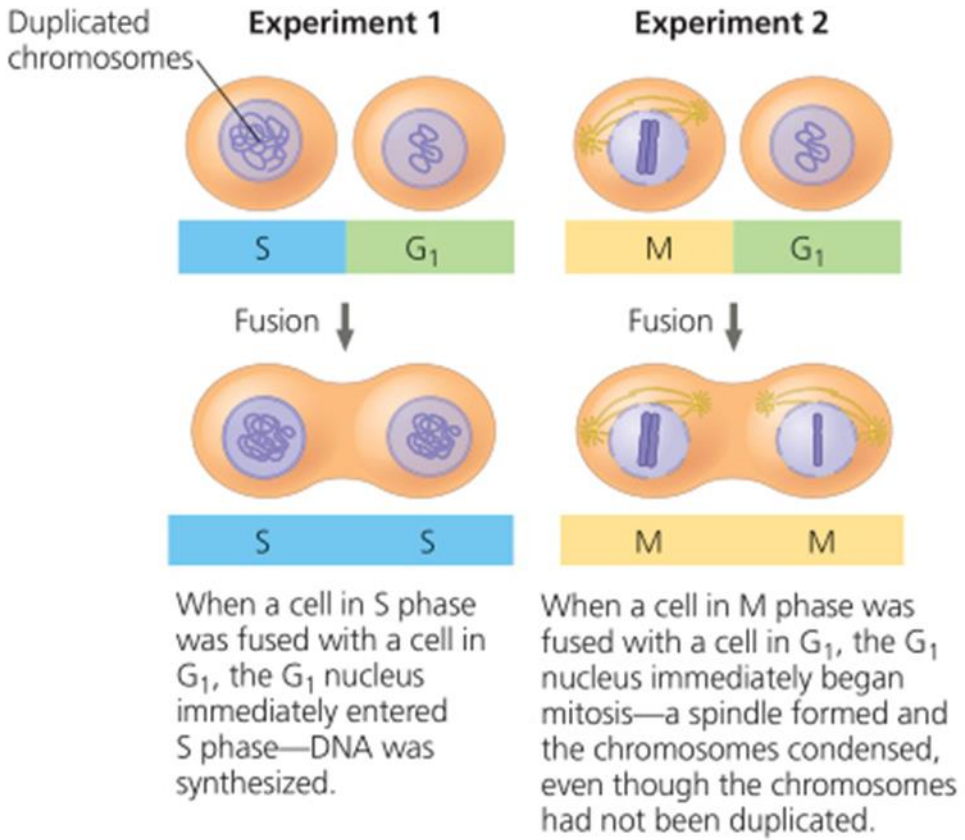
- The cell cycle appears to be driven by specific chemical signals present in the cytoplasm

- Some evidence for this hypothesis comes from experiments in which cultured mammalian cells at different phases of the cell cycle were fused to form a single cell with two nuclei

Do molecular signals in the cytoplasm regulate the cell cycle?

Experiment Researchers at the University of Colorado wondered whether a cell's progression through the cell cycle is controlled by cytoplasmic molecules. They induced cultured mammalian cells at different phases of the cell cycle to fuse. Two experiments are shown.

Results



When a cell in S phase was fused with a cell in G_1, the G_1 nucleus immediately entered S phase—DNA was synthesized.

When a cell in M phase was fused with a cell in G_1, the G_1 nucleus immediately began mitosis—a spindle formed and the chromosomes condensed, even though the chromosomes had not been duplicated.

Conclusion The results of fusing a G_1 cell with a cell in the S or M phase of the cell cycle suggest that molecules present in the cytoplasm during the S or M phase control the progression to those phases.

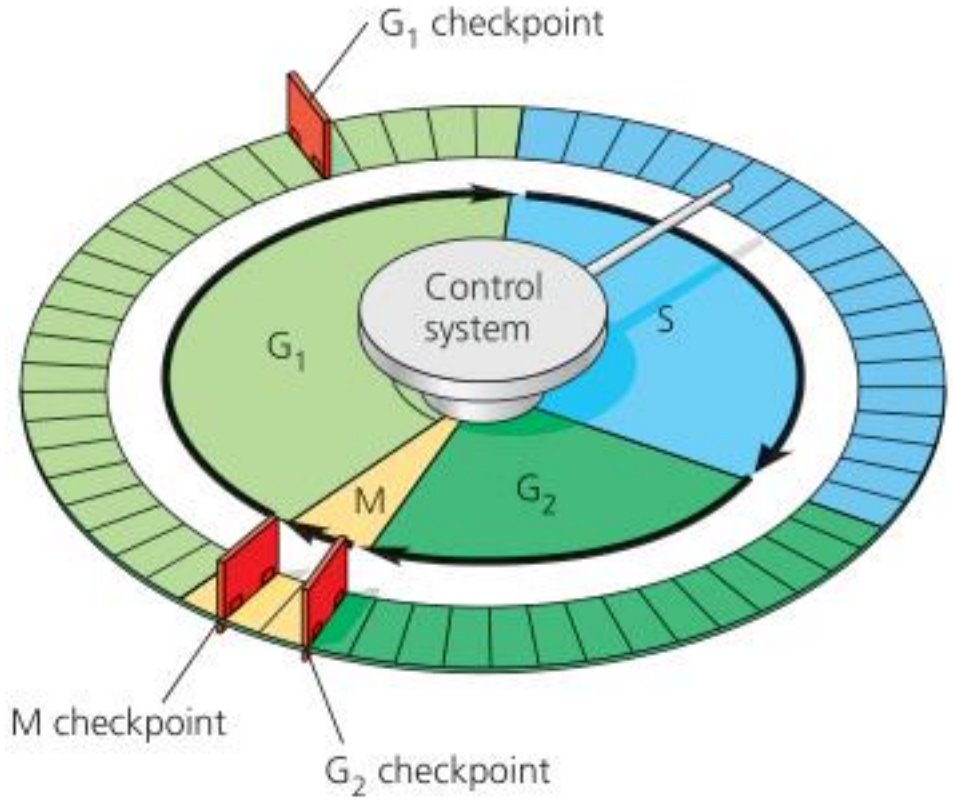
The Cell Cycle Control System

- The sequential events of the cell cycle are directed by a distinct cell cycle control system, which is similar to a clock

- The cell cycle control system is regulated by both internal and external controls

- The clock has specific checkpoints where the cell cycle stops until a go-ahead signal is received

▼ Figure 12.15 Mechanical analogy for the cell cycle control system. In this diagram, the flat "stepping stones" around the perimeter represent sequential events. Like the control device of a washing machine, the cell cycle control system proceeds on its own, driven by a built-in clock. However, the system is subject to internal and external regulation at various checkpoints; three important checkpoints are shown (red).

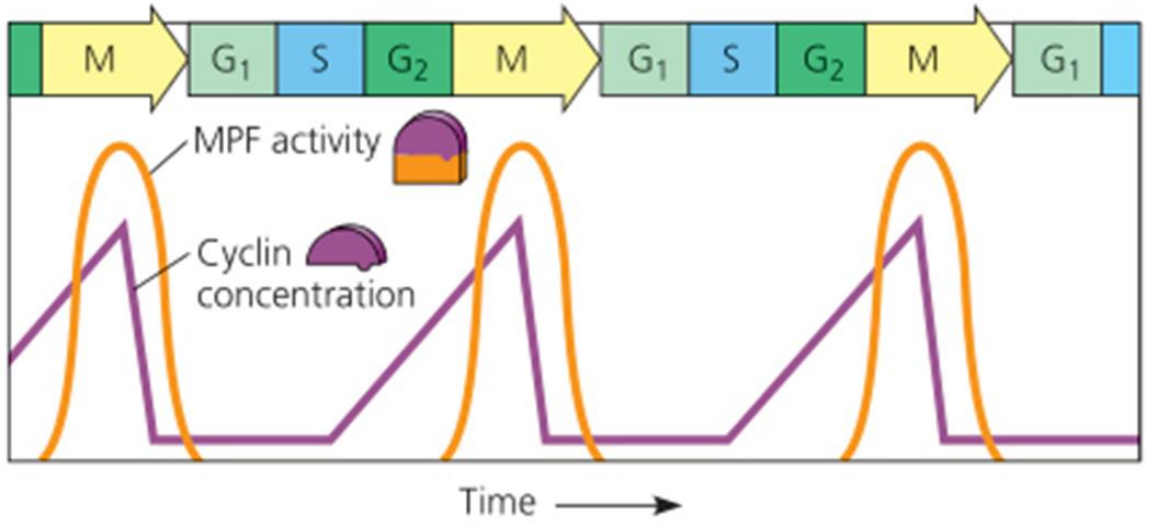


The Cell Cycle Clock: Cyclins and Cyclin-Dependent Kinases

Changes in the abundance and activity of cell cycle control molecules regulate the sequential events of the cell cycle.

- Two types of regulatory proteins are involved in cell cycle control: cyclins and cyclin-dependent kinases (Cdks) (protein kinases are enzymes that activate or inactivate other proteins by phosphorylating them.) (To be active, Cdks must be attached to a cyclin.)
- MPF (maturation-promoting factor) is a cyclin-Cdk complex that triggers a cell's passage past the G2 checkpoint into the M phase. When cyclins that accumulate during G2 associate with Cdk molecules, the resulting MPF complex is active—it phosphorylates a variety of proteins, initiating mitosis.

▼ Figure 12.16 Molecular control of the cell cycle at the G_2 checkpoint. The steps of the cell cycle are timed by rhythmic fluctuations in the activity of cyclin-dependent kinases (Cdks). Here we focus on a cyclin-Cdk complex in animal cells called MPF, which acts at the G_2 checkpoint as a go-ahead signal, triggering the events of mitosis.



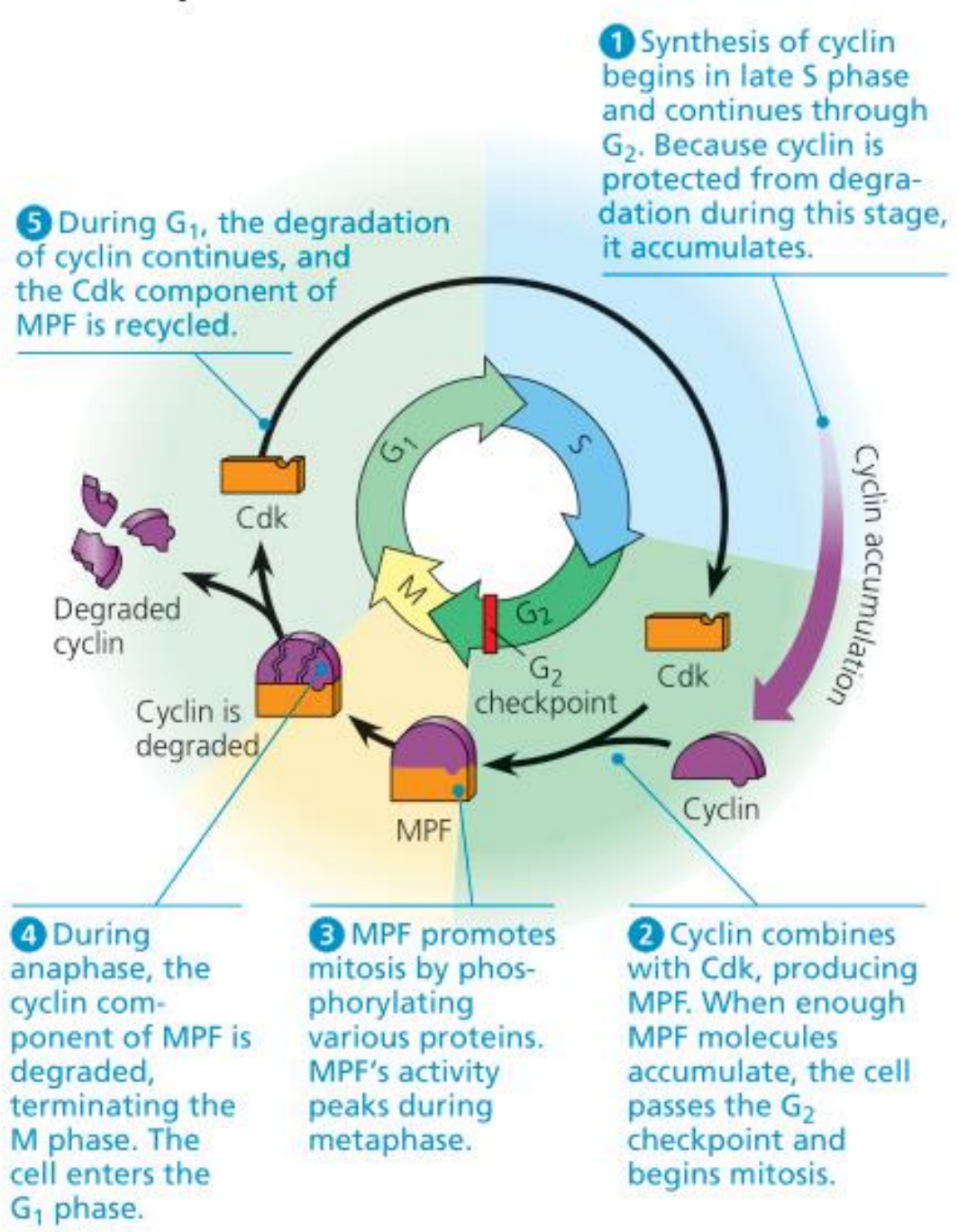
(a) Fluctuation of MPF activity and cyclin concentration during the cell cycle

- The activity of a Cdk rises and falls with changes in concentration of its cyclin partner.

The cyclin level rises during the S and G2 phases and then falls abruptly during M phase.

The Cell Cycle Clock:

For example, MPF causes phosphorylation of various proteins of the nuclear lamina, which promotes fragmentation of the nuclear envelope during prometaphase of mitosis.

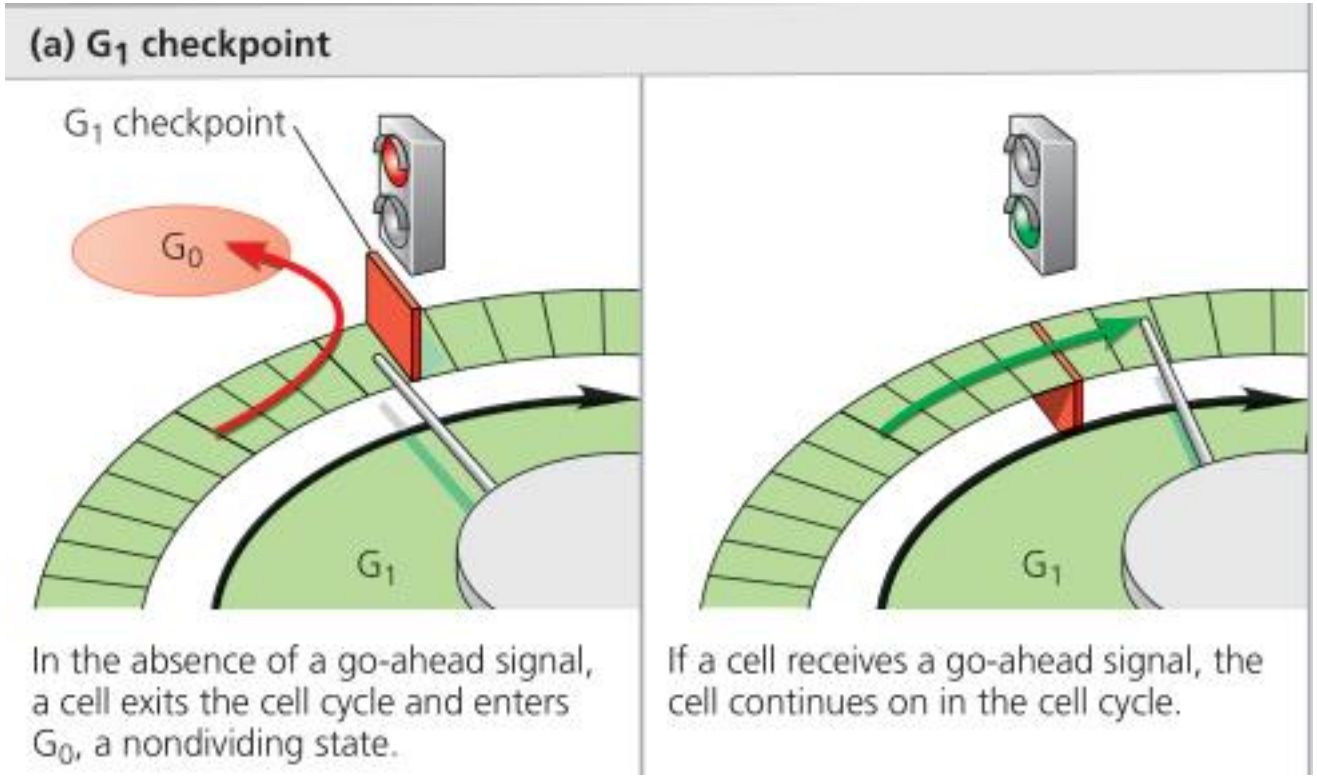


(b) Molecular mechanisms that help regulate the cell cycle

Stop and Go Signs: Internal and External Signals at the Checkpoints

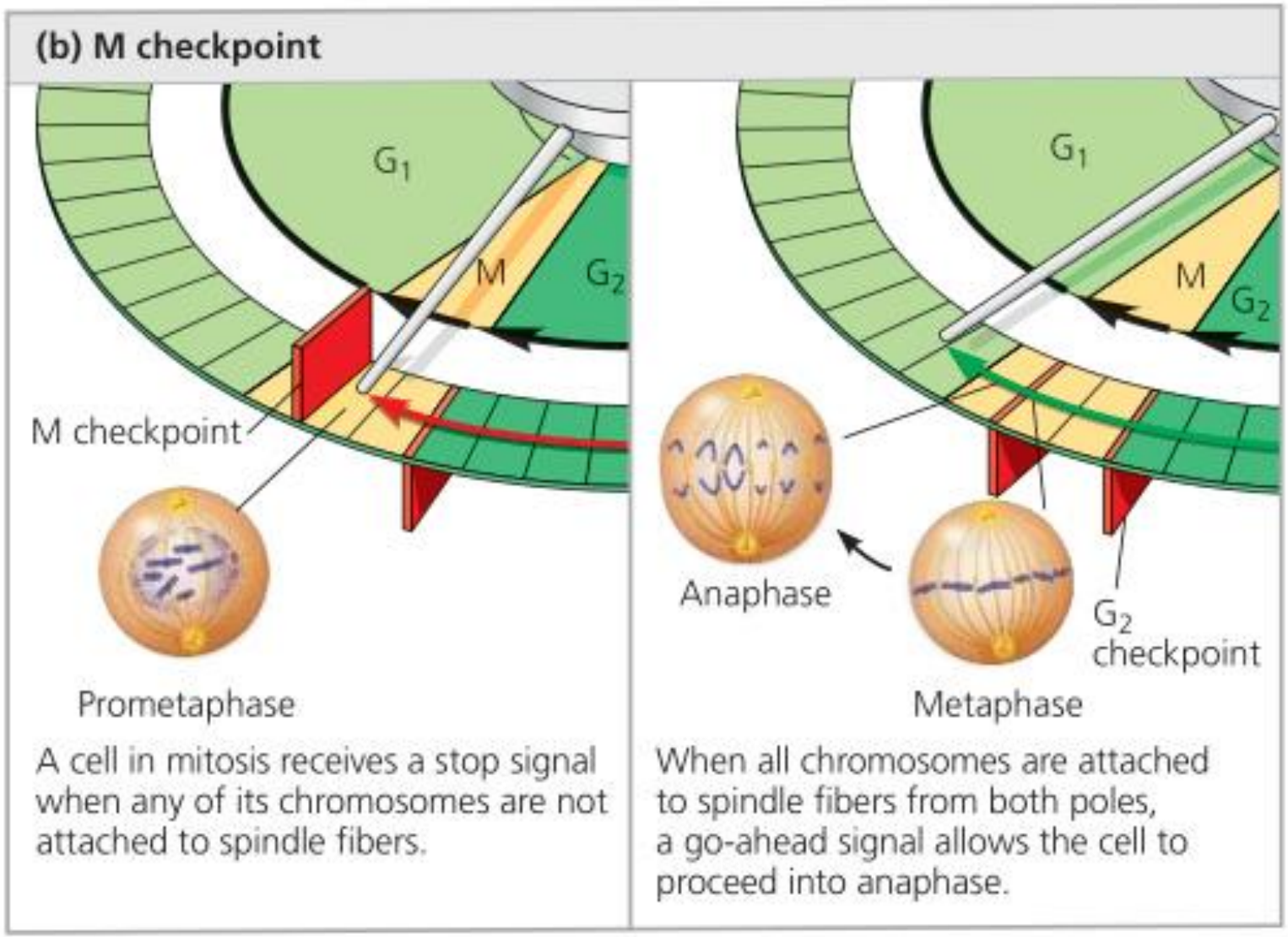
- Three important checkpoints are those in the G1, G2, and M phases

- For many cells, the G1 checkpoint seems to be the most important
- If a cell receives a go-ahead signal at the G1 checkpoint, it will usually complete the S, G2, and M phases and divide
- If the cell does not receive the go-ahead signal, it will exit the cycle, switching into a nondividing state called the G0 phase



Most cells of the human body are actually in the G0 phase. Mature nerve cells and muscle cells never divide. Other cells, such as liver cells, can be “called back” from the G0 phase to the cell cycle by external cues, such as growth factors released during injury.

Stop and Go Signs: Internal and External Signals at the Checkpoints



Anaphase, the separation of sister chromatids, does not begin until all the chromosomes are properly attached to the spindle at the metaphase plate. as long as some kinetochores are unattached to spindle microtubules, the sister chromatids remain together, delaying anaphase. Only when the kinetochores of all the chromosomes are properly attached to the spindle does the appropriate regulatory protein complex become activated. Once activated, the complex sets off a chain of molecular events that activates the enzyme separase, which cleaves the cohesins, allowing the sister chromatids to separate. This mechanism ensures that daughter cells do not end up with missing or extra chromosomes.

Stop and Go Signs: Internal and External Signals at the Checkpoints

There are checkpoints in addition to those in G1, G2, and M.

A checkpoint in S phase stops cells with DNA damage from proceeding in the cell cycle.

Another checkpoint between anaphase and telophase ensures anaphase is completed and the chromosomes are well separated before cytokinesis can begin, thus avoiding chromosomal damage.

Stop and Go Signs: Internal and External Signals at the Checkpoints

- Animal cells generally have built-in stop signals that halt the cell cycle at checkpoints until overridden by go-ahead signals.

- Many signals registered at checkpoints come from cellular surveillance mechanisms within the cell. These signals report whether crucial cellular processes that should have occurred by that point have in fact been completed correctly and thus whether or not the cell cycle should proceed.

- An example of an internal signal is that cells will not begin anaphase until all chromosomes are properly attached to the spindle at the metaphase plate

- This mechanism ensures that daughter cells have the correct number of chromosomes

Stop and Go Signs: Internal and External Signals at the Checkpoints

- Checkpoints also register signals from outside the cell

- External factors that influence cell division include specific **growth factors** (a growth factor is a protein released by certain cells that stimulates other cells to divide.)

- Platelet-derived growth factor (PDGF) is made by blood cell fragments called platelets. When an injury occurs, platelets release PDGF in the vicinity. The PDGF molecules bind to receptor tyrosine kinases on injured fibroblasts triggering a signal transduction pathway that allows the cells to pass the G1 checkpoint and divide. Proliferation of fibroblasts helps heal the wound.

- In **density-dependent inhibition**, crowded cells stop dividing. the binding of a cell-surface protein to its counterpart on an adjoining cell sends a signal to both cells that inhibits cell division, preventing them from moving forward in the cell cycle, even in the presence of growth factors.

Stop and Go Signs: Internal and External Signals at the Checkpoints

- Most cells also exhibit **anchorage dependence**—to divide, they must be attached to a substratum (the extracellular matrix of a tissue) anchorage is signaled to the cell cycle control system via pathways involving plasma membrane proteins and elements of the cytoskeleton linked to them.

- Density-dependent inhibition and anchorage dependence check the growth of cells at an **optimal density**

- **Cancer cells** do no exhibit regulation of their division.

Loss of Cell Cycle Controls in Cancer Cells

- Cancer cells do not respond normally to the body's control mechanisms

- Cancer cells do not need growth factors to grow and divide which suggests either of the following scenarios:

- They may make their own growth factor

- They may convey a growth factor's signal without the presence of the growth factor

- They may have an abnormal cell cycle control system

Loss of Cell Cycle Controls in Cancer Cells

▪ There are other important differences between normal cells and cancer cells that reflect malfunctions of the cell cycle.

▪ cancer cells stop dividing (if they do) at random points in the cycle, rather than at the normal check-points.

▪ Cancer cells can go on dividing indefinitely in culture if they are given a continual supply of nutrients.

A normal cell is converted to a cancerous cell by a process called **transformation (**causing them to divide indefinitely) (nearly all normal, non-transformed mammalian cells growing in culture divide only about 20 to 50 times before they stop dividing, age, and die.)

- Cancer cells evade the normal controls that trigger a cell to undergo programmed cell death when something is wrong—for example, when an irreparable mistake has occurred during DNA replication preceding mitosis.

Loss of Cell Cycle Controls in Cancer Cells

- Cancer cells that are not eliminated by the immune system form tumors (masses of abnormal cells within otherwise normal tissue)

- If abnormal cells remain only at the original site, the lump is called a **benign tumor**

- **Malignant tumors** invade surrounding tissues and can undergo **metastasis** (the spread of cancer cells to other parts of the body) where they may form additional tumors

▪ Localized tumors may be treated with high-energy radiation, which damages the DNA in the cancer cells

▪ To treat metastatic cancers, chemotherapies that target the cell cycle may be used

Meiosis and Sexual Life Cycles

Living organisms are distinguished by their ability to reproduce their own kind

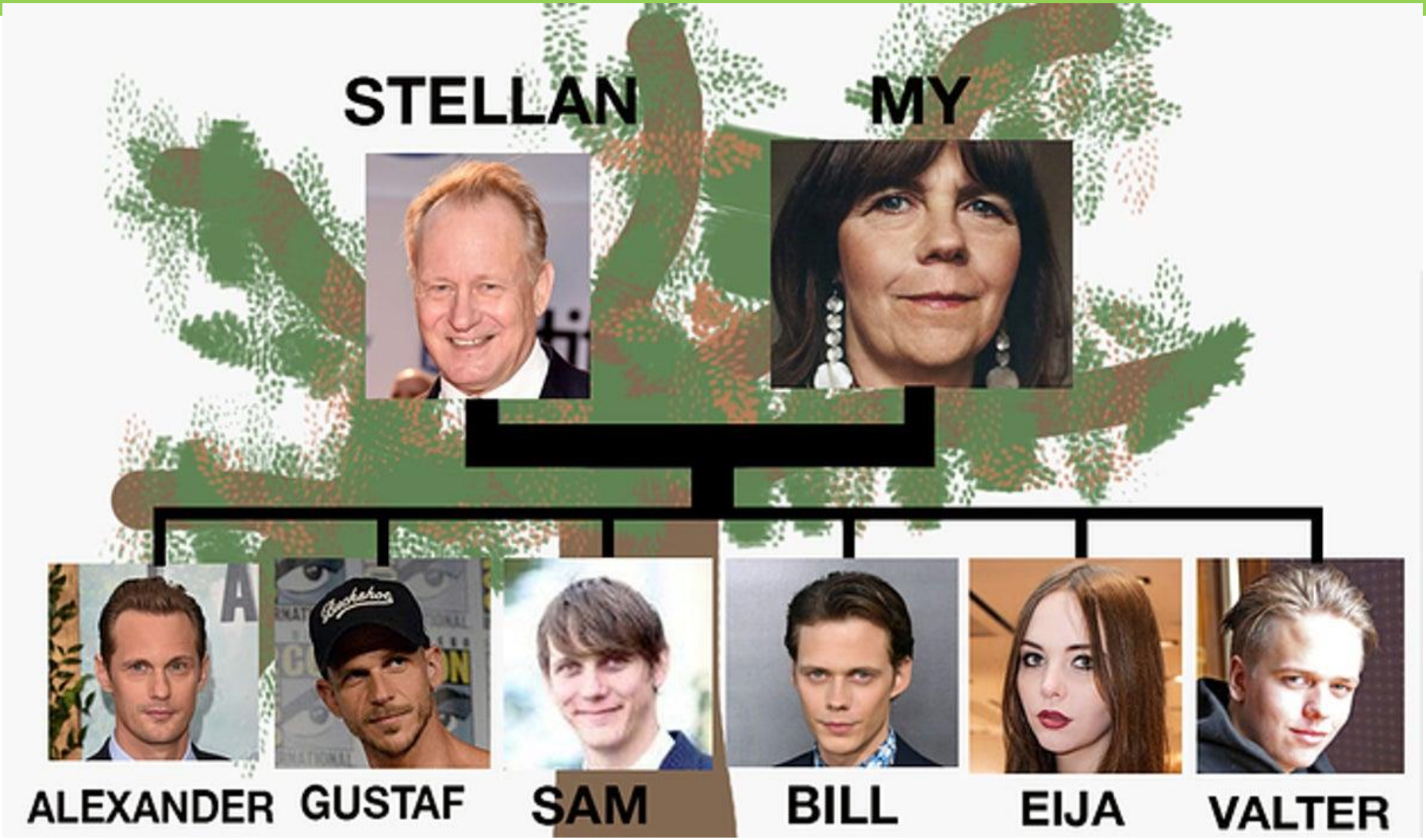
Heredity (Inheritance): transmission of traits from one generation to the next

Our genes are responsible for specific traits that emerge as we develop from fertilized eggs into adults.

Variation: the differences in appearance that offspring show from parents and siblings.

Genetics: scientific study of heredity and variation

Offspring acquire genes from parents by inheriting chromosomes.

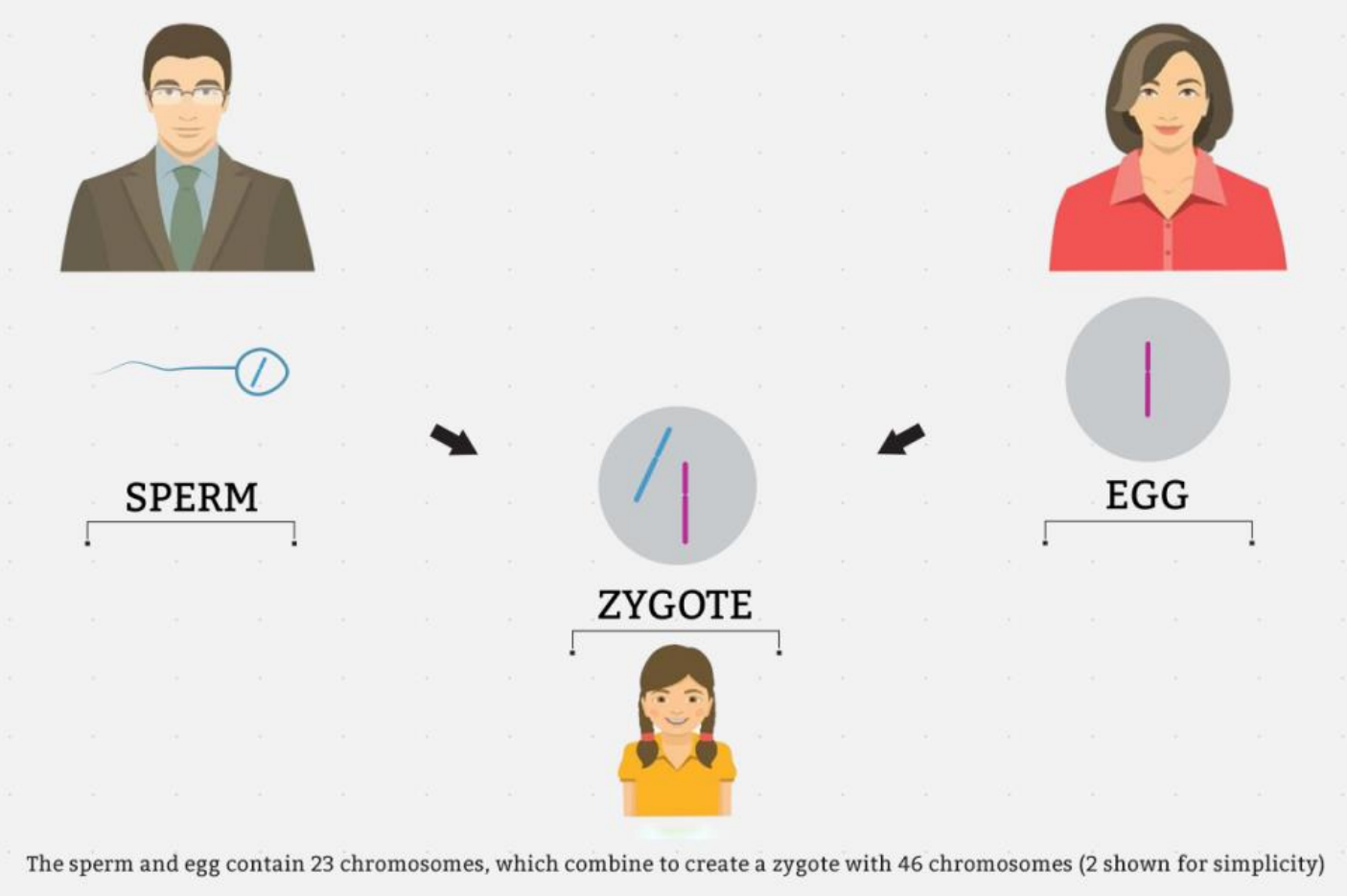


The genes we inherit from our mothers and fathers are our genetic link to our parents, and they account for family resemblances such as shared eye color or freckles.

Offspring acquire genes from parents by inheriting chromosomes.

- Genes are the units of heredity, and are made up of segments of DNA

- In animals and plants, genes are passed to the next generation through reproductive cells called **gametes** (sperm and eggs)
During fertilization, male and female gametes (sperm and eggs) unite, passing on
genes of both parents to their offspring.



The sperm and egg contain 23 chromosomes, which combine to create a zygote with 46 chromosomes (2 shown for simplicity)

Offspring acquire genes from parents by inheriting chromosomes.

- Every species has a characteristic number of chromosomes.

Humans have 46 chromosomes in their **somatic cells**, all cells of the body except gametes and their precursors.

Each chromosome consists of a single long DNA molecule, elaborately coiled in association with various proteins. One chromosome includes several hundred to a few thousand genes, each gene being a precise sequence of nucleotides along the DNA molecule.

Organism	Chromosomes in Gamete	Chromosomes in Somatic Cell
Human	23	46
Mouse	20	40
Flatworm	8	16
Corn	10	20

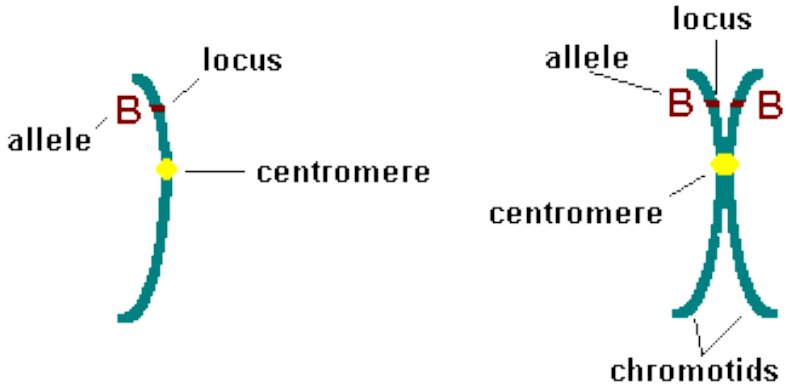
Offspring acquire genes from parents by inheriting chromosomes.

- **locus:** A gene's specific location along the length of a chromosome.

- Except for small amounts of DNA in mitochondria and chloroplasts, most DNA is packaged into chromosomes.

- One set of chromosomes is inherited from each parent.

Our genome consists of the genes and other DNA that make up the chromosomes we inherited from our parents.

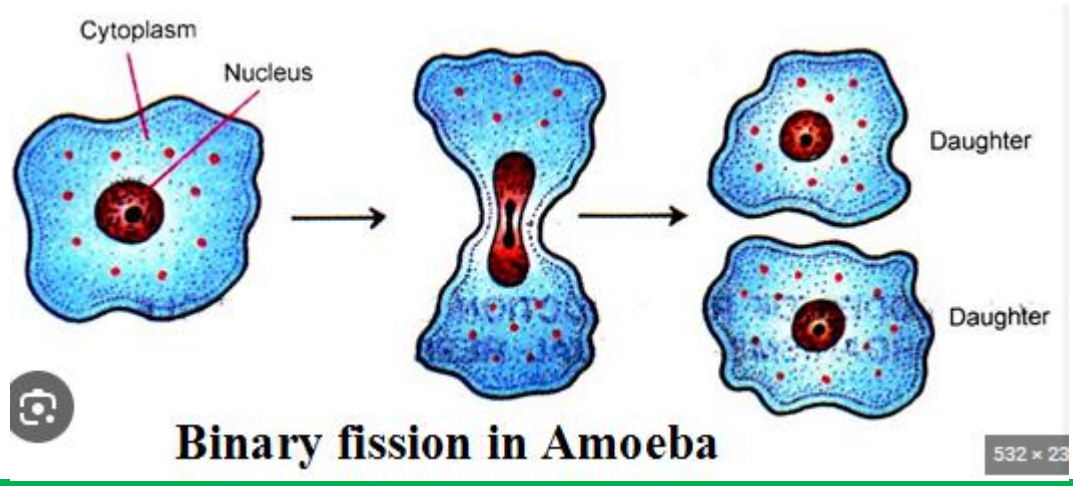


Comparison of Asexual and Sexual Reproduction

• In **asexual reproduction**, one parent produces genetically identical offspring by mitosis e.g. A yeast cell or an amoeba is the sole parent and passes copies of all its genes to its offspring without the fusion of gametes.

• Asexual reproduction gives rise to **clones**, a group of genetically identical individuals

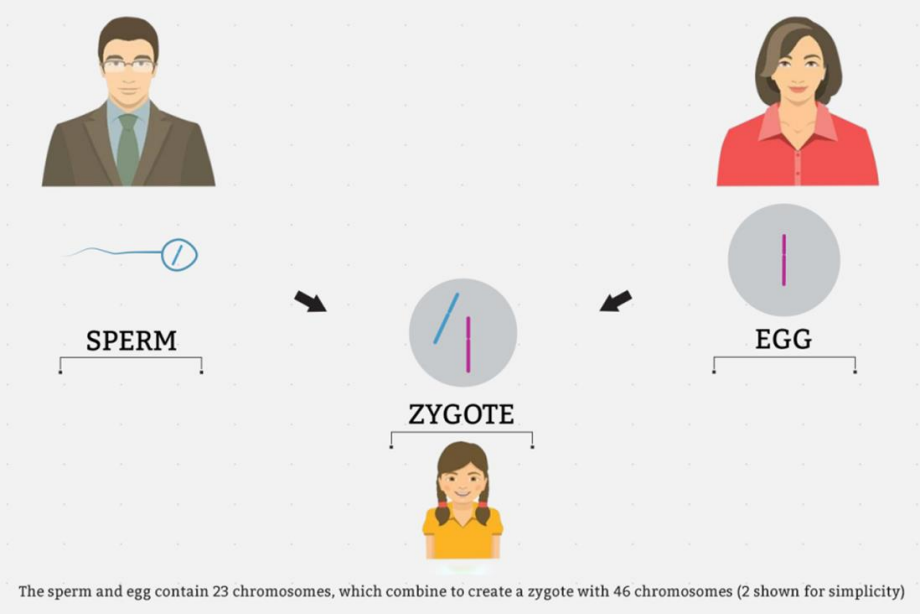
• Genetic differences occasionally arise in asexually reproducing organisms as a result of changes in the DNA called **mutations**.



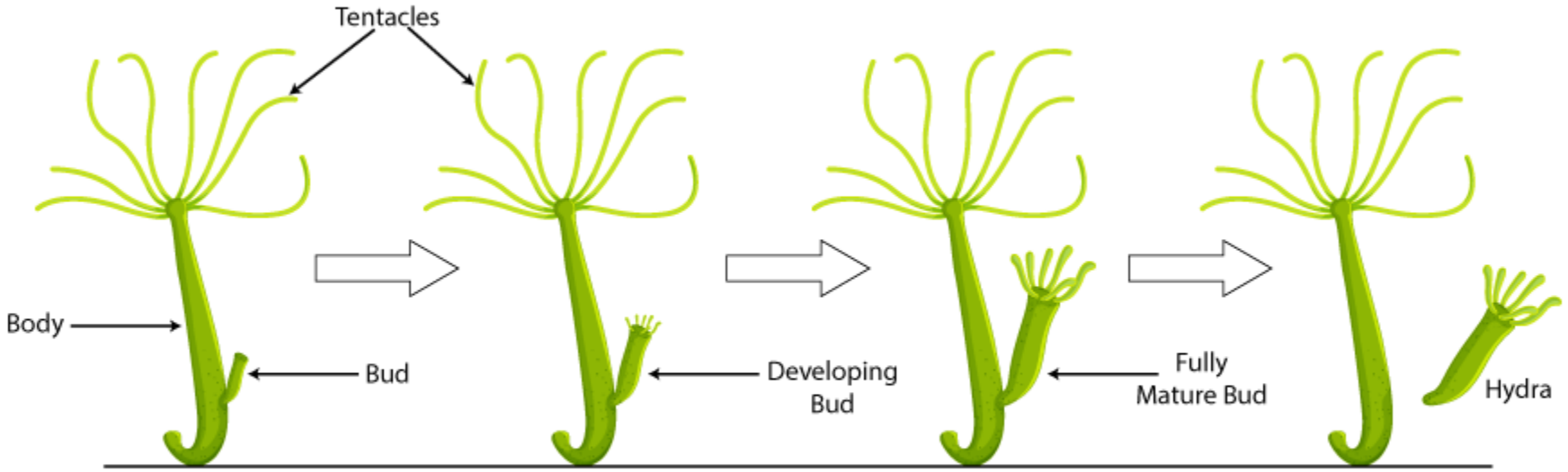
Comparison of Asexual and Sexual Reproduction

- In **sexual reproduction**, two parents give rise to offspring that have unique combinations of genes inherited from the two parents

- Offspring of sexual reproduction **vary genetically** from their parents and siblings



Asexual Reproduction in Multicellular Organisms



Budding In Hydra

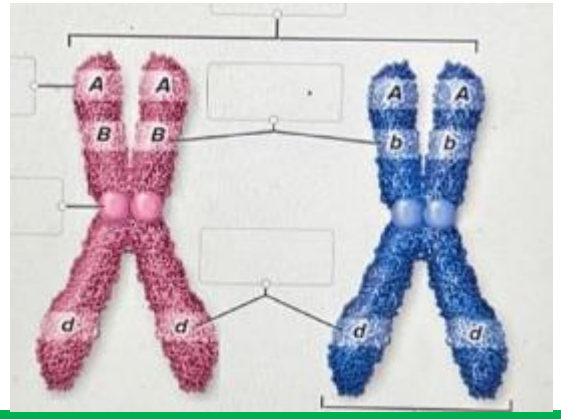
Some multicellular organisms are also capable of reproducing asexually.
This relatively simple animal, a hydra, reproduces by budding.
The bud, a localized mass of mitotically dividing cells, develops into a small hydra, which detaches from the parent.

• Human somatic cells (any cell other than a gamete) have **23 pairs** of chromosomes.

• The two chromosomes in each pair are called homologous chromosomes, or homologs.

• Chromosomes in a homologous pair are the same length, centromere position, and staining pattern and carry genes controlling the same inherited characters.

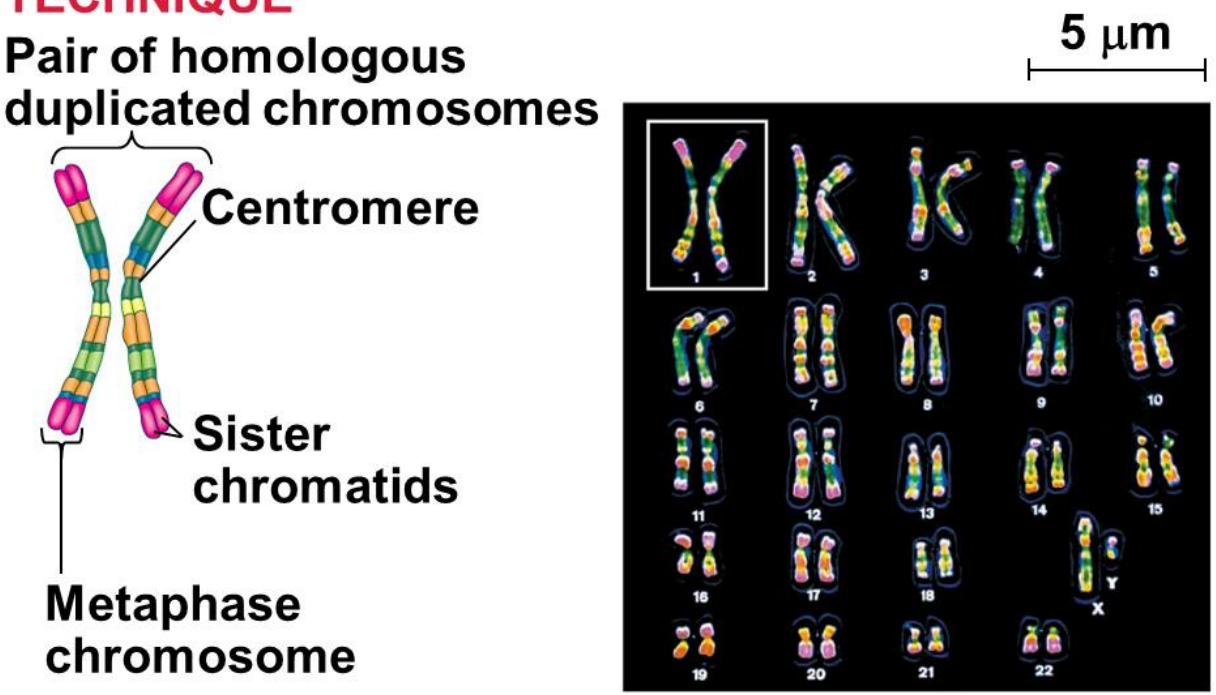
• Specific Chromosomes are identified based on : size, the position of their centromeres, and the pattern of colored bands produced by certain chromatin-binding stains.



APPLICATION



TECHNIQUE



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- **karyotype:** a display of condensed chromosomes arranged in pairs. Karyotyping can be used to screen for defective chromosomes or abnormal numbers of chromosomes associated with certain congenital disorders, such as Down syndrome

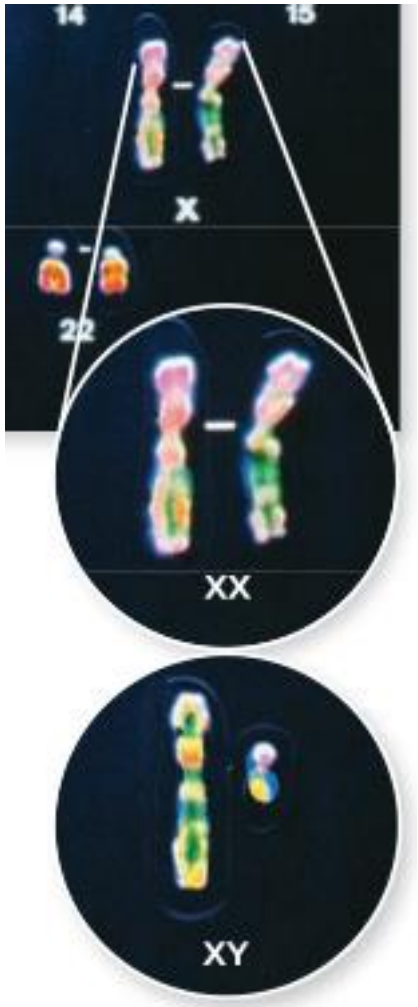
The **sex chromosomes** are called X and Y

- Human females have a homologous pair of X chromosomes (XX)

- Human males have one X and one Y chromosome (XY)

The X and Y chromosomes are not homologous except in small regions.

- The 22 pairs of chromosomes that do not determine sex are called **autosomes**.

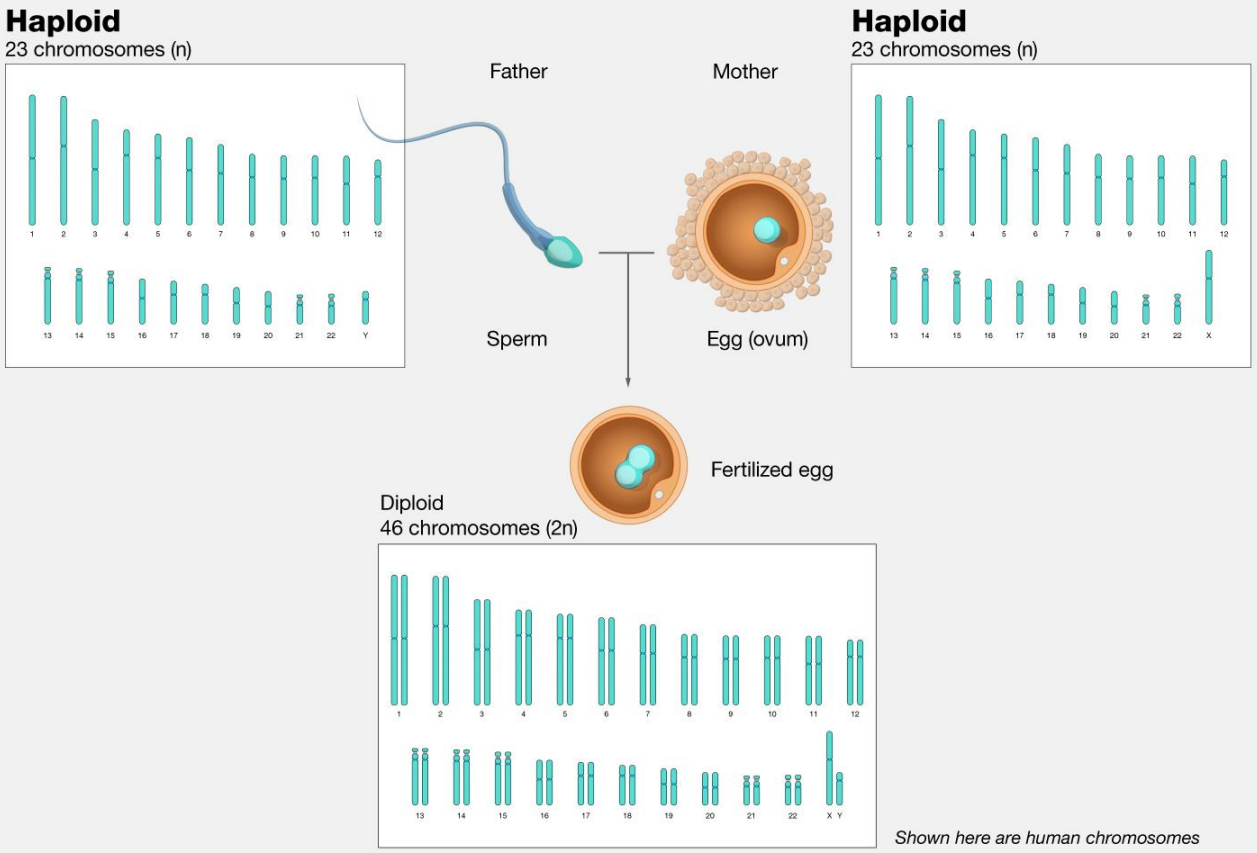


• Each pair of homologous chromosomes arises from sexual reproduction and includes one chromosome from each parent

• The 46 chromosomes in a human somatic cell are actually two sets of 23:

one from the mother (maternal) and one from the father (paternal)

• The number of chromosomes in a single set is represented by n.

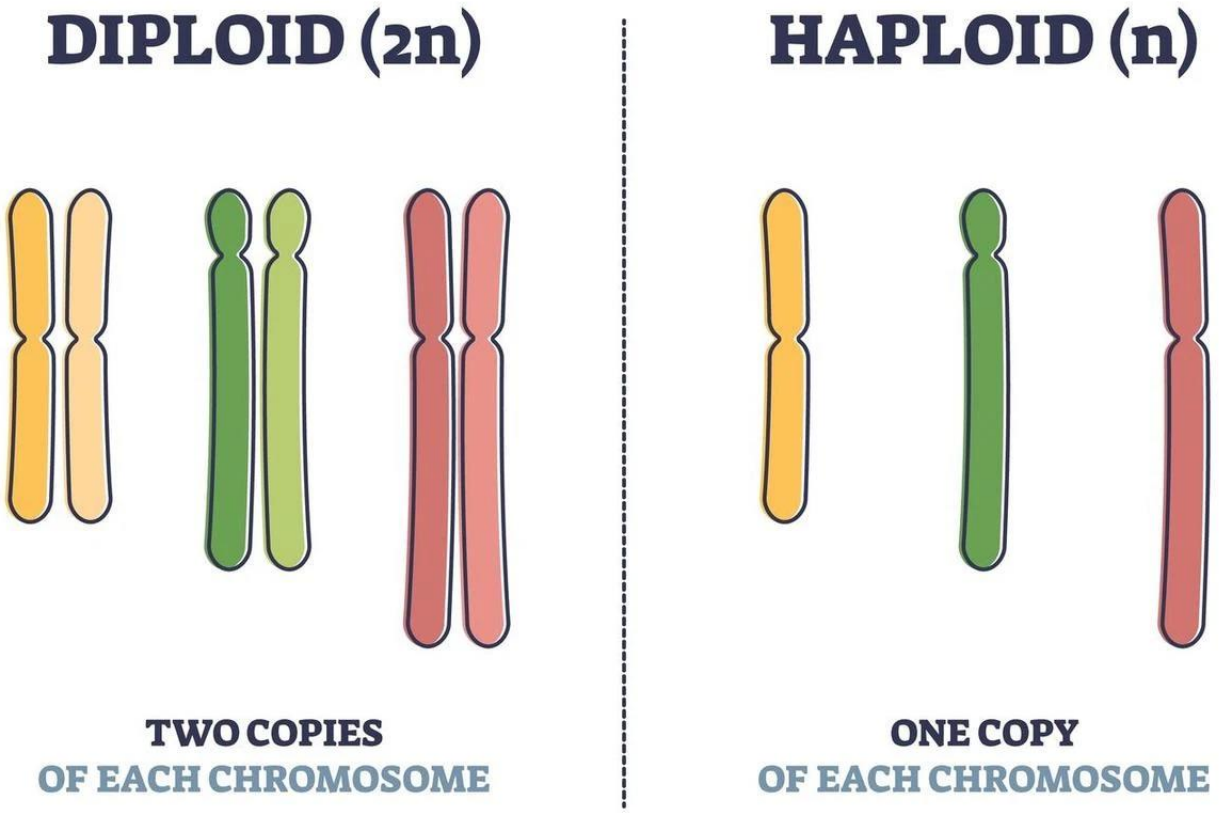


Somatic Cells

- **A diploid cell (2n)** has two sets of chromosomes

- For humans, the diploid number is 46 (2n = 46)

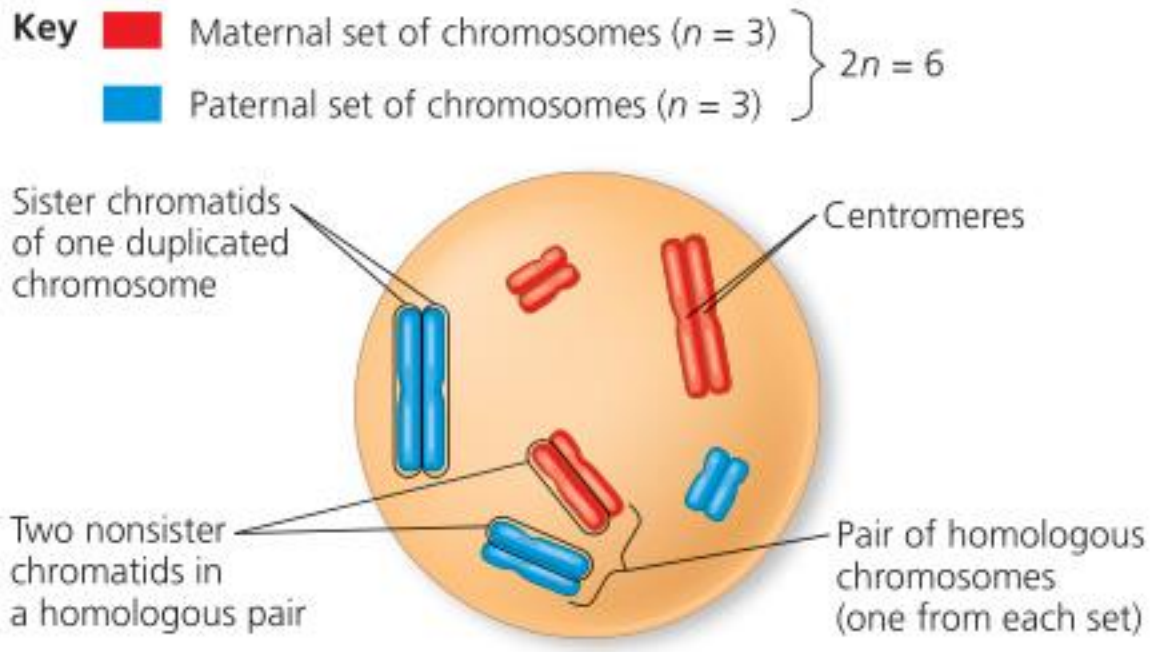
The chromosome number generally does not correlate with the size or complexity of a species' genome; it simply reflects how many linear pieces of DNA make up the genome, which is a function of the evolutionary history of that species



Describing Chromosomes

- In a cell in which DNA synthesis has occurred, each chromosome is replicated
- Each replicated chromosome consists of two identical sister chromatids

▼ Figure 13.4 Describing chromosomes. A cell from an organism with a diploid number of 6 ($2n = 6$) is depicted here following chromosome duplication and condensation. Each of the six duplicated chromosomes consists of two sister chromatids associated closely along their lengths. Each homologous pair is composed of one chromosome from the maternal set (red) and one from the paternal set (blue). Each set is made up of three chromosomes in this example (long, medium, and short). Together, one maternal and one paternal chromatid in a pair of homologous chromosomes are called nonsister chromatids.



Reproductive cell-Gametes (Sperm and Eggs)

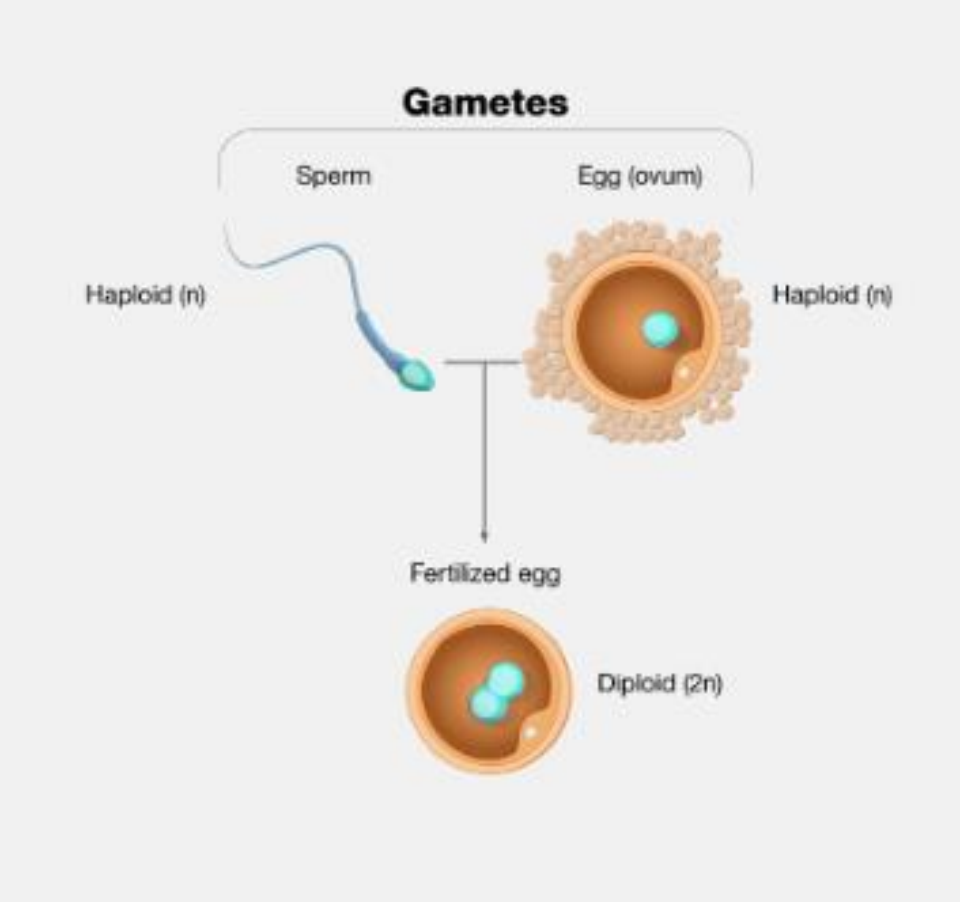
A gamete (sperm or egg) contains a **single set** of chromosomes, and is **haploid** (n)

For humans, the haploid number is 23 (n = 23)

Each set of 23 consists of 22 autosomes and a single sex chromosome

In an unfertilized egg (ovum), the sex chromosome is X

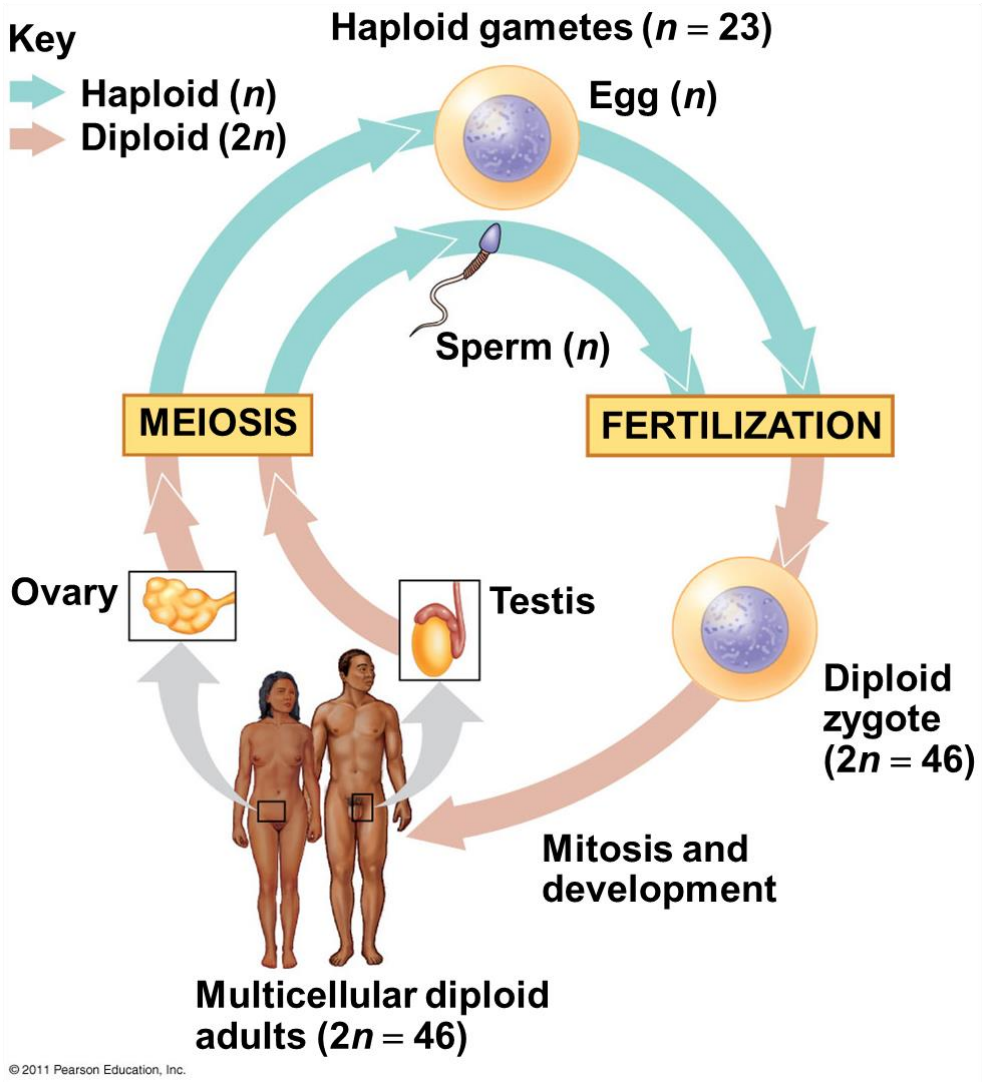
In a sperm cell, the sex chromosome may be either X or Y



Behavior of Chromosome Sets in the Human Life Cycle

- At sexual maturity, the ovaries and testes produce haploid gametes
- Gametes are the only types of human cells produced by meiosis
- Meiosis results in one set of chromosomes in each gamete (n=23)
- Fertilization and meiosis alternate in sexual life cycles to maintain chromosome number

- **Fertilization** is the union of gametes (the sperm and the egg)
- The fertilized egg is called a **zygote** and has one set of chromosomes from each parent (diploid)
- The zygote produces somatic cells by mitosis and develops into an adult



The Variety of Sexual Life Cycles

• **life cycle**: generation-to-generation sequence of stages in the reproductive history of an organism from conception to production of its own offspring.

Fertilization and meiosis alternate in sexual life cycles, maintaining a constant number of chromosomes in a species from one generation to the next.

Although the alternation of meiosis and fertilization is common to all organisms that reproduce sexually, the timing of these two events in the life cycle varies, depending on the species.

The Variety of Sexual Life Cycles

- 3 types of life cycles in different organisms are- Animal Sexual Life cycle, Plants and Algae life cycle and Fungi and Protista life cycle.

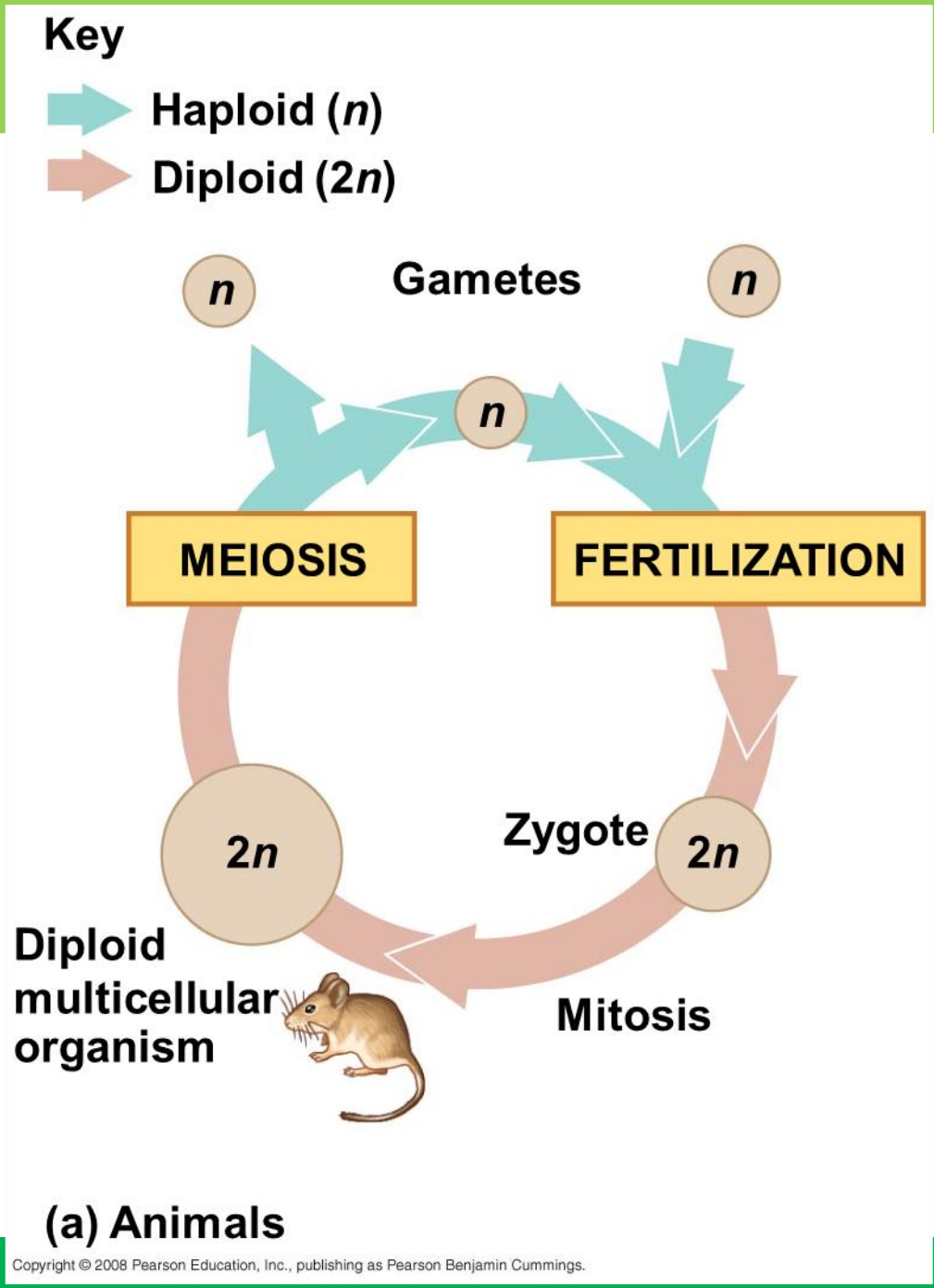
- Depending on the type of life cycle, either haploid or diploid cells can divide by mitosis

- However, only diploid cells can undergo meiosis

- In all three life cycles, the halving and doubling of chromosomes contributes to genetic variation in offspring

1- Animal Sexual Life Cycle

- In animals, meiosis produces gametes, which undergo no further cell division before fertilization
- Gametes are the only types of human cells produced by meiosis, which results in 1 set of chromosomes in each gamete (Gametes are the only haploid cells in animals)
- Fertilization and meiosis alternate in sexual life cycles to maintain chromosome number
- Gametes fuse to form a diploid zygote that divides by mitosis to develop into a multicellular organism



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2-Plants and Algae Life Cycle

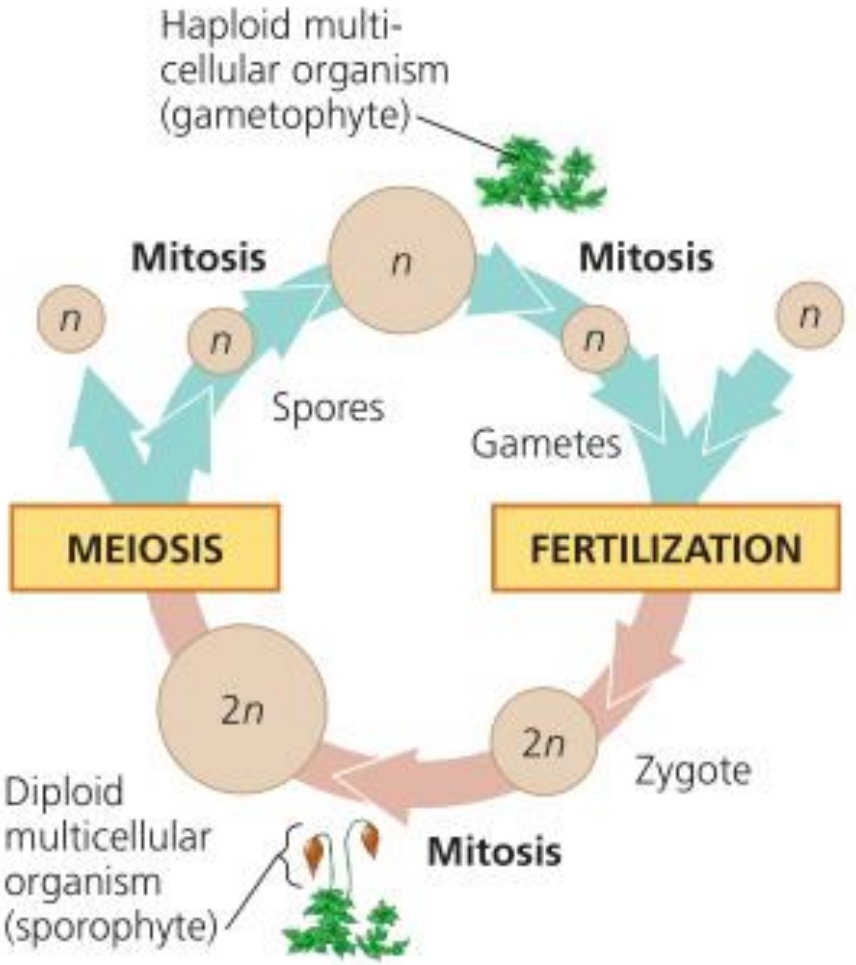
Plants and some algae exhibit an **alternation of generations**
This life cycle includes both a diploid and haploid multicellular stage
The diploid zygote, called the sporophyte, makes haploid spores by meiosis

Each spore grows by mitosis into a haploid organism called a gametophyte

A gametophyte makes haploid gametes by mitosis

Fusion of 2 haploid gametes at fertilization=a diploid zygote

The sporophyte generation produces a gametophyte offspring, which produces the next sporophyte generation.



(b) Plants and some algae

3-Fungi and Protists Life Cycle

- In most fungi and some protists, the only diploid stage is the single-celled zygote; there is no multicellular diploid stage
- The zygote produces haploid cells by meiosis
- Each haploid cell grows by mitosis into a haploid multicellular organism
- The haploid adult produces gametes by mitosis

